

CompTIA N10-009 - CompTIA Network+ Exam

Question #1 (Topic: Exam A)

SIMULATION

A network technician replaced an access layer switch and needs to reconfigure it to allow the connected devices to connect to the correct networks.

INSTRUCTIONS

Click on the appropriate port(s) on Switch 1 and Switch 3 to verify or reconfigure the correct settings:

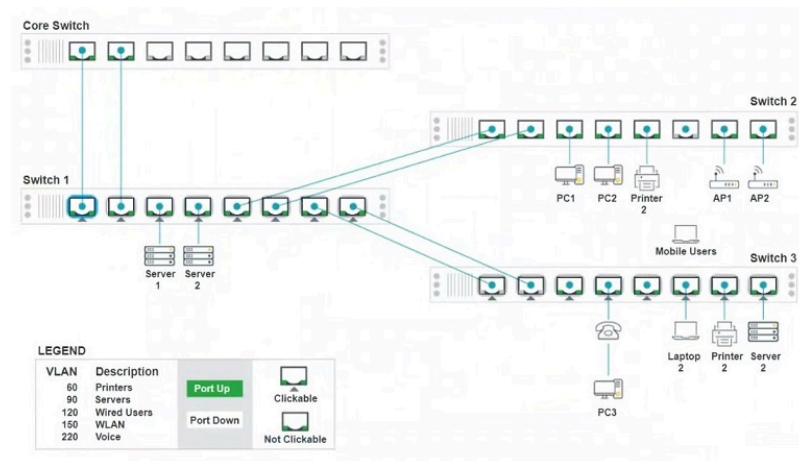
Ensure each device accesses only its correctly associated network.

Disable all unused switchports.

Require fault-tolerant connections between the switches.

Only make necessary changes to complete the above requirements.

If at any time you would like to bring back the initial state of the simulation, please click the Reset All button.



Switch 1 - Port 1 Configuration

Status

Port ☒ Enabled

LACP ☒ Enabled

Wired

Speed ☐ Auto ☐ 100 ☒ 1000

Duplex ☐ Auto ☐ Half ☒ Full

VLAN Configuration

VLAN 1

VLAN 60

VLAN 90

VLAN 120

VLAN 150

VLAN 220

VLAN60

Port Tagging

Tagged

Tagged

UnTagged

VLAN90

Port Tagging

Tagged

Tagged

UnTagged

VLAN120

Port Tagging

Tagged

Tagged

UnTagged

VLAN150

Port Tagging

Tagged

Tagged

UnTagged

VLAN220

Port Tagging

Tagged

Tagged

UnTagged

Reset to Default

Save

Close

Switch 1 - Port 2 Configuration

Status

Port ☒ Enabled

LACP ☒ Enabled

Wired

Speed ☐ Auto ☐ 100 ☒ 1000

Duplex ☐ Auto ☐ Half ☒ Full

VLAN Configuration

+ Add VLAN

VLAN 1

VLAN 60

VLAN 90

VLAN 120

VLAN 150

VLAN 220

VLAN60

Port Tagging

Tagged

Tagged

UnTagged

VLAN90

Port Tagging

Tagged

Tagged

UnTagged

VLAN120

Port Tagging

Tagged

Tagged

UnTagged

VLAN150

Port Tagging

Tagged

Tagged

UnTagged

VLAN220

Port Tagging

Tagged

Tagged

UnTagged

Reset to Default

Save

Close

Switch 1 - Port 3 Configuration ✕

Status

Port ☒ Enabled

LACP ☐ Disabled

Wired

Speed ☐ Auto ☐ 100 ☒ 1000

Duplex ☐ Auto ☐ Half ☒ Full

VLAN Configuration

+

Add VLAN

VLAN 1

VLAN 60

VLAN 90

VLAN 120

VLAN 150

VLAN 220

VLAN90

✕

Port Tagging

UnTagged

Tagged

UnTagged

Reset to Default

Save

Close

Switch 1 - Port 4 Configuration ✕

Status

Port ☒ Enabled

LACP ☐ Disabled

Wired

Speed ☐ Auto ☐ 100 ☒ 1000

Duplex ☐ Auto ☐ Half ☒ Full

VLAN Configuration

+

Add VLAN

VLAN 1

VLAN 60

VLAN 90

VLAN 120

VLAN 150

VLAN 220

VLAN90

✕

Port Tagging

UnTagged

Tagged

UnTagged

Reset to Default

Save

Close

Switch 1 - Port 5 Configuration ✕

Status

Port ☒ Enabled

LACP ☒ Enabled

Wired

Speed ☐ Auto ☐ 100 ☒ 1000

Duplex ☐ Auto ☐ Half ☒ Full

VLAN Configuration

+

Add VLAN

VLAN 1

VLAN 60

VLAN 90

VLAN 120

VLAN 150

VLAN 220

VLAN60

✕

Port Tagging

Tagged

Tagged

UnTagged

VLAN120

✕

Port Tagging

Tagged

Tagged

UnTagged

VLAN150

✕

Port Tagging

Tagged

Tagged

UnTagged

Reset to Default

Save

Close

Switch 1 - Port 6 Configuration ✕

Status

Port ☒ Enabled

LACP ☒ Enabled

Wired

Speed ☐ Auto ☐ 100 ☒ 1000

Duplex ☐ Auto ☐ Half ☒ Full

VLAN Configuration

+

Add VLAN

VLAN 1

VLAN 60

VLAN 90

VLAN 120

VLAN 150

VLAN 220

VLAN60

✕

Port Tagging

Tagged

Tagged

UnTagged

VLAN120

✕

Port Tagging

Tagged

Tagged

UnTagged

VLAN150

✕

Port Tagging

Tagged

Tagged

UnTagged

Reset to Default

Save

Close

Switch 1 - Port 7 Configuration

Status

Port ☒ Enabled

LACP ☒ Enabled

Wired

Speed ☐ Auto ☐ 100 ☒ 1000

Duplex ☐ Auto ☐ Half ☒ Full

VLAN Configuration

+ Add VLAN

VLAN 1

VLAN 60

VLAN 90

VLAN 120

VLAN 150

VLAN 220

VLAN60

Port Tagging

Tagged

Tagged

UnTagged

VLAN90

Port Tagging

Tagged

Tagged

UnTagged

VLAN120

Port Tagging

Tagged

Tagged

UnTagged

VLAN220

Port Tagging

Tagged

Tagged

UnTagged

Reset to Default

Save

Close

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6/254

Switch 1 - Port 8 Configuration

Status

Port ☒ Enabled

LACP ☒ Enabled

Wired

Speed ☐ Auto ☐ 100 ☒ 1000

Duplex ☐ Auto ☐ Half ☒ Full

VLAN Configuration

+ Add VLAN

VLAN 1

VLAN 60

VLAN 90

VLAN 120

VLAN 150

VLAN 220

VLAN60

Port Tagging

Tagged

Tagged

UnTagged

VLAN90

Port Tagging

Tagged

Tagged

UnTagged

VLAN120

Port Tagging

Tagged

Tagged

UnTagged

VLAN220

Port Tagging

Tagged

Tagged

UnTagged

Reset to Default

Save

Close

Switch 3 - Port 3 Configuration

Status

Port

☒ Enabled

LACP

☐ Disabled

Wired

Speed

☐ Auto ☐ 100 ☒ 1000

Duplex

☐ Auto ☐ Half ☒ Full

VLAN Configuration

+ Add VLAN

VLAN 1

VLAN 60

VLAN 90

VLAN 120

VLAN 150

VLAN 220

VLAN1

Port Tagging

UnTagged

Tagged

UnTagged

Reset to Default

Save

Close

Switch 3 - Port 4 Configuration

Status

Port

☒ Enabled

LACP

☐ Disabled

Wired

Speed

☐ Auto ☐ 100 ☒ 1000

Duplex

☐ Auto ☐ Half ☒ Full

VLAN Configuration

+ Add VLAN

VLAN 1

VLAN 60

VLAN 90

VLAN 120

VLAN 150

VLAN 220

VLAN1

Port Tagging

UnTagged

Tagged

UnTagged

Reset to Default

Save

Close

Switch 3 - Port 5 Configuration

Status

Port

☒ Enabled

LACP

☐ Disabled

Wired

Speed

☐ Auto ☐ 100 ☒ 1000

Duplex

☐ Auto ☐ Half ☒ Full

VLAN Configuration

+ Add VLAN

VLAN 1

VLAN 60

VLAN 90

VLAN 120

VLAN 150

VLAN 220

VLAN1

Port Tagging

UnTagged

Tagged

UnTagged

Reset to Default

Save

Close

Switch 3 - Port 6 Configuration ✕

Status

Port ☒ Enabled

LACP ☐ Disabled

Wired

Speed ☐ Auto ☐ 100 ☒ 1000

Duplex ☐ Auto ☐ Half ☒ Full

VLAN Configuration

+

Add VLAN

VLAN 1

VLAN 60

VLAN 90

VLAN 120

VLAN 150

VLAN 220

VLAN1

✕

Port Tagging

UnTagged

Tagged

UnTagged

Reset to Default

Save

Close

Switch 3 - Port 7 Configuration ✕

Status

Port ☒ Enabled

LACP ☐ Disabled

Wired

Speed ☐ Auto ☐ 100 ☒ 1000

Duplex ☐ Auto ☐ Half ☒ Full

VLAN Configuration

+

Add VLAN

VLAN 1

VLAN 60

VLAN 90

VLAN 120

VLAN 150

VLAN 220

VLAN1

✕

Port Tagging

UnTagged

Tagged

UnTagged

Reset to Default

Save

Close

Answer: Analysis and Configuration Recommendations:

Switch 1 Configurations:

- **Port 1 to Port 7:**
 - Ports should have VLANs set according to the devices they connect to. For instance, if a port is connecting to servers, only the VLAN for servers (e.g., VLAN 90) should be enabled and tagged if required.
 - Ensure unused VLANs are not active or set to untagged on these ports to prevent unauthorized network access.
- **Port 8:**
 - This port's configuration needs to align with the devices it connects to. Based on your first image, adjust the VLAN tagging accordingly. If it connects to printers, VLAN 60 should be tagged, and all other VLANs should be disabled or untagged.

Switch 3 Configurations:

- **Ports 2 to 8:**
 - The configurations here must also match the connected devices. For mobile user connections, only VLANs relevant to user access (like VLAN 150 for WLAN) should be enabled and set to tagged or untagged based on network policies.
 - Any VLAN not associated with the connected devices should be disabled to secure the network.

Fault Tolerant Connections:

- **Ensure redundancy:** If these switches are connected via multiple ports, configure Link Aggregation Control Protocol (LACP) if not already set up to provide redundancy and increased bandwidth.
- **Check duplex and speed settings:** Ensure that duplex settings are set to Auto to avoid duplex mismatch which can cause performance issues.

Disabling Unused Ports:

- Any port not connected or not planned to be used should be disabled to prevent unauthorized access or network breaches.

Final Checks:

- **Verify settings:** After configuration changes, verify all settings to ensure they adhere to network policies and the devices are performing as expected.
- **Documentation:** Update network documentation to reflect changes for future troubleshooting and audits.

Answer: Analysis and Configuration Recommendations:**Switch 1 Configurations:**

- **Port 1 to Port 7:**
 - Ports should have VLANs set according to the devices they connect to. For instance, if a port is connecting to servers, only the VLAN for servers (e.g., VLAN 90) should be enabled and tagged if required.
 - Ensure unused VLANs are not active or set to untagged on these ports to prevent unauthorized network access.
- **Port 8:**
 - This port's configuration needs to align with the devices it connects to. Based on your first image, adjust the VLAN tagging accordingly. If it connects to printers, VLAN 60 should be tagged, and all other VLANs should be disabled or untagged.

Switch 3 Configurations:

- **Ports 2 to 8:**
 - The configurations here must also match the connected devices. For mobile user connections, only VLANs relevant to user access (like VLAN 150 for WLAN) should be enabled and set to tagged or untagged based on network policies.
 - Any VLAN not associated with the connected devices should be disabled to secure the network.

Fault Tolerant Connections:

- **Ensure redundancy:** If these switches are connected via multiple ports, configure Link Aggregation Control Protocol (LACP) if not already set up to provide redundancy and increased bandwidth.
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Disabling Unused Ports:

- Any port not connected or not planned to be used should be disabled to prevent unauthorized access or network breaches.

Final Checks:

- **Verify settings:** After configuration changes, verify all settings to ensure they adhere to network policies and the devices are performing as expected.
- **Documentation:** Update network documentation to reflect changes for future troubleshooting and audits.

Question #2 (Topic: Exam A)**SIMULATION**

A network technician was recently onboarded to a company. A manager has tasked the technician with documenting the network and has provided the technician with partial information from previous documentation.

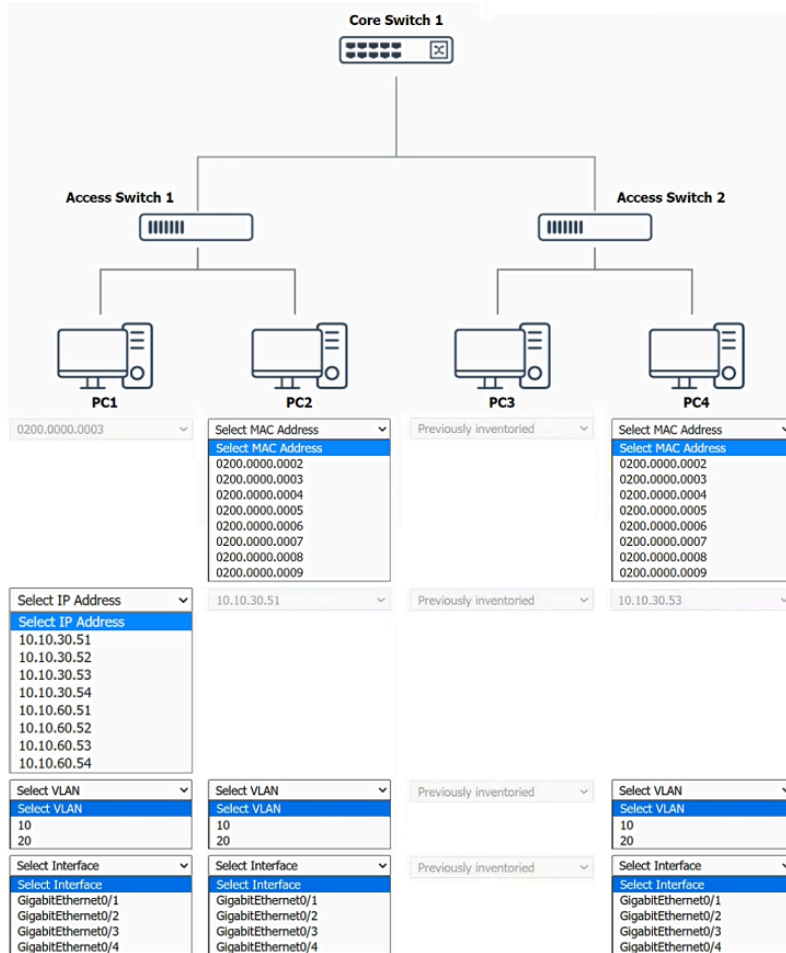
INSTRUCTIONS

Click on each switch to perform a network discovery by entering commands into the terminal. Type help to view a list of available commands.

Fill in the missing information using the drop-down menus provided.

If at any time you would like to bring back the initial state of the simulation, please

click the Reset All button.



Answer: Steps for Network Documentation:

1. **Network Discovery:**
 - Use discovery commands on each switch to collect data about connected devices. Common commands include `show mac-address-table` to view MAC addresses associated with the ports and `show ip interface brief` to check the IP configuration of switch interfaces.
2. **Document MAC and IP Addresses:**
 - From the MAC address table, document the MAC addresses linked to each port. In your provided image, select the appropriate MAC address for each PC (e.g., PC1, PC2, PC3, PC4) based on their connection to Access Switch 1 or 2.
 - Assign and document IP addresses for each device from the range available in the dropdown (e.g., 10.10.30.51 for PC1).
3. **VLAN Assignment:**
 - Assign and document VLANs for each PC according to network segmentation policies. It seems VLAN 10 and 20 are options; ensure each PC is assigned to the correct VLAN based on their network usage (e.g., regular users, admin, guest network).
4. **Interface Configuration:**
 - Document the interface each PC is connected to on their respective Access Switch. Ensure the interface matches the physical connection layout.

Final Verification:

- Use commands like `show vlan` to verify that VLAN assignments are correct across devices.
- Use `show running-config` to verify the overall configuration and ensure it aligns with security and operational policies.

Answer: Steps for Network Documentation:

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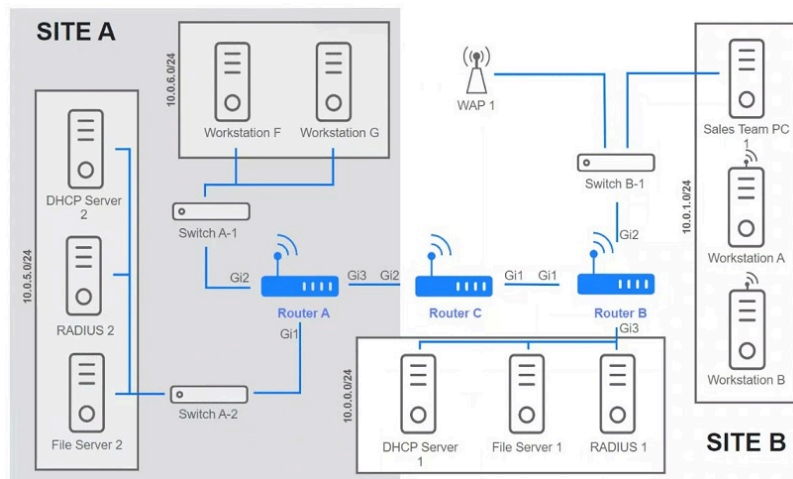
Question #3 (Topic: Exam A)**SIMULATION**

Users are unable to access files on their department share located on file server 2. The network administrator has been tasked with validating routing between networks hosting workstation A and file server 2.

INSTRUCTIONS

Click on each router to review output, identify any issues, and configure the appropriate solution.

If at any time you would like to bring back the initial state of the simulation, please click the Reset All button.



Router A

Routing Table Routing Configuration

```
Router-A# show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
        n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        H - NHRP, G - NHRP registered, g - NHRP registration summary
        o - ODR, P - periodic downloaded static route, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

S*   0.0.0.0/0 is directly connected, GigabitEthernet3
      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C     10.0.4.0/22 is directly connected, GigabitEthernet2
C     10.0.6.0/24 is directly connected, GigabitEthernet2
L     10.0.6.1/32 is directly connected, GigabitEthernet2
      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C     172.16.27.0/30 is directly connected, GigabitEthernet3
L     172.16.27.1/32 is directly connected, GigabitEthernet3
```

Reset to Default Save Close

Router A

Routing Table Routing Configuration

Was a problem found?: ☐ Yes ☒ No

Install Static Route

Destination Prefix:

Destination Prefix Mask:

Interface:

Reset to Default Save Close

Router A

Routing Table Routing Configuration

Was a problem found?: ☒ Yes ☐ No

Install Static Route

Destination Prefix:

Destination Prefix Mask:

Interface:

Gi1
Gi2
Gi3
Gi4

Reset to Default Save Close

Router B

Routing Table Routing Configuration

```
Router-B# show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
        n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        H - NHRP, G - NHRP registered, g - NHRP registration summary
        o - ODR, P - periodic downloaded static route, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

S*   0.0.0.0/0 is directly connected, GigabitEthernet1
      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C     10.0.0.0/22 is directly connected, GigabitEthernet3
L     10.0.0.1/32 is directly connected, GigabitEthernet3
      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C     172.16.27.4/30 is directly connected, GigabitEthernet1
L     172.16.27.5/32 is directly connected, GigabitEthernet1
```

Reset to Default Save Close

Router B ✕

Routing Table

Routing Configuration

Was a problem found?: ☐ Yes ☒ No

Install Static Route

Destination Prefix:

Destination Prefix Mask:

Interface:

Reset to Default

Save

Close

Router B ✕

Routing Table

Routing Configuration

Was a problem found?: ☒ Yes ☐ No

Install Static Route

Destination Prefix:

Destination Prefix Mask:

Interface:

Gi1

Gi2

Gi3

Gi4

Reset to Default

Save

Close

Router C

Routing Table **Routing Configuration**

```
Router-C# show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from Pfr

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
S    10.0.0.0/22 [1/0] via GigabitEthernet1
S    10.0.4.0/22 [1/0] via GigabitEthernet2
172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.16.27.0/30 is directly connected, GigabitEthernet2
L    172.16.27.2/32 is directly connected, GigabitEthernet2
C    172.16.27.4/30 is directly connected, GigabitEthernet1
L    172.16.27.6/32 is directly connected, GigabitEthernet1
```

Reset to Default **Save** **Close**

Router C

Routing Table **Routing Configuration**

Was a problem found?: ☐ Yes ☒ No

Install Static Route

Destination Prefix:

Destination Prefix Mask:

Interface:

Reset to Default **Save** **Close**

Router C

Routing Table | **Routing Configuration**

Was a problem found?: ☒ Yes ☐ No

Install Static Route

Destination Prefix:

Destination Prefix Mask:

Interface:

Gi1
Gi2
Gi3
Gi4

Reset to Default Save Close

Answer: **Issue Identified:** Workstation A cannot access File Server 2 due to missing routing information.

Solution:

1. **Review Router Configuration:**
 - o Analyze the routing tables on Routers A, B, and C to confirm whether routes to the File Server 2 subnet are present.
2. **Add Static Route:**
 - o Install a static route on the router that connects Workstation A's network to File Server 2's network:
 - **Destination Prefix:** Enter the network address of File Server 2.
 - **Destination Prefix Mask:** Enter the subnet mask for File Server 2's network.
 - **Interface:** Select the router interface that faces towards File Server 2's network.
3. **Action:**
 - o Configure the static route using the routing configuration panel on the router that requires the update.
 - o Save the changes and test connectivity from Workstation A to File Server 2 to ensure the issue is resolved.

Answer: **Issue Identified:** Workstation A cannot access File Server 2 due to missing routing information.

Solution:

1. **Review Router Configuration:**
 - o Analyze the routing tables on Routers A, B, and C to confirm whether routes to the File Server 2 subnet are present.
2. **Add Static Route:**
 - o Install a static route on the router that connects Workstation A's network to File Server 2's network:
 - **Destination Prefix:** Enter the network address of File Server 2.
 - **Destination Prefix Mask:** Enter the subnet mask for File Server 2's network.
 - **Interface:** Select the router interface that faces towards File Server 2's network.
3. **Action:**
 - o Configure the static route using the routing configuration panel on the router that requires the update.
 - o Save the changes and test connectivity from Workstation A to File Server 2 to ensure the issue is resolved.

Question #4 (Topic: Exam A)

SIMULATION

A network administrator has been tasked with configuring a network for a new corporate office. The office consists of two buildings, separated by 50 feet with no physical connectivity. The configuration must meet the following requirements:

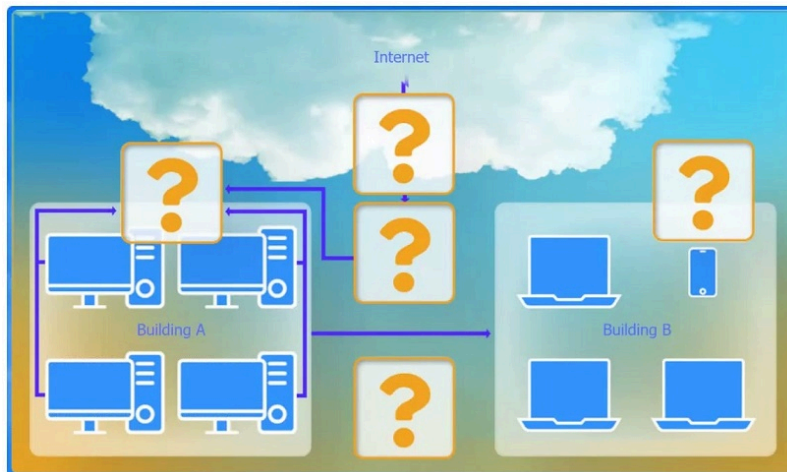
Devices in both buildings should be able to access the Internet.
Security insists that all Internet traffic be inspected before entering the network.
Desktops should not see traffic destined for other devices.

INSTRUCTIONS

Select the appropriate network device for each location. If applicable, click on the magnifying glass next to any device which may require configuration updates and make any necessary changes.

Not all devices will be used, but all locations should be filled.

If at any time you would like to bring back the initial state of the simulation, please click the Reset All button.



Switch
WAP
Hub
Wireless range extender
Router
Firewall

Answer: Central Device Connecting to the Internet (marked with a question mark at the top middle connecting to the Internet):

- **Device:** Firewall
- **Reason:** To inspect and filter all internet traffic before it enters the network, ensuring security compliance.

Device between Building A and the Internet (first question mark inside Building A):

- **Device:** Router
- **Reason:** To route traffic between the internal networks (Building A, Building B) and the internet via the firewall.

Devices inside Building A (two question marks linked to desktops and server infrastructure):

- **Device:** Switch
- **Reason:** To connect all desktops and servers within Building A, ensuring that each device can communicate with others within the building without seeing traffic destined for other devices.

Device connecting Building A to Building B (question mark between the two buildings):

- **Device:** Wireless range extender
- **Reason:** Given the 50 feet separation and no physical connectivity, a wireless range extender can bridge this gap by extending the wireless signal from Building A to Building B, facilitating network access across both buildings.

Device inside Building B (question mark inside Building B):

- **Device:** Switch
- **Reason:** To provide connectivity for the devices in Building B, allowing them to connect to the network and access resources in Building A as well as the internet.

Answer: Central Device Connecting to the Internet (marked with a question mark at the top middle connecting to the Internet):

- **Device:** Firewall
- **Reason:** To inspect and filter all internet traffic before it enters the network, ensuring security compliance.

Device between Building A and the Internet (first question mark inside Building A):

- **Device:** Router
- **Reason:** To route traffic between the internal networks (Building A, Building B) and the internet via the firewall.

Devices inside Building A (two question marks linked to desktops and server infrastructure):

- **Device:** Switch
- **Reason:** To connect all desktops and servers within Building A, ensuring that each device can communicate with others within the building without seeing traffic destined for other devices.

Device connecting Building A to Building B (question mark between the two buildings):

- **Device:** Wireless range extender
- **Reason:** Given the 50 feet separation and no physical connectivity, a wireless range extender can bridge this gap by extending the wireless signal from Building A to Building B, facilitating network access across both buildings.

Device inside Building B (question mark inside Building B):

- **Device:** Switch
- **Reason:** To provide connectivity for the devices in Building B, allowing them to connect to the network and access resources in Building A as well as the internet.

Question #5 (Topic: Exam A)

SIMULATION

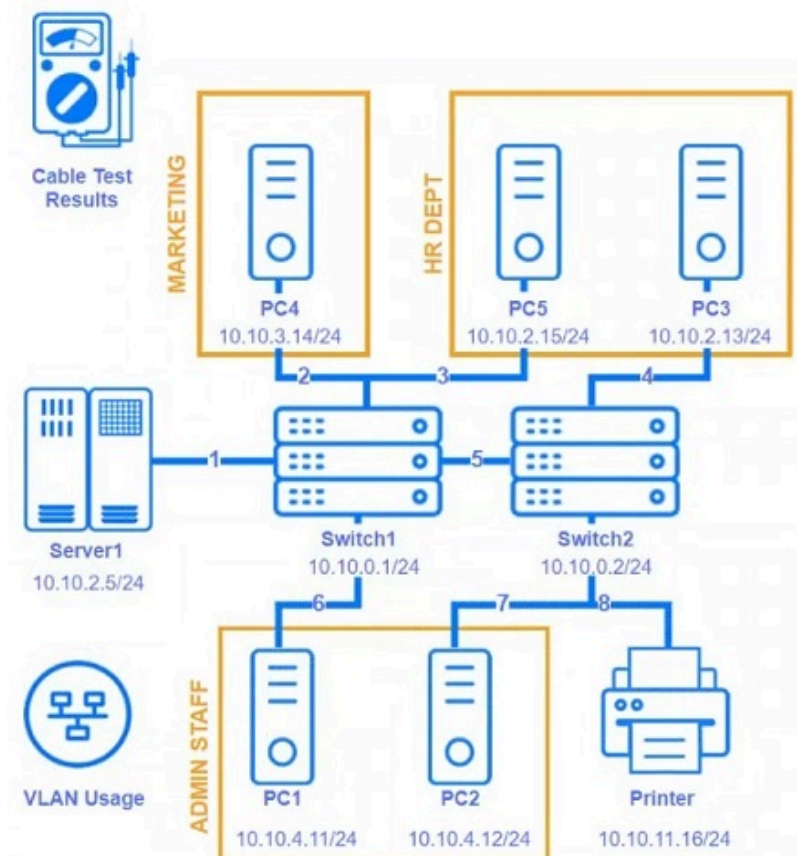
A network technician needs to resolve some issues with a customer's SOHO network. The customer reports that some of the devices are not connecting to the network, while others appear to work as intended.

INSTRUCTIONS

Troubleshoot all the network components and review the cable test results by clicking on each device and cable. Type help in each terminal to view a list of available commands.

Diagnose the appropriate component(s) by identifying any components with a problem and recommend a solution to correct each problem.

If at any time you would like to bring back the initial state of the simulation, please click the Reset All button.



Cable Test Results							
Cable 1	Cable 2	Cable 3	Cable 4	Cable 5	Cable 6	Cable 7	Cable 8
Length: 22M				1	2	3	6
VLAN: VLAN 2				1	2	3	6
Speed: 1000 FDX				4	5	7	8
Port: GigabitEthernet0/1				4	5	7	8

Cable Test Results							
Cable 1	Cable 2	Cable 3	Cable 4	Cable 5	Cable 6	Cable 7	Cable 8
Length: 103M				1	2	3	6
VLAN: VLAN 3				1	2	3	6
Speed: 1000 FDX				4	5	7	8
Port: GigabitEthernet0/4				4	5	7	8

Cable Test Results							
Cable 1	Cable 2	Cable 3	Cable 4	Cable 5	Cable 6	Cable 7	Cable 8
Length: 18M				1	2	3	6
VLAN: VLAN 2				1	2	3	6
Speed: 1000 FDX				4	5	7	8
Port: GigabitEthernet0/3				4	5	7	8

Cable Test Results

Cable 1	Cable 2	Cable 3	Cable 4	Cable 5	Cable 6	Cable 7	Cable 8
Length: 20M			1 2 3 6	4 5 7 8			
VLAN: VLAN 1			1 2 3 6	4 5 7 8			
Speed: 1000 FDX			1 2 3 6	4 5 7 8			
Port: GigabitEthernet0/2			1 2 3 6	4 5 7 8			

Cable Test Results

Cable 1	Cable 2	Cable 3	Cable 4	Cable 5	Cable 6	Cable 7	Cable 8
Length: 16M			1 2 3 6	4 5 7 8			
VLAN: VLAN 1			1 2 3 6	4 5 7 8			
Speed: 1000 FDX			1 2 3 6	4 5 7 8			
Port: GigabitEthernet0/5			1 2 3 6	4 5 7 8			

Cable Test Results

Cable 1	Cable 2	Cable 3	Cable 4	Cable 5	Cable 6	Cable 7	Cable 8
Length: 42M			1 2 3 6	4 5 7 8			
VLAN: VLAN 4			1 2 3 6	4 5 7 8			
Speed: 1000 FDX			1 2 3 6	4 5 7 8			
Port: GigabitEthernet0/2			1 2 3 6	4 5 7 8			

Cable Test Results

Cable 1	Cable 2	Cable 3	Cable 4	Cable 5	Cable 6	Cable 7	Cable 8
Length: 12M			1 2 3 6	4 5 7 8			
VLAN: VLAN 1			1 2 3 6	4 5 7 8			
Speed: 1000 FDX			1 2 3 6	4 5 7 8			
Port: GigabitEthernet0/1			1 2 3 6	4 5 7 8			

Cable Test Results

Cable 1	Cable 2	Cable 3	Cable 4	Cable 5	Cable 6	Cable 7	Cable 8
Length: 90M			1 2 3 6	4 5 7 8			
VLAN: VLAN 1			1 2 3 6	4 5 7 8			
Speed: 1000 FDX			1 2 3 6	4 5 7 8			
Port: GigabitEthernet0/3			1 2 3 6	4 5 7 8			

VLAN Usage



VLAN1:	Default	
VLAN2:	Server/HR	10.10.2.0/24
VLAN3:	Marketing	10.10.3.0/24
VLAN4:	Admin Staff	10.10.4.0/24
VLAN10:	PCs	10.10.10.0/24
VLAN11:	Printer	10.10.11.0/24

Printer



HP Network Configuration Page

Model: HP Officejet Pro 8610

General Information

Network Status	Ready
Active Connection Type	Wired
URL(s) for Embedded Web Server	http://HP4D30EC, http://192.168.2.9
Firmware Revision	FDP1CN1347AR
Hostname	HP4D30EC
Serial Number	CN3AO1KG42
Internet	Not Connected

802.3 Wired

Hardware Address (MAC)	9c:b6:54:4d:30:ec
Link Configuration	None

IPv4

IP Address	10.10.11.56
Subnet Mask	255.255.255.0
Default Gateway	10.10.11.1
Configuration Source	DHCP
Primary DNS Server	8.8.8.8
Secondary DNS Server	8.8.4.4
Total Packets Transmitted	15655
Total Packets Received	394068

Remediation

Select Device/Cable 



Select Device/Cable

- PC1
- PC2
- PC3
- PC4
- PC5
- Printer
- Server1
- Switch1
- Switch2
- Cable1
- Cable2
- Cable3
- Cable4
- Cable5
- Cable6
- Cable7
- Cable8

Remediation

Select Device/Cable 



PC3 

Problem

Select a problem 

Select a problem

- Bad name server
- Bad reverse IP address
- Bad subnet
- Bad IP address
- Bad gateway address

Solution

Select a solution 

Select a solution

- Change IP address
- Change gateway address
- Change subnet mask
- Change DNS address
- Flush ARP cache
- Release and renew IP address

Answer: Router and Firewall Configuration:

- Ensure that all routers are configured with correct static routes to facilitate proper network traffic routing between different VLANs and external networks.
- Install a firewall at the network's entry point to inspect and filter Internet traffic as required by the security policy.

Switch Configuration:

- Configure switches to ensure PCs in different VLANs do not see each other's traffic. Implement VLAN tagging on the switches to segregate traffic between the different departments as indicated.

Cable Test Results:

- Address any faulty cables as indicated by the test results. For example, any cable showing physical connection issues (crossed pairs, etc.) should be replaced.
- Ensure that cables are appropriately labeled and meet the length requirements to avoid potential transmission issues over long distances.

Device Specific Issues:

- For devices showing incorrect configuration details such as bad IP addresses, subnet masks, or DNS configurations, apply appropriate corrections. For instance, resetting IP configurations or updating DNS settings where mismatches are found.

Printer Configuration:

- Check the printer configuration to ensure it aligns with the network settings, particularly the IP address, subnet, and gateway to guarantee connectivity for all users needing access.

Overall Network Health:

- Continuously monitor the network for any emerging issues, perform regular updates to the firmware of network devices, and ensure compliance with the latest security standards to maintain network integrity and performance.

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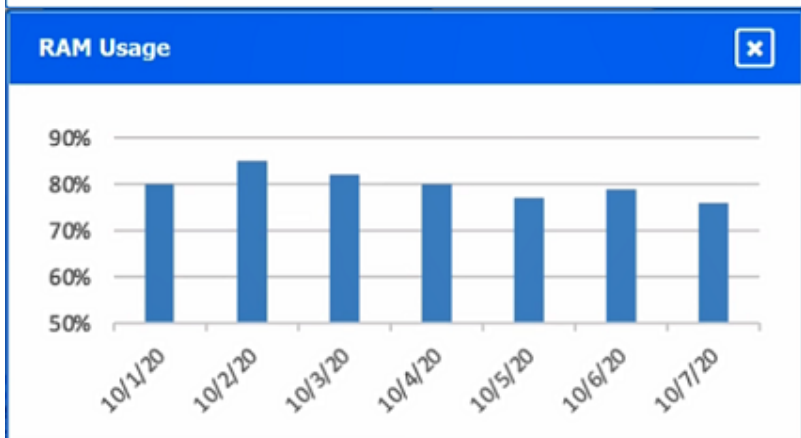
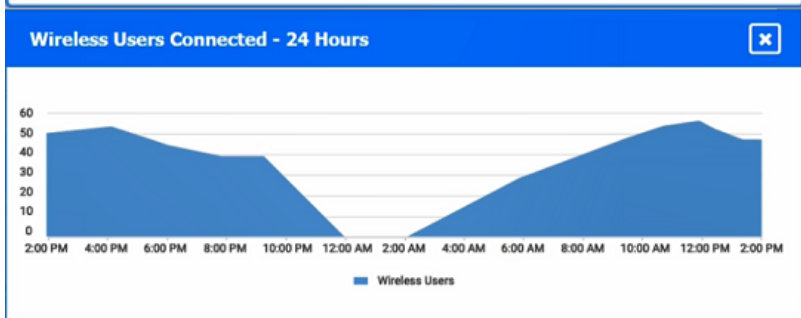
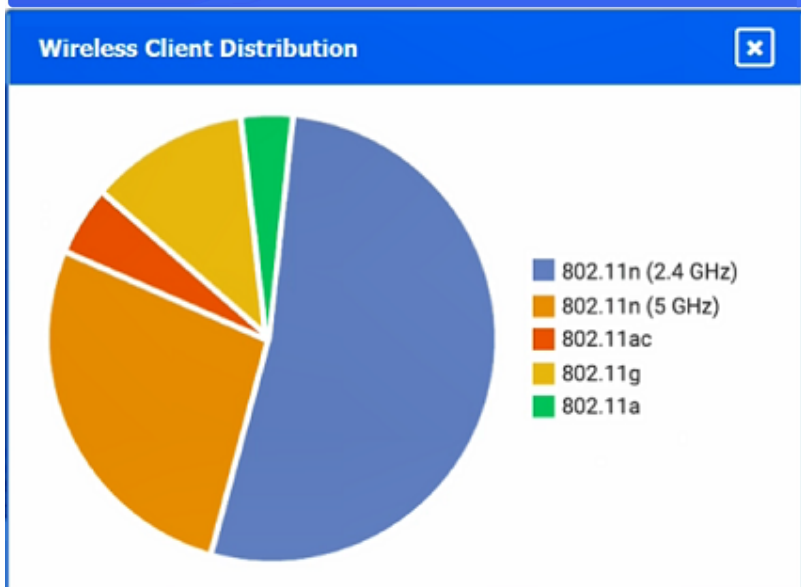
Question #6 (Topic: Exam A)**SIMULATION**

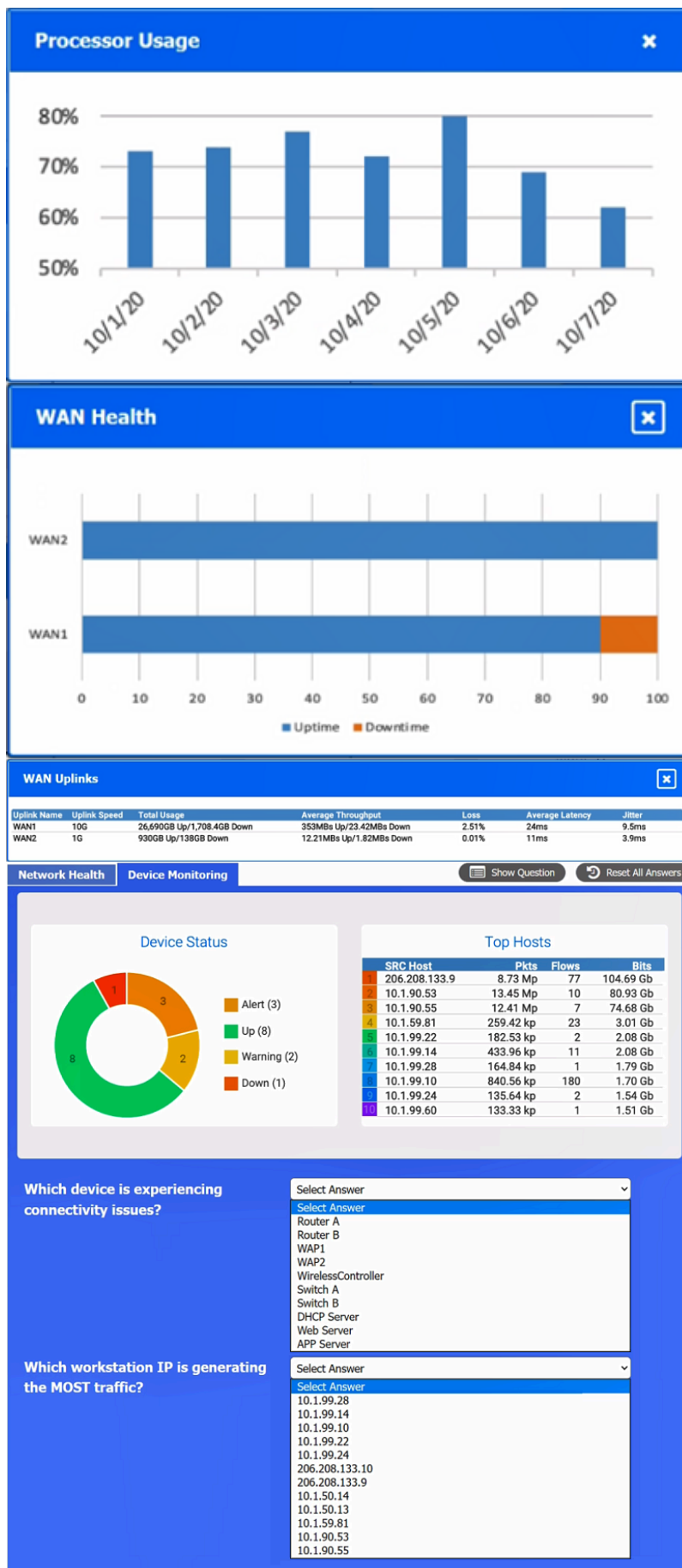
After a recent power outage, users are reporting performance issues accessing the application servers. Wireless users are also reporting intermittent Internet issues.

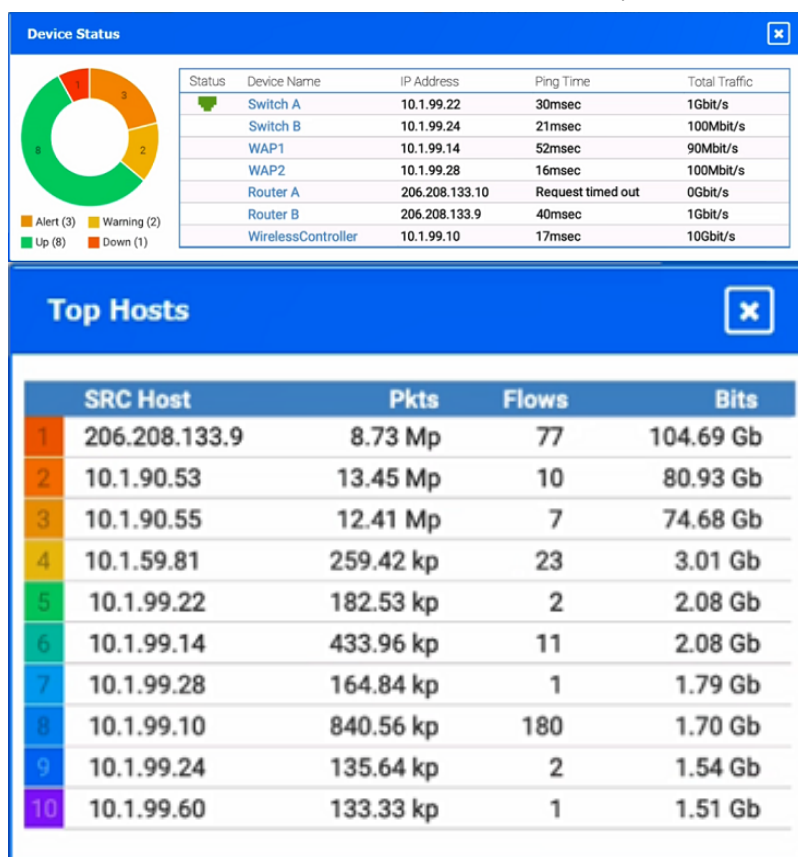
INSTRUCTIONS

Click on each tab at the top of the screen. Select a widget to view information, then use the drop-down menus to answer the associated questions.

If at any time you would like to bring back the initial state of the simulation, please click the Reset All button.







Answer: WAN 1:

- Uplink Speed: 10G
- Total Usage: 26.969GB Up / 1.748GB Down
- Average Throughput: 353MBps Up / 23.42MBps Down
- Loss: 2.51%
- Average Latency: 24ms
- Jitter: 9.5ms

WAN 2:

- Uplink Speed: 1G
- Total Usage: 930GB Up / 138GB Down
- Average Throughput: 12.21MBps Up / 1.82MBps Down
- Loss: 0.01%
- Average Latency: 11ms
- Jitter: 3.9ms

Although WAN 1 offers greater bandwidth and throughput, it exhibits higher latency and jitter, which could detract from VoIP performance. In contrast, WAN 2, despite its lower bandwidth, offers much lower latency, nearly negligible packet loss, and reduced jitter. These characteristics are essential for VoIP applications, which demand consistent and quick packet delivery.

Therefore, for VoIP communications, WAN 2 is recommended due to its enhanced stability in latency and jitter handling, which are pivotal for the clarity and reliability of voice calls. WAN 1 could be better utilized for other tasks that require high bandwidth and are less affected by latency variations, such as large-scale data transfers.

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Question #7 (Topic: Exam A)

Which of the following steps of the troubleshooting methodology would most likely include checking through each level of the OSI model after the problem has been identified?

A. Establish a theory.

B. Implement the solution.

C. Create a plan of action.

D. Verify functionality.

Answer: A

Answer: A

Question #8 (Topic: Exam A)

While troubleshooting a VoIP handset connection, a technician's laptop is able to successfully connect to network resources using the same port. The technician needs to identify the port on the switch. Which of the following should the technician use to determine the switch and port?

A. LLDP

B. IKE

C. VLAN

D. netstat

Answer: **A**

Answer: **A**

Question #9 (Topic: Exam A)

A network administrator needs to set up a file server to allow user access. The organization uses DHCP to assign IP addresses. Which of the following is the best solution for the administrator to set up?

A. A separate scope for the file server using a /32 subnet

B. A reservation for the server based on the MAC address

C. A static IP address within the DHCP IP range

D. A SLAAC for the server

Answer: **B**

Answer: **B**

Question #10 (Topic: Exam A)

Which of the following technologies are X.509 certificates most commonly associated with?

A. PKI

B. VLAN tagging

C. LDAP

D. MFA

Answer: **A**

Answer: **A**

Question #11 (Topic: Exam A)

A network administrator wants to implement an authentication process for temporary access to an organization's network. Which of the following technologies would facilitate this process?

A. Captive portal

B. Enterprise authentication

C. Ad hoc network

D. WPA3

Answer: **A**

Answer: **A**

Question #12 (Topic: Exam A)

A user is unable to navigate to a website because the provided URL is not resolving to the correct IP address. Other users are able to navigate to the intended website without issue. Which of the following is most likely causing this issue?

A. Hosts file

B. Self-signed certificate

C. Nameserver record

D. IP helper

Answer: **A**

Answer: **A**

Question #13 (Topic: Exam A)

A network administrator is planning to host a company application in the cloud, making the application available for all internal and third-party users. Which of the following concepts describes this arrangement?

A. Multitenancy

B. VPC

C. NFV

D. SaaS

Answer: **A**

Answer: **A**

Question #14 (Topic: Exam A)

Which of the following should be used to obtain remote access to a network appliance that has failed to start up properly?

A. Crash cart

B. Jump box

C. Secure Shell

D. Out-of-band management

Answer: **D**

Answer: **D**

Question #15 (Topic: Exam A)

Which of the following attacks utilizes a network packet that contains multiple network tags?

A. MAC flooding

B. VLAN hopping

C. DNS spoofing

D. ARP poisoning

Answer: **B**

Answer: **B**

Question #16 (Topic: Exam A)

A network administrator is configuring a new switch and wants to connect two ports to the core switch to ensure redundancy. Which of the following configurations would meet this requirement?

A. Full duplex

B. 802.1Q tagging

C. Native VLAN

D. Link aggregation

Answer: **D**

Answer: **D**

Question #17 (Topic: Exam A)

Which of the following ports is used for secure email?

A. 25

B. 110

C. 143

D. 587

Answer: **D**

Answer: **D**

Question #18 (Topic: Exam A)

A client wants to increase overall security after a recent breach. Which of the following would be best to implement? (Choose two.)

A. Least privilege network access

B. Dynamic inventories

C. Central policy management

D. Zero-touch provisioning

E. Configuration drift prevention

F. Subnet range limits

Answer: **AC**

Answer: **AC**

Question #19 (Topic: Exam A)

Which of the following is a cost-effective advantage of a split-tunnel VPN?

A. Web traffic is filtered through a web filter.

B. More bandwidth is required on the company's internet connection.

C. Monitoring detects insecure machines on the company's network.

D. Cloud-based traffic flows outside of the company's network.

Answer: **D**

Answer: **D**

Question #20 (Topic: Exam A)

A network technician is troubleshooting a web application's poor performance. The office has two internet links that share the traffic load. Which of the following tools

should the technician use to determine which link is being used for the web application?

A. netstat

B. nslookup

C. ping

D. tracer

Answer: **D**

Answer: **D**

Question #21 (Topic: Exam A)

Which of the following attacks can cause users who are attempting to access a company website to be directed to an entirely different website?

A. DNS poisoning

B. Denial-of-service

C. Social engineering

D. ARP spoofing

Answer: **A**

Answer: **A**

Question #22 (Topic: Exam A)

As part of an attack, a threat actor purposefully overflows the content-addressable memory (CAM) table on a switch. Which of the following types of attacks is this scenario an example of?

A. ARP spoofing

B. Evil twin

C. MAC flooding

D. DNS poisoning

Answer: **C**

Answer: **C**

Question #23 (Topic: Exam A)

A company's office has publicly accessible meeting rooms equipped with network ports. A recent audit revealed that visitors were able to access the corporate network by plugging personal laptops into open network ports. Which of the following should the company implement to prevent this in the future?

A. URL filters

B. VPN

C. ACLs

D. NAC

Answer: **D**

Answer: **D**

Question #24 (Topic: Exam A)

A user notifies a network administrator about losing access to a remote file server. The network administrator is able to ping the server and verifies the current firewall rules do not block access to the network fileshare. Which of the following tools would help identify which ports are open on the remote file server?

A. dig

B. nmap

C. tracert

D. nslookup

Answer: **B**

Answer: **B**

Question #25 (Topic: Exam A)

Which of the following technologies is the best choice to listen for requests and distribute user traffic across web servers?

A. Router

B. Switch

C. Firewall

D. Load balancer

Answer: **D**

Answer: **D**

Question #26 (Topic: Exam A)

A company is hosting a secure server that requires all connections to the server to be encrypted. A junior administrator needs to harden the web server. The following ports on the web server are open:

443

80

22

587

Which of the following ports should be disabled?

A. 22

B. 80

C. 443

D. 587

Answer: **B**

Answer: **B**

Question #27 (Topic: Exam A)

Which of the following is the next step to take after successfully testing a root cause theory?

- A. Determine resolution steps.
- B. Duplicate the problem in a lab.
- C. Present the theory for approval.
- D. Implement the solution to the problem.

Answer: **A**

Answer: **A**

Question #28 (Topic: Exam A)

A network administrator is configuring a new switch and wants to ensure that only assigned devices can connect to the switch. Which of the following should the administrator do?

- A. Configure ACLs.
- B. Implement a captive portal.
- C. Enable port security.
- D. Disable unnecessary services.

Answer: **C**

Answer: **C**

Question #29 (Topic: Exam A)

A customer needs six usable IP addresses. Which of the following best meets this requirement?

- A. 255.255.255.128

B. 255.255.255.192

C. 255.255.255.224

D. 255.255.255.240

Answer: **D**

Answer: **D**

Question #30 (Topic: Exam A)

A user reports having intermittent connectivity issues to the company network. The network configuration for the user reveals the following:

IP address: 192.168.1.10

Subnet mask: 255.255.255.0

Default gateway: 192.168.1.254

The network switch shows the following ARP table:

MAC address	IP address	Interface	VLAN
0c00.1134.0001	192.168.1.10	eth4	10
0c00.1983.210a	192.168.2.13	eth5	11
0c00.1298.d239	192.168.1.10	eth6	10
0c00.a291.c113	192.168.2.12	eth7	11
0c00.923b.2391	192.168.1.11	eth8	10
feff.2391.1022	192.168.1.254	eth1	10

Which of the following is the most likely cause of the user's connection issues?

A. A port with incorrect VLAN assigned

B. A switch with spanning tree conflict

C. Another PC with manually configured IP

D. A router with overlapping route tables

Answer: **C**

Answer: **C**

Question #31 (Topic: Exam A)

Which of the following is created to illustrate the effectiveness of wireless networking coverage in a building?

A. Logical diagram

B. Layer 3 network diagram

C. Service-level agreement

D. Heat map

Answer: **D**

Answer: **D**

Question #32 (Topic: Exam A)

Which of the following cloud deployment models is most commonly associated with multitenancy and is generally offered by a service provider?

A. Private

B. Community

C. Public

D. Hybrid

Answer: **C**

Answer: **C**

Question #33 (Topic: Exam A)

A network administrator needs to create an SVI on a Layer 3-capable device to separate voice and data traffic. Which of the following best explains this use case?

A. A physical interface used for trunking logical ports

B. A physical interface used for management access

C. A logical interface used for the routing of VLANs

D. A logical interface used when the number of physical ports is insufficient

Answer: **C**

Answer: **C**

Question #34 (Topic: Exam A)

A network administrator is performing a refresh of a wireless environment. As the APs are being placed, they overlap a little bit with each other. Which of the following 2.4GHz channels should be selected to ensure that they do not conflict?

A. 1, 3, 5

B. 1, 6, 11

C. 2, 6, 10

D. 3, 6, 9

Answer: **B**

Answer: **B**

Question #35 (Topic: Exam A)

Which of the following network cables involves bouncing light off of protective cladding?

A. Twinaxial

B. Coaxial

C. Single-mode

D. Multimode

Answer: **D**

Answer: **D**

Question #36 (Topic: Exam A)

Which of the following would allow a network administrator to analyze attacks coming from the internet without affecting latency?

A. IPS

B. IDS

C. Load balancer

D. Firewall

Answer: **B**

Answer: **B**

Question #37 (Topic: Exam A)

A technician is troubleshooting wireless connectivity near a break room. Whenever a user turns on the microwave, connectivity to the user's laptop is lost. Which of the following frequency bands is the laptop most likely using?

A. 2.4GHz

B. 5GHz

C. 6GHz

D. 900MHz

Answer: **A**

Answer: **A**

Question #38 (Topic: Exam A)

A network administrator needs to implement routing capabilities in a hypervisor. Which of the following should the administrator most likely implement?

A. VPC

B. Firewall

C. NFV

D. IaaS

Answer: **C**

Answer: **C**

Question #39 (Topic: Exam A)

A network technician needs to install patch cords from the UTP patch panel to the access switch for a newly occupied set of offices. The patch panel is not labeled for easy jack identification. Which of the following tools provides the easiest way to identify the appropriate patch panel port?

A. Toner

B. Laptop

C. Cable tester

D. Visual fault locator

Answer: **A**

Answer: **A**

Question #40 (Topic: Exam A)

A network administrator received complaints of intermittent network connectivity issues. The administrator investigates and finds that the network design contains potential loop scenarios. Which of the following should the administrator do?

A. Enable spanning tree.

B. Configure port security.

C. Change switch port speed limits.

D. Enforce 802.1Q tagging.

Answer: **A**

Answer: **A**

Question #41 (Topic: Exam A)

Which of the following routing technologies uses a successor and a feasible successor?

A. IS-IS

B. OSPF

C. BGP

D. EIGRP

Answer: **D**

Answer: **D**

Question #42 (Topic: Exam A)

Which of the following best describes what an organization would use port address translation for?

A. VLANs on the perimeter

B. Public address on the perimeter router

C. Non-routable address on the perimeter router

D. Servers on the perimeter

Answer: **C**

Answer: **C**

Question #43 (Topic: Exam A)

A company is implementing a wireless solution in a high-density environment. Which of the following 802.11 standards is used when a company is concerned about device saturation and coverage?

A. 802.11ac

B. 802.11ax

C. 802.11g

D. 802.11n

Answer: **B**

Answer: **B**

Question #44 (Topic: Exam A)

Which of the following network traffic types is sent to all nodes on the network?

A. Unicast

B. Broadcast

C. Multicast

D. Anycast

Answer: **B**

Answer: **B**

Question #45 (Topic: Exam A)

Which of the following cable types provides the highest possible transmission speed?

A. Plenum

B. Ethernet

C. Fiber-optic

D. DAC

Answer: **C**

Answer: **C**

Question #46 (Topic: Exam A)

Which of the following layers of the OSI model is responsible for end-to-end encryption?

A. Presentation

B. Application

C. Session

D. Transport

Answer: **A**

Answer: **A**

Question #47 (Topic: Exam A)

A network security administrator needs to monitor the contents of data sent between a secure network and the rest of the company. Which of the following monitoring methods will accomplish this task?

A. Port mirroring

B. Flow data

C. Syslog entries

D. SNMP traps

Answer: **A**

Answer: **A**

Question #48 (Topic: Exam A)

Which of the following does a full-tunnel VPN provide?

- A. Lower bandwidth requirements
- B. The ability to reset local computer passwords
- C. Corporate inspection of all network traffic
- D. Access to blocked sites

Answer: **C**

Answer: **C**

Question #49 (Topic: Exam A)

Which of the following does a hash provide?

- A. Non-repudiation
- B. Integrity
- C. Confidentiality
- D. Availability

Answer: **B**

Answer: **B**

Question #50 (Topic: Exam A)

A company wants to implement a disaster recovery site for non-critical applications, which can tolerate a short period of downtime. Which of the following types of sites should the company implement to achieve this goal?

A. Hot

B. Cold

C. Warm

D. Passive

Answer: **C**

Answer: **C**

Question #51 (Topic: Exam A)

Which of the following is an XML-based security concept that works by passing sensitive information about users, such as log-in information and attributes, to providers?

A. IAM

B. MFA

C. RADIUS

D. SAML

Answer: **D**

Answer: **D**

Question #52 (Topic: Exam A)

Which of the following routing technologies uses unequal cost load balancing and port 88?

A. EIGRP

B. BGP

C. RIP

D. OSPF

Answer: **A**

Answer: **A**

Question #53 (Topic: Exam A)

A wireless network consultant is deploying a large number of WAPs and wants to centrally control them from one wireless LAN controller. Which of the following network types should the consultant employ?

A. Mesh

B. Infrastructure

C. Point-to-point

D. Ad hoc

Answer: **B**

Answer: **B**

Question #54 (Topic: Exam A)

Which of the following network topologies involves sending all traffic through a single point?

A. Mesh

B. Hybrid

C. Hub-and-spoke

D. Point-to-point

Answer: **C**

Answer: **C**

Question #55 (Topic: Exam A)

Which of the following functions is used to prioritize network traffic based on the type of traffic?

A. QoS

B. VPN

C. CDN

D. TTL

Answer: **A**

Answer: **A**

Question #56 (Topic: Exam A)

Which of the following most likely determines the size of a rack for installation? (Choose two.)

A. KVM size

B. Switch depth

C. Hard drive size

D. Cooling fan speed

E. Outlet amperage

F. Server height

Answer: **BF**

Answer: **BF**

Question #57 (Topic: Exam A)

A network administrator is planning to implement device monitoring to enhance network visibility. The security team requires that the solution provides authentication

and encryption. Which of the following meets these requirements?

A. SIEM

B. Syslog

C. NetFlow

D. SNMPv3

Answer: **D**

Answer: **D**

Question #58 (Topic: Exam A)

A network administrator needs to change where the outside DNS records are hosted. Which of the following records should the administrator change at the registrar to accomplish this task?

A. NS

B. SOA

C. PTR

D. CNAME

Answer: **A**

Answer: **A**

Question #59 (Topic: Exam A)

A support agent receives a report that a remote user's wired devices are constantly disconnecting and have slow speeds. Upon inspection, the support agent sees that the user's coaxial modem has a signal power of -97dB. Which of the following should the support agent recommend to troubleshoot the issue?

A. Removing any splitters connected to the line

B. Switching the devices to wireless

C. Moving the devices closer to the modem

D. Lowering the network speed

Answer: **A**

Answer: **A**

Question #60 (Topic: Exam A)

Which of the following does OSPF use to communicate routing updates?

A. Unicast

B. Anycast

C. Multicast

D. Broadcast

Answer: **C**

Answer: **C**

Question #61 (Topic: Exam A)

A storage network requires reduced overhead and increased efficiency for the amount of data being sent. Which of the following should an engineer most likely configure to meet these requirements?

A. Link speed

B. Jumbo frames

C. QoS

D. 802.1q tagging

Answer: **B**

Answer: **B**

Question #62 (Topic: Exam A)

A security administrator is creating a new firewall object for a device with IP address 192.168.100.1/25. However, the firewall software only uses dotted decimal notation in configuration fields. Which of the following is the correct subnet mask to use?

A. 255.255.254.0

B. 255.255.255.1

C. 255.255.255.128

D. 255.255.255.192

Answer: **C**

Answer: **C**

Question #63 (Topic: Exam A)

Which of the following disaster recovery metrics describes the average length of time a piece of equipment can be expected to operate normally?

A. RPO

B. RTO

C. MTTR

D. MTBF

Answer: **D**

Answer: **D**

Question #64 (Topic: Exam A)

A network administrator logs on to a router and sees an interface with an IP address of 10.61.52.34 255.255.255.252. Which of the following best describes how this interface

IP address is being used?

- A. As a point-to-point connection
- B. To connect to the internet
- C. As a virtual address for redundancy
- D. For out-of-band management

Answer: **A**

Answer: **A**

Question #65 (Topic: Exam A)

A network technician is troubleshooting a faulty NIC and tests the theory. Which of the following should the technician do next?

- A. Develop a theory.
- B. Establish a plan of action.
- C. Implement the solution.
- D. Document the findings.

Answer: **B**

Answer: **B**

Question #66 (Topic: Exam A)

A network administrator is configuring access points for installation in a dense environment where coverage is often overlapping. Which of the following channel widths should the administrator choose to help minimize interference in the 2.4GHz spectrum?

- A. 11MHz
- B. 20MHz

C. 40MHz

D. 80MHz

E. 160MHz

Answer: **B**

Answer: **B**

Question #67 (Topic: Exam A)

A network manager wants to view network traffic for devices connected to a switch. A network engineer connects an appliance to a free port on the switch and needs to configure the switch port connected to the appliance. Which of the following is the best option for the engineer to enable?

A. Trunking

B. Port mirroring

C. Full duplex

D. SNMP

Answer: **B**

Answer: **B**

Question #68 (Topic: Exam A)

A network administrator is in the process of installing 35 PoE security cameras. After the administrator installed and tested the new cables, the administrator installed the cameras. However, a small number of the cameras do not work. Which of the following is the most likely reason?

A. Incorrect wiring standard

B. Power budget exceeded

C. Signal attenuation

D. Wrong voltage

Answer: **B**

Answer: **B**

Question #69 (Topic: Exam A)

A network administrator is troubleshooting an application issue after a firewall change. The administrator has confirmed that the port and protocol are accessible to the user, but the application is still having issues. Which of the following tools allows the administrator to look at traffic on the application layer of the OSI model?

A. ifconfig

B. tcpdump

C. nslookup

D. traceroute

Answer: **B**

Answer: **B**

Question #70 (Topic: Exam A)

Which of the following ports should a network administrator enable for encrypted login to a network switch?

A. 22

B. 23

C. 80

D. 123

Answer: **A**

Answer: **A**

Question #71 (Topic: Exam A)

Which of the following is used to stage copies of a website closer to geographically dispersed users?

A. VPN

B. CDN

C. SAN

D. SDN

Answer: **B**

Answer: **B**

Question #72 (Topic: Exam A)

Which of the following appliances provides users with an extended footprint that allows connections from multiple devices within a designated WLAN?

A. Router

B. Switch

C. Access point

D. Firewall

Answer: **C**

Answer: **C**

Question #73 (Topic: Exam A)

An administrator is configuring a switch that will be placed in an area of the office that is accessible to customers. Which of the following is the best way for the administrator to mitigate unknown devices from connecting to the network?

A. SSE

B. ACL

C. Perimeter network

D. 802.1X

Answer: **D**

Answer: **D**

Question #74 (Topic: Exam A)

Which of the following diagrams would most likely include specifications about fiber connector types?

A. Logical

B. Physical

C. Rack

D. Routing

Answer: **B**

Answer: **B**

Question #75 (Topic: Exam A)

Which of the following is the most likely reason an insurance brokerage would enforce VPN usage?

A. To encrypt sensitive data in transit

B. To secure the endpoints

C. To maintain contractual agreements

D. To comply with data retention requirements

Answer: **A**

Answer: **A**

Question #76 (Topic: Exam A)

An organization moved its DNS servers to new IP addresses. After this move, customers are no longer able to access the organization's website. Which of the following DNS entries should be updated?

A. AAA

B. CNAME

C. MX

D. NS

Answer: **D**

Answer: **D**

Question #77 (Topic: Exam A)

Which of the following are environmental factors that should be considered when installing equipment in a building? (Choose two.)

A. Fire suppression system

B. UPS location

C. Humidity control

D. Power load

E. Floor construction type

F. Proximity to nearest MDF

Answer: **AC**

Answer: **AC**

Question #78 (Topic: Exam A)

A customer lost the connection to the telephone system. The administration console is configured with multiple network interfaces and is connected to multiple switches. The network administrator troubleshoots and verifies the following:

The support team is able to connect remotely to the administration console.

Rebooting the switch shows solid link and activity lights even on unused ports.

Rebooting the telephone system does not bring the system back online.

The console is able to connect directly to individual modules successfully.

Which of the following is the most likely reason the customer lost the connection?

A. A switch failed.

B. The console software needs to be reinstalled.

C. The cables to the modules need to be replaced.

D. A module failed.

Answer: **A**

Answer: **A**

Question #79 (Topic: Exam A)

A systems administrator is configuring a new device to be added to the network. The administrator is planning to perform device hardening prior to connecting the device.

Which of the following should the administrator do first?

A. Update the network ACLs.

B. Place the device in a screened subnet.

C. Enable content filtering.

D. Change the default admin passwords.

Answer: **D**

Answer: **D**

Question #80 (Topic: Exam A)

A network administrator needs to connect two network closets that are 492ft (150m) away from each other. Which of the following cable types should the administrator install between the closets?

A. Single-mode fiber

B. Coaxial

C. DAC

D. STP

Answer: **A**

Answer: **A**

Question #81 (Topic: Exam A)

An IT manager needs to connect ten sites in a mesh network. Each needs to be secured with reduced provisioning time. Which of the following technologies will best meet this requirement?

A. SD-WAN

B. VXLAN

C. VPN

D. NFV

Answer: **A**

Answer: **A**

Question #82 (Topic: Exam A)

Which of the following is most likely responsible for the security and handling of personal data in Europe?

A. GDPR

B. SCADA

C. SAML

D. PCI DSS

Answer: **A**

Answer: **A**

Question #83 (Topic: Exam A)

An organization has a security requirement that all network connections can be traced back to a user. A network administrator needs to identify a solution to implement on the wireless network. Which of the following is the best solution?

A. Implementing enterprise authentication

B. Requiring the use of PSKs

C. Configuring a captive portal for users

D. Enforcing wired equivalent protection

Answer: **A**

Answer: **A**

Question #84 (Topic: Exam A)

A customer is adding fiber connectivity between adjacent buildings. A technician terminates the multimode cable to the fiber patch panel. After the technician connects the fiber patch cable, the indicator light does not turn on. Which of the following should a technician try first to troubleshoot this issue?

A. Reverse the fibers.

B. Rerterminate the fibers.

C. Verify the fiber size.

D. Examine the cable runs for visual faults.

Answer: **A**

Answer: **A**

Question #85 (Topic: Exam A)

Users cannot connect to an internal website with an IP address 10.249.3.76. A network administrator runs a command and receives the following output:

```
...  
1  3ms  2ms  3ms  192.168.25.234  
2  2ms  3ms  1ms  192.168.3.100  
3  4ms  5ms  2ms  10.249.3.1  
4  *    *    *  
5  *    *    *  
6  *    *    *  
7  *    *    *  
...
```

Which of the following command-line tools is the network administrator using?

A. tracer

B. netstat

C. tcpdump

D. nmap

Answer: **A**

Answer: **A**

Question #86 (Topic: Exam A)

After running a Cat 8 cable using passthrough plugs, an electrician notices that connected cables are experiencing a lot of cross talk. Which of the following troubleshooting steps should the electrician take first?

A. Inspect the connectors for any wires that are touching or exposed.

B. Restore default settings on the connected devices.

C. Terminate the connections again.

D. Check for radio frequency interference in the area.

Answer: **A**

Answer: **A**

Question #87 (Topic: Exam A)

A technician is troubleshooting a user's laptop that is unable to connect to a corporate server. The technician thinks the issue pertains to routing. Which of the following commands should the technician use to identify the issue?

A. tcpdump

B. dig

C. tracert

D. arp

Answer: **C**

Answer: **C**

Question #88 (Topic: Exam A)

After installing a series of Cat 8 keystones, a data center architect notices higher than normal interference during tests. Which of the following steps should the architect take to troubleshoot the issue?

A. Check to see if the end connections were wrapped in copper tape before terminating.

B. Use passthrough modular crimping plugs instead of traditional crimping plugs.

C. Connect the RX/TX wires to different pins.

D. Run a speed test on a device that can only achieve 100Mbps speeds.

Answer: **A**

Answer: **A**

Question #89 (Topic: Exam A)

A small business is deploying new phones, and some of the phones have full HD videoconferencing features. The Chief Information Officer is concerned that the network might not be able to handle the traffic if the traffic reaches a certain threshold. Which of the following can the network engineer configure to help ease these concerns?

A. A VLAN with 100Mbps speed limits

B. An IP helper to direct VoIP traffic

C. A smaller subnet mask

D. Full duplex on all user ports

Answer: **A**

Answer: **A**

Question #90 (Topic: Exam A)

A virtual machine has the following configuration:

IPv4 address: 169.254.10.10

Subnet mask: 255.255.0.0

The virtual machine can reach collocated systems but cannot reach external addresses on the internet. Which of the following is most likely the root cause?

A. The subnet mask is incorrect.

B. The DHCP server is offline.

C. The IP address is an RFC1918 private address.

D. The DNS server is unreachable.

Answer: **B**

Answer: **B**

Question #91 (Topic: Exam A)

Which of the following is used to estimate the average life span of a device?

A. RPO

B. RTO

C. MTTR

D. MTBF

Answer: **D**

Answer: **D**

Question #92 (Topic: Exam A)

Which of the following is the most secure way to provide site-to-site connectivity?

A. VXLAN

B. IKE

C. GRE

D. IPSec

Answer: **D**

Answer: **D**

Question #93 (Topic: Exam A)

A network technician is terminating a cable to a fiber patch panel in the MDF. Which of the following connector types is most likely in use?

A. F-type

B. RJ11

C. BNC

D. SC

Answer: **D**

Answer: **D**

Question #94 (Topic: Exam A)

A network administrator is connecting two Layer 2 switches in a network. These switches must transfer data in multiple networks. Which of the following would fulfill this requirement?

A. Jumbo frames

B. 802.1Q tagging

C. Native VLAN

D. Link aggregation

Answer: **B**

Answer: **B**

Question #95 (Topic: Exam A)

A network administrator wants users to be able to authenticate to the corporate network using a port-based authentication framework when accessing both wired and wireless devices. Which of the following is the best security feature to accomplish this task?

A. 802.1X

B. Access control list

C. Port security

D. MAC filtering

Answer: **A**

Answer: **A**

Question #96 (Topic: Exam A)

Which of the following is most closely associated with a dedicated link to a cloud environment and may not include encryption?

A. Direct Connect

B. Internet gateway

C. Captive portal

D. VPN

Answer: **A**

Answer: **A**

Question #97 (Topic: Exam A)

A systems administrator is investigating why users cannot reach a Linux web server with a browser but can ping the server IP. The server is online, the web server process is running, and the link to the switch is up. Which of the following commands should the administrator run on the server first?

A. traceroute

B. netstat

C. tcpdump

D. arp

Answer: **B**

Answer: **B**

Question #98 (Topic: Exam A)

Which of the following devices can operate in multiple layers of the OSI model?

A. Hub

B. Switch

C. Transceiver

D. Modem

Answer: **B**

Answer: **B**

Question #99 (Topic: Exam A)

Before using a guest network, an administrator requires users to accept the terms of use. Which of the following is the best way to accomplish this goal?

A. Pre-shared key

B. Autonomous access point

C. Captive portal

D. WPA2 encryption

Answer: **C**

Answer: **C**

Question #100 (Topic: Exam A)

A network administrator for a small office is adding a passive IDS to its network switch for the purpose of inspecting network traffic. Which of the following should the administrator use?

A. SNMP trap

B. Port mirroring

C. Syslog collection

D. API integration

Answer: **B**

Answer: **B**

Question #101 (Topic: Exam A)

Which of the following most likely requires the use of subinterfaces?

A. A router with only one available LAN port

B. A firewall performing deep packet inspection

C. A hub utilizing jumbo frames

D. A switch using Spanning Tree Protocol

Answer: **A**

Answer: **A**

Question #102 (Topic: Exam A)

A company receives a cease-and-desist order from its ISP regarding prohibited torrent activity. Which of the following should be implemented to comply with the cease-and-desist order?

A. MAC security

B. Content filtering

C. Screened subnet

D. Perimeter network

Answer: **B**

Answer: **B**

Question #103 (Topic: Exam A)

Which of the following requires network devices to be managed using a different set of IP addresses?

A. Console

B. Split tunnel

C. Jump box

D. Out of band

Answer: **D**

Answer: **D**

Question #104 (Topic: Exam A)

A network administrator wants to configure a backup route in case the primary route fails. A dynamic routing protocol is not installed on the router. Which of the following routing features should the administrator choose to accomplish this task?

A. Neighbor adjacency

B. Link state flooding

C. Administrative distance

D. Hop count

Answer: **C**

Answer: **C**

Question #105 (Topic: Exam A)

Which of the following steps of the troubleshooting methodology should a technician take to confirm a theory?

A. Duplicate the problem.

B. Identify the symptoms.

C. Gather information.

D. Determine any changes.

Answer: **A**

Answer: **A**

Question #106 (Topic: Exam A)

A network administrator is deploying a new switch and wants to make sure that the default priority value was set for a spanning tree. Which of the following values would the network administrator expect to see?

A. 4096

B. 8192

C. 32768

D. 36684

Answer: **C**

Answer: **C**

Question #107 (Topic: Exam A)

A network administrator is configuring a wireless network with an ESSID. Which of the following is a user benefit of ESSID compared to SSID?

A. Stronger wireless connection

B. Roaming between access points

C. Advanced security

D. Increased throughput

Answer: **B**

Answer: **B**

Question #108 (Topic: Exam A)

A network administrator needs to connect two routers in a point-to-point configuration and conserve IP space. Which of the following subnets should the administrator use?

A. /24

B. /26

C. /28

D. /30

Answer: **D**

Answer: **D**

Question #109 (Topic: Exam A)

An attacker follows an employee through a badge-secured door before the door closes. Which of the following types of attacks is occurring?

A. Shoulder surfing

B. Tailgating

C. Phishing

D. On-path

Answer: **B**

Answer: **B**

Question #110 (Topic: Exam A)

A research facility is expecting to see an exponential increase in global network traffic in the near future. The offices are equipped with 2.5Gbps fiber connections from the ISP, but the facility is currently only utilizing 1Gbps connections. Which of the following would need to be configured in order to use the ISP's connection speed?

A. 802.1Q tagging

B. Network address translation

C. Port duplex

D. Link aggregation

Answer: **D**

Answer: **D**

Question #111 (Topic: Exam A)

A VoIP phone is plugged in to a port but cannot receive calls. Which of the following needs to be done on the port to address the issue?

A. Trunk all VLANs on the port.

B. Configure the native VLAN.

C. Tag the traffic to voice VLAN.

D. Disable VLANs.

Answer: **C**

Answer: **C**

Question #112 (Topic: Exam A)

Which of the following can support a jumbo frame?

A. Access point

B. Bridge

C. Hub

D. Switch

Answer: **D**

Answer: **D**

Question #113 (Topic: Exam A)

A network manager wants to implement a SIEM system to correlate system events. Which of the following protocols should the network manager verify?

A. NTP

B. DNS

C. LDAP

D. DHCP

Answer: **A**

Answer: **A**

Question #114 (Topic: Exam A)

A network engineer is designing a secure communication link between two sites. The entire data stream needs to remain confidential. Which of the following will achieve this goal?

A. GRE

B. IKE

C. ESP

D. AH

Answer: **C**

Answer: **C**

Question #115 (Topic: Exam A)

Which of the following protocols has a default administrative distance value of 90?

A. RIP

B. EIGRP

C. OSPF

D. BGP

Answer: **B**

Answer: **B**

Question #116 (Topic: Exam A)

An office is experiencing intermittent connection issues. A network engineer identifies that the issue occurs whenever someone uses the fax machine that is connected to a switch. Which of the following should the engineer do first to resolve the issue?

A. Run a new Cat 5 line.

B. Enable 802.1Q tagging.

C. Change the MTU.

D. Configure a VLAN.

Answer: **D**

Answer: **D**

Question #117 (Topic: Exam A)

A network engineer receives a vendor alert regarding a vulnerability in a router CPU. Which of the following should the engineer do to resolve the issue?

A. Update the firmware.

B. Replace the system board.

C. Patch the OS.

D. Isolate the system.

Answer: **A**

Answer: **A**

Question #118 (Topic: Exam A)

Which of the following fiber connector types is the most likely to be used on a network interface card?

A. LC

B. SC

C. ST

D. MPO

Answer: **A**

Answer: **A**

Question #119 (Topic: Exam A)

Which of the following connectors provides console access to a switch?

A. ST

B. RJ45

C. BNC

D. SFP

Answer: **B**

Answer: **B**

Question #120 (Topic: Exam A)

A network administrator notices interference with industrial equipment in the 2.4GHz range. Which of the following technologies would most likely mitigate this issue? (Choose two.)

A. Mesh network

B. 5GHz frequency

C. Omnidirectional antenna

D. Non-overlapping channel

E. Captive portal

F. Ad hoc network

Answer: **BD**

Answer: **BD**

Question #121 (Topic: Exam A)

Which of the following network topologies contains a direct connection between every node in the network?

A. Mesh

B. Star

C. Hub-and-spoke

D. Point-to-point

Answer: **A**

Answer: **A**

Question #122 (Topic: Exam A)

A network architect needs to create a wireless field network to provide reliable service to public safety vehicles. Which of the following types of networks is the best solution?

A. Mesh

B. Ad hoc

C. Point-to-point

D. Infrastructure

Answer: **A**

Answer: **A**

Question #123 (Topic: Exam A)

A manager is evaluating the environmental design of a data center. Which of the following setups should the manager implement if the maximum thermal dissipation is the highest concern?

A. A blue-green

B. A hot-cold

C. An east-west

D. An active-passive

Answer: **B**

Answer: **B**

Question #124 (Topic: Exam A)

A network administrator configured a router interface as 10.0.0.95 255.255.255.240. The administrator discovers that the router is not routing packets to a web server with IP

10.0.0.81/28. Which of the following is the best explanation?

- A. The web server is in a different subnet.
- B. The router interface is a broadcast address.
- C. The IP address space is a class A network.
- D. The subnet is in a private address space.

Answer: **B**

Answer: **B**

Question #125 (Topic: Exam A)

A network administrator is notified that a user cannot access resources on the network. The network administrator checks the physical connections to the workstation labeled User 3 and sees the Ethernet is properly connected. However, the network interface's indicator lights are not blinking on either the computer or the switch. Which of the following is the most likely cause?

- A. The switch failed.
- B. The default gateway is wrong.
- C. The port is shut down.
- D. The VLAN assignment is incorrect.

Answer: **C**

Answer: **C**

Question #126 (Topic: Exam A)

Which of the following internal routing protocols is best characterized as having fast convergence and being loop-free?

- A. BGP

B. STP

C. OSPF

D. RIP

Answer: **C**

Answer: **C**

Question #127 (Topic: Exam A)

An administrator is setting up an SNMP server for use in the enterprise network and needs to create device IDs within a MIB. Which of the following describes the function of a MIB?

A. DHCP relay device

B. Policy enforcement point

C. Definition file for event translation

D. Network access controller

Answer: **C**

Answer: **C**

Question #128 (Topic: Exam A)

A critical infrastructure switch is identified as end-of-support. Which of the following is the best next step to ensure security?

A. Apply the latest patches and bug fixes.

B. Decommission and replace the switch.

C. Ensure the current firmware has no issues.

D. Isolate the switch from the network.

Answer: **B**

Answer: **B**

Question #129 (Topic: Exam A)

A company's marketing team created a new application and would like to create a DNS record for newapplication.comptia.org that always resolves to the same address as www.comptia.org. Which of the following records should the administrator use?

A. SOA

B. MX

C. CNAME

D. NS

Answer: **C**

Answer: **C**

Question #130 (Topic: Exam A)

A network administrator wants to implement security zones in the corporate network to control access to only individuals inside of the corporation. Which of the following security zones is the best solution?

A. Extranet

B. Trusted

C. VPN

D. Public

Answer: **B**

Answer: **B**

Question #131 (Topic: Exam A)

Which of the following network devices converts wireless signals to electronic signals?

A. Router

B. Firewall

C. Access point

D. Load balancer

Answer: **C**

Answer: **C**

Question #132 (Topic: Exam A)

A network administrator deployed wireless networking in the office area. When users visit the outdoor patio and try to download emails with large attachments or stream training videos, they notice buffering issues. Which of the following is the most likely cause?

A. Network congestion

B. Wireless interference

C. Signal degradation

D. Client disassociation

Answer: **C**

Answer: **C**

Question #133 (Topic: Exam A)

Which of the following activities would have groups from different departments evaluate the disaster recovery process?

A. Validation test

B. SLA alignment

C. Tabletop exercises

D. Active-active approach

Answer: **C**

Answer: **C**

Question #134 (Topic: Exam A)

Which of the following routing protocols uses an autonomous system number?

A. IS-IS

B. OSPF

C. BGP

D. EIGRP

Answer: **C**

Answer: **C**

Question #135 (Topic: Exam A)

Which of the following is a characteristic of the application layer?

A. It relies upon other layers for packet delivery.

B. It checks independently for packet loss.

C. It encrypts data in transit.

D. It performs address translation.

Answer: **A**

Answer: **A**

Question #136 (Topic: Exam A)

Which of the following IP transmission types encrypts all of the transmitted data?

A. ESP

B. AH

C. GRE

D. UDP

E. TCP

Answer: **A**

Answer: **A**

Question #137 (Topic: Exam A)

Which of the following should be configured so users can authenticate to a wireless network using company credentials?

A. SSO

B. SAML

C. MFA

D. RADIUS

Answer: **D**

Answer: **D**

Question #138 (Topic: Exam A)

A network engineer performed a migration to a new mail server. The engineer changed the MX record, verified the change was accurate, and confirmed the new mail server was reachable via the IP address in the A record. However, users are not receiving email. Which of the following should the engineer have done to prevent the issue from occurring?

A. Change the email client configuration to match the MX record.

B. Reduce the TTL record prior to the MX record change.

C. Perform a DNS zone transfer prior to the MX record change.

D. Update the NS record to reflect the IP address change.

Answer: **B**

Answer: **B**

Question #139 (Topic: Exam A)

Which of the following should a network administrator configure when adding OT devices to an organization's architecture?

A. Honeynet

B. Data-at-rest encryption

C. Time-based authentication

D. Network segmentation

Answer: **D**

Answer: **D**

Question #140 (Topic: Exam A)

To reduce costs and increase mobility, a Chief Technology Officer (CTO) wants to adopt cloud services for the organization and its affiliates. To reduce the impact for users, the CTO wants key services to run from the on-site data center and enterprise services to run in the cloud. Which of the following deployment models is the best choice for the organization?

A. Public

B. Hybrid

C. SaaS

D. Private

Answer: **B**

Answer: **B**

Question #141 (Topic: Exam A)

Which of the following is used to describe the average duration of an outage for a specific service?

A. RPO

B. MTTR

C. RTO

D. MTBF

Answer: **B**

Answer: **B**

Question #142 (Topic: Exam A)

A network engineer is setting up a new VoIP network for a customer. The current network is segmented only for computers and servers. No additional switch ports can be used in the new network. Which of the following does the engineer need to do to configure the network correctly? (Choose two.)

A. Change network translation definitions.

B. Enable 802.1Q.

C. Implement a routing protocol.

D. Set up voice VLANs.

E. Reconfigure the DNS.

F. Place devices in the perimeter network.

Answer: **BD**

Answer: **BD**

Question #143 (Topic: Exam A)

A network rack has four servers and four switches with dual power supplies. Only one intelligent PDU is installed in the rack. Which of the following is the reason to add a second PDU?

A. Power redundancy

B. Failed PSU monitoring

C. Surge protection

D. Electricity conservation

Answer: **A**

Answer: **A**

Question #144 (Topic: Exam A)

Which of the following allows a user to connect to an isolated device on a stand-alone network?

A. Jump box

B. API gateway

C. Secure Shell

D. Clientless VPN

Answer: **A**

Answer: **A**

Question #145 (Topic: Exam A)

Early in the morning, an administrator installs a new DHCP server. In the afternoon, some users report they are experiencing network outages. Which of the following is the most likely issue?

A. The administrator did not provision enough IP addresses.

B. The administrator configured an incorrect default gateway.

C. The administrator did not provision enough routes.

D. The administrator did not provision enough MAC addresses.

Answer: **A**

Answer: **A**

Question #146 (Topic: Exam A)

A company is implementing a policy that will not allow employees to bring personal devices to the office and connect to the wireless network. Which of the following is the best way to enforce this policy?

A. MAC filtering

B. 802.1X

C. Port security

D. ACL

Answer: **B**

Answer: **B**

Question #147 (Topic: Exam A)

A network engineer is troubleshooting an issue with a VoIP network. After performing analysis, the engineer found that when users upload large files, voice quality is severely degraded. When no other traffic is on the network, voice quality is fine. Which of the following should the engineer do to most likely resolve the issue?

A. Enable MAC filtering.

B. Reduce the VLAN database.

C. Configure QoS on the router.

D. Implement ACLs.

Answer: **C**

Answer: **C**

Question #148 (Topic: Exam A)

Which of the following disaster recovery metrics is used to describe the amount of data that is lost since the last backup?

A. MTTR

B. RTO

C. RPO

D. MTBF

Answer: **C**

Answer: **C**

Question #149 (Topic: Exam A)

A network administrator needs to improve network monitoring. Which of the following should the administrator do first?

A. Establish baseline metrics.

B. Implement a SIEM.

C. Perform regular packet captures.

D. Conduct availability monitoring.

Answer: **A**

Answer: **A**

Question #150 (Topic: Exam A)

A network administrator is troubleshooting issues with a DHCP server at a university. More students have recently arrived on campus, and the users are unable to obtain an IP address. Which of the following should the administrator do to address the issue?

A. Enable IP helper.

B. Change the subnet mask.

C. Increase the scope size.

D. Add address exclusions.

Answer: **C**

Answer: **C**

Question #151 (Topic: Exam A)

Which of the following is designed to distribute network traffic among devices based on the quantity of traffic?

A. Load balancer

B. Router

C. Proxy

D. Switch

Answer: **A**

Answer: **A**

Question #152 (Topic: Exam A)

A user recently moved a workstation to a different part of the office. The user is able to access the internet and print but is unable to access server resources. Which of the following is the most likely cause of the issue?

A. Incorrect default gateway

B. Wrong VLAN assignment

C. Error-disabled port

D. Duplicate IP address

Answer: **B**

Answer: **B**

Question #153 (Topic: Exam A)

A technician is planning an equipment installation into a rack in a data center that practices hot aisle/cold aisle ventilation. Which of the following directions should the equipment exhaust face when installed in the rack?

A. Sides

B. Top

C. Front

D. Rear

Answer: **D**

Answer: **D**

Question #154 (Topic: Exam A)

A network administrator needs to add 255 useable IP addresses to the network. A /24 is currently in use. Which of the following prefixes would fulfill this need?

A. /23

B. /25

C. /29

D. /32

Answer: **A**

Answer: **A**

Question #155 (Topic: Exam A)

Which of the following allows an organization to map multiple internal devices to a single external-facing IP address?

A. NAT

B. BGP

C. OSPF

D. FHRP

Answer: **A**

Answer: **A**

Question #156 (Topic: Exam A)

Which of the following steps in the troubleshooting methodology includes checking logs for recent changes?

A. Identify the problem.

B. Document the findings and outcomes.

C. Test the theory to determine cause.

D. Establish a plan of action.

Answer: **A**

Answer: **A**

Question #157 (Topic: Exam A)

Which of the following configurations exempts traffic to the internet from traversing a VPN?

A. Client-to-site

B. Active-passive

C. Split-tunnel

D. Out-of-band

Answer: **C**

Answer: **C**

Question #158 (Topic: Exam A)

Which of the following ports creates a secure connection to a directory service?

A. 22

B. 389

C. 445

D. 636

Answer: **D**

Answer: **D**

Question #159 (Topic: Exam A)

A network engineer configures a new switch and connects it to an existing switch for expansion and redundancy. Users immediately lose connectivity to the network. The network engineer notes the following spanning tree information from both switches.

Switch1

Port	State	Cost
1	Forward	2
2	Forward	2

Switch2

Port	State	Cost
1	Forward	2
2	Forward	2

Which of the following best describes the issue?

A. The port cost should not be equal.

B. The ports should use link aggregation.

C. A root bridge needs to be identified.

D. The switch should be configured for RSTP.

Answer: **C**

Answer: **C**

Question #160 (Topic: Exam A)

A network administrator is implementing security zones for each department. Which of the following should the administrator use to accomplish this task?

A. ACLs

B. Port security

C. Content filtering

D. NAC

Answer: **A**

Answer: **A**

Question #161 (Topic: Exam A)

Which of the following protocols is used to route traffic on the public internet?

A. BGP

B. OSPF

C. EIGRP

D. RIP

Answer: **A**

Answer: **A**

Question #162 (Topic: Exam A)

Which of the following protocols provides remote access utilizing port 22?

A. SSH

B. Telnet

C. TLS

D. RDP

Answer: **A**

Answer: **A**

Question #163 (Topic: Exam A)

After changes were made to a firewall, users are no longer able to access a web server. A network administrator wants to ensure that ports 80 and 443 on the web server are still accessible from the user IP space. Which of the following commands is best suited to perform this testing?

A. dig

B. ifconfig

C. ping

D. nmap

Answer: **D**

Answer: **D**

Question #164 (Topic: Exam A)

Which of the following attacks would most likely cause duplicate IP addresses in a network?

A. Rogue DHCP server

B. DNS poisoning

C. Social engineering

D. Denial-of-service

Answer: **A**

Answer: **A**

Question #165 (Topic: Exam A)

An organization wants better network visibility. The organization's requirements include:

- Multivendor/OS-monitoring capabilities
- Real-time collection
- Data correlation

Which of the following meets these requirements?

A. SNMP

B. SIEM

C. Nmap

D. Syslog

Answer: **B**

Answer: **B**

Question #166 (Topic: Exam A)

An administrator is evaluating the use of authentication within SNMP. Which of the following is the most secure way of authenticating devices using only SNMP?

A. Use version 1 of SNMP and use a community string to serve as a PSK

B. Use version 3 of SNMP and set up trap messages for critical events on the network

C. Use version 2c of SNMP and use informs to validate device authentication

D. Use version 2u of SNMP to authenticate devices for network monitoring

Answer: **B**

Answer: **B**

Question #167 (Topic: Exam A)

A company has observed increased user traffic to gambling websites and would like to limit this behavior on work computers. Which of the following should the company most likely implement?

A. ACLS

B. Content filter

C. Port security

D. Screened subnet

Answer: **B**

Answer: **B**

Question #168 (Topic: Exam A)

Following a fire in a data center, the cabling was replaced. Soon after an administrator notices network issues. Which of the following are the most likely causes of the network issues? (Choose two.)

A. The switches are not the correct voltage

B. The HVAC system was not verified as fully functional after the fire

C. The VLAN database was not deleted before the equipment was brought back online

D. The RJ45 cables were replaced with unshielded cables

E. The wrong transceiver type was used for the new termination

F. The new RJ45 cables are a higher category than the old ones

Answer: **DE**

Answer: **DE**

Question #169 (Topic: Exam A)

Cloud computing has the capability to meet increased and decreased demands for computing infrastructure for a short period of time. Which of the following best describes this characteristic?

A. Efficiency

B. Elasticity

C. Multitenancy

D. Scalability

Answer: **B**

Answer: **B**

Question #170 (Topic: Exam A)

Which of the following can also provide a security feature when implemented?

A. NAT

B. BGP

C. FHRP

D. EIGRP

Answer: **A**

Answer: **A**

Question #171 (Topic: Exam A)

Which of the following best describes the transmission format that occurs at the transport layer over connectionless communication?

A. Datagram

B. Segment

C. Frames

D. Packets

Answer: **A**

Answer: **A**

Question #172 (Topic: Exam A)

Which of the following is most likely to work with an FC connection and offers more scalability?

A. DAS

B. SAN

C. SSE

D. NAS

Answer: **B**

Answer: **B**

Question #173 (Topic: Exam A)

A company is moving to a hybrid cloud model. As part of this move the mail server will be moved to the cloud. The systems administrator needs to ensure the mail server continues to receive email. Which of the following types of DNS records should the systems administrator update?

A. NS

B. PTR

C. MX

D. A

Answer: **C**

Answer: **C**

Question #174 (Topic: Exam A)

A network engineer is configuring network ports in a public office. To increase security, the engineer wants the ports to allow network connections only after authentication. Which of the following security features should the engineer use?

A. Port security

B. 802.1X

C. MAC filtering

D. Access control list

Answer: **B**

Answer: **B**

Question #175 (Topic: Exam A)

A network administrator is expanding a network and wants to ensure no unauthorized redundant links are present. Which of the following should the administrator use to identify and block redundant links?

A. SDN

B. STP

C. LLDP

D. OSPF

Answer: **B**

Answer: **B**

Question #176 (Topic: Exam A)

A network administrator notices uncommon communication between VMs on ephemeral ports on the same subnet. The administrator is concerned about that traffic moving laterally within the network. Which of the following describes the type of traffic flow the administrator is analyzing?

A. East-west

B. Point-to-point

C. Horizontal-scaling

D. Hub-and-spoke

Answer: **A**

Answer: **A**

Question #177 (Topic: Exam A)

Which of the following would be violated if an employee accidentally deleted a customer's data?

A. Integrity

B. Confidentiality

C. Vulnerability

D. Availability

Answer: **D**

Answer: **D**

Question #178 (Topic: Exam A)

A network engineer is now in charge of all SNMP management in the organization. The engineer must use a SNMP version that does not utilize plaintext data. Which of the following is the minimum version of SNMP that supports this requirement?

A. v1

B. v2c

C. v2u

D. v3

Answer: **D**

Answer: **D**

Question #179 (Topic: Exam A)

A company recently acquired a number of sites and no documentation was provided. A network administrator needs to identify and document all of the digital assets in use. Which of the following is the best method for the administrator to use?

A. Availability monitoring

B. Packet capture

C. Physical inventory

D. Network discovery

Answer: **D**

Answer: **D**

Question #180 (Topic: Exam A)

A network administrator recently upgraded a wireless infrastructure with new APs. Users are reporting that, when stationary, the wireless connection drops and reconnects after 20 to 30 seconds. While reviewing the logs, the administrator notices that the APs are changing channels. Which of the following is the most likely reason for the service interruptions?

A. Channel interference

B. Roaming misconfiguration

C. Network congestion

D. Insufficient wireless coverage

Answer: **A**

Answer: **A**

Question #181 (Topic: Exam A)

A network administrator needs to set up a multicast network for audio and video broadcasting. Which of the following networks would be the most appropriate for this application?

A. 172.16.0.0/24

B. 192.168.0.0/24

C. 224.0.0.0/24

D. 240.0.0.0/24

Answer: **C**

Answer: **C**

Question #182 (Topic: Exam A)

A user connects to a corporate VPN via a web browser and is able to use TLS to access the internal financial system to input a time card. Which of the following best describes how the VPN is being used?

A. Clientless

B. Client-to-site

C. Full tunnel

D. Site-to-site

Answer: **A**

Answer: **A**

Question #183 (Topic: Exam A)

A network administrator needs to fail over services to an off-site environment. This process will take four weeks to become fully operational. Which of the following DR concepts does this describe?

A. Hot site

B. Warm site

C. Cold site

D. Active-active approach

Answer: **C**

Answer: **C**

Question #184 (Topic: Exam A)

A security analyst wants to control internet access based on site reputation and categorization. The analyst needs a solution that does not require traffic flow changes. Which of the following solutions would most likely meet these requirements?

A. Proxy

B. DNS filtering

C. Router ACL

D. Load balancer

Answer: **B**

Answer: **B**

Question #185 (Topic: Exam A)

A wireless technician wants to implement a technology that will allow user devices to automatically navigate to the best available frequency standard. Which of the following technologies should the technician use?

A. Band steering

B. Wireless LAN controller

C. Directional antenna

D. Autonomous access point

Answer: **A**

Answer: **A**

Question #186 (Topic: Exam A)

A user called the help desk after business hours to complain that files on a device are inaccessible and the wallpaper was changed. The network administrator thinks that this issue is an isolated incident, but the security analyst thinks the issue might be a ransomware attack. Which of the following troubleshooting steps should be taken first?

A. Identify the problem.

B. Establish a theory.

C. Document findings.

D. Create a plan of action.

Answer: **A**

Answer: **A**

Question #187 (Topic: Exam A)

A data center interconnect using a VXLAN was recently implemented. A network engineer observes slow performance and fragmentation on the interconnect. Which of the following technologies will resolve the issue?

A. 802.1Q tagging

B. Spanning tree

C. Link aggregation

D. Jumbo frames

Answer: **D**

Answer: **D**

Question #188 (Topic: Exam A)

A new server is deployed in a trusted zone and is validated to be online with all appropriate services running. However, users in a perimeter network cannot access the server. Which of the following should a network administrator do to resolve the reported issue?

A. Create a TXT record on the DNS server for the new server.

B. Update the firewall ACL to allow access to the new server.

C. Configure a static route for the user VLAN to the new server.

D. Insert a static ARP entry for the new server on the Layer 3 switch.

Answer: **B**

Answer: **B**

Question #189 (Topic: Exam A)

Network administrators are using the Telnet protocol to administer network devices that are on the 192.168.1.0/24 subnet. Which of the following tools should the administrator use to best identify the devices?

A. dig

B. nmap

C. tracert

D. telnet

Answer: **B**

Answer: **B**

Question #190 (Topic: Exam A)

A network administrator recently updated configurations on a Layer 3 switch. Following the updates, users report being unable to reach a specific file server. Which of the following is the most likely cause?

- A. Incorrect ACLs
- B. Switching loop
- C. Duplicate IP addresses
- D. Wrong default route

Answer: **A**

Answer: **A**

Question #191 (Topic: Exam A)

A network engineer wants to implement a solution where all web servers will send event data over port 514. Which of the following services will accomplish this task?

- A. Syslog
- B. SMTP
- C. DNS
- D. DHCP

Answer: **A**

Answer: **A**

Question #192 (Topic: Exam A)

A security administrator wants to discover zero-day attacks before they can be used on the company's network. Which of the following can the administrator use to accomplish this task?

A. Central repository

B. Honeypot

C. Next-generation firewall

D. Evil twin

Answer: **B**

Answer: **B**

Question #193 (Topic: Exam A)

Users at a satellite office are experiencing issues when using videoconferencing. Which of the following should a technician focus on first to rectify these issues?

A. Quality of service

B. Network signal

C. Time to live

D. Load balancing

Answer: **A**

Answer: **A**

Question #194 (Topic: Exam A)

A network administrator is trying to locate the switch interface a PC is connected to. The administrator accesses the local switch, pings the PC IP address, and then uses the command show arp. Which of the following commands should the administrator use next?

A. show route

B. show mac-address-table

C. show vlan

D. show interface

Answer: **B**

Answer: **B**

Question #195 (Topic: Exam A)

A user attempts to log in to a corporate website by utilizing a shortcut. The shortcut has been used many times before. The user then notices some discrepancies on the company website. Which of the following is most likely the reason for this issue?

A. VLAN hopping

B. ARP spoofing

C. DNS poisoning

D. MAC flooding

Answer: **C**

Answer: **C**

Question #196 (Topic: Exam A)

Which of the following is the correct order of components in a bottom-up approach for the three-tier hierarchical model?

A. Access, distribution, and core

B. Core, root, and distribution

C. Core, spine, and leaf

D. Access, core, and root

Answer: **A**

Answer: **A**

Question #197 (Topic: Exam A)

A client with a 2.4GHz wireless network has stated that the entire office is experiencing intermittent issues with laptops after the WAP was moved. Which of the following is the most likely reason for these issues?

A. The network uses a non-overlapping channel.

B. The signal is reflecting too much.

C. The network has excessive noise.

D. A microwave is in the office.

Answer: **D**

Answer: **D**

Question #198 (Topic: Exam A)

Which of the following SD-WAN features allows a router to be shipped directly to the installation site and does not require site-level configuration?

A. Zero-touch provisioning

B. Application aware

C. Transport agnostic

D. Central policy management

Answer: **A**

Answer: **A**

Question #199 (Topic: Exam A)

A network technician is troubleshooting network latency and has determined the issue to be occurring between two network switches (Switch10 and Switch11). Symptoms

reported include poor video performance and slow file copying. Given the following information:

Switch10 "show interfaces" output:

```
MTU 1600 bytes, Speed 1000/full, VLAN10
12912 packets input, 2398471 bytes, 0 runts, 0 giants
19328 packets output, 9832984 bytes, 0 runts, 0 giants
```

Switch11 "show interfaces" output:

```
MTU 1500 bytes, Speed Auto Negotiate, VLAN10
19328 packets input, 9832984 bytes, 0 runts, 2394 giants
12912 packets output, 2398471 bytes, 0 runts, 0 giants
```

Which of the following should the technician most likely do to resolve the issue?

A. Disable automatic negotiation on Switch11.

B. Modify Switch10 MTU value to 1500.

C. Configure STP on both switches.

D. Change the native VLAN on the ports.

Answer: **B**

Answer: **B**

Question #200 (Topic: Exam A)

Which of the following types of routes will take precedence when building a routing table for a given subnet?

A. Static

B. BGP

C. OSPF

D. Default

Answer: **A**

Answer: **A**

Question #201 (Topic: Exam A)

Which of the following steps in the troubleshooting methodology comes after using a top-to-bottom examination of the OSI model to determine cause?

A. Test the theory.

B. Establish a plan of action.

C. Verify full system functionality.

D. Identify the problem.

Answer: **A**

Answer: **A**

Question #202 (Topic: Exam A)

Which of the following would an adversary do while conducting an evil twin attack?

A. Trick users into using an AP with an SSID that is identical to a legitimate network.

B. Manipulate address resolution to point devices to a malicious endpoint.

C. Present an identical MAC to gain unauthorized access to network resources.

D. Capture data in transit between two legitimate endpoints to steal data.

Answer: **A**

Answer: **A**

Question #203 (Topic: Exam A)

An ISP provided a company with an IP range of 98.174.235.142/28. A network technician configured a router with a subnet mask of 255.255.255.224 and default gateway of 98.174.235.129. After the configuration was set up, the company is unable to connect to the ISP. Which of the following would resolve this issue?

A. Change the subnet mask to .240.

B. Change the Ethernet cable from the external interface to the modem.

C. Change the CIDR to /29.

D. Change the static IP to 98.174.235.128/28.

Answer: **A**

Answer: **A**

Question #204 (Topic: Exam A)

A network administrator is troubleshooting a connection between two switches that is experiencing intermittent errors. The administrator needs to determine which port on the remote switch the faulty cable is connected to. Which of the following is the best tool to use to identify the error?

A. tracer

B. LLDP run

C. nmap

D. ping

Answer: **B**

Answer: **B**

Question #205 (Topic: Exam A)

Which of the following is a difference between EOL and EOS?

A. EOL discontinues the product but may offer support.

B. EOS replaces free support with a subscription model.

C. EOS only applies to physical products.

D. EOL still guarantees warranty service.

Answer: **A**

Answer: **A**

Question #206 (Topic: Exam A)

A network architect is implementing an off-premises computing facility and needs to ensure that operation will not be impacted by major outages. Which of the following should the architect consider?

A. Hot site

B. DCI

C. Direct Connect

D. Active-passive approach

Answer: **A**

Answer: **A**

Question #207 (Topic: Exam A)

A network engineer is testing a website to ensure it is compatible with IPv6. After attempting to ping the website by its IPv6 address, the engineer determines that the DNS has not been set up properly. Which of the following should the network engineer complete to resolve this issue?

A. Enable a PTR record.

B. Update the existing TXT record.

C. Add a new AAA record.

D. Configure a secondary NS record.

Answer: **C**

Answer: **C**

Question #208 (Topic: Exam A)

A small coffee shop wants to set up multiple 2.4GHz wireless access points. The access points will support a large number of users, and the network technician wants to mitigate interference as much as possible. Which of the following is the number of 22MHz channels that the equipment can support?

A. 1

B. 2

C. 3

D. 4

Answer: **C**

Answer: **C**

Question #209 (Topic: Exam A)

A company is purchasing a 40Gbps broadband connection service from an ISP. Which of the following should most likely be configured on the 10G switch to take advantage of the new service?

A. 802.11Q tagging

B. Jumbo frames

C. Half duplex

D. Link aggregation

Answer: **D**

Answer: **D**

Question #210 (Topic: Exam A)

Which of the following are the main differences between ESP and AH? (Choose two.)

A. AH provides confidentiality through the use of encryption.

B. ESP provides authentication for IP headers and their payloads.

C. ESP provides confidentiality through the use of encryption.

D. AH provides authentication for IP headers and their payloads.

E. AH provides data origin authorization over shared secret.

F. ESP provides data origin authorization over shared secret.

Answer: **CD**

Answer: **CD**

Question #211 (Topic: Exam A)

An organization wants to ensure that incoming emails were sent from a trusted source. Which of the following DNS records is used to verify the source?

A. TXT

B. AAA

C. CNAME

D. MX

Answer: **A**

Answer: **A**

Question #212 (Topic: Exam A)

Which of the following is the most closely associated with segmenting compute resources within a single cloud account?

A. Network security group

B. IaaS

C. VPC

D. Hybrid cloud

Answer: **C**

Answer: **C**

Question #213 (Topic: Exam A)

Which of the following is the most likely function to be decremented when a tool is used to measure the distance between two locations on the internet?

A. GRE

B. TTL

C. VPN

D. QoS

Answer: **B**

Answer: **B**

Question #214 (Topic: Exam A)

A user's home mesh wireless network is experiencing latency issues. A technician has:

- Performed a speed test.
- Rebooted the devices.
- Performed a site survey.
- Performed a wireless packet capture.

The technician reviews the following information:

SSID	MAC	RSSI	Channel
Home	AAAAAA111112	-83dBm	11
Home	AAAAAA111113	-91dBm	11
Home	AAAAAA111114	-70dBm	11

The technician notices in the packet capture that frames were retransmitted. Which of the following is the most likely cause of the user's network issue?

A. The SSIDs should not be the same.

B. The network has too much overlap.

C. The devices are incompatible with the mesh network.

D. The nodes are underpowered.

Answer: **B**

Answer: **B**

Question #215 (Topic: Exam A)

A network engineer implements a 192.168.100.0/25 subnet for a building without obtaining sizing requirements. It is later determined that the building will house 700 people. Which of the following subnet masks will most efficiently support that number of people?

A. 255.255.252.0

B. 255.255.254.0

C. 255.255.255.0

D. 255.255.255.128

Answer: **A**

Answer: **A**

Question #216 (Topic: Exam A)

A company is building a new campus that will have wired and wireless devices. The network will be designed to support casual internet usage for guests and break rooms for employees. Which of the following authentication methods is best suited for guests given the requirements of this scenario?

A. ESP

B. PSK

C. WPA

D. IKE

Answer: **B**

Answer: **B**

Question #217 (Topic: Exam A)

Following the troubleshooting methodology, a technician implements a solution that resolves the reported issue. Which of the following should the technician do next?

A. Document the findings.

B. Create a plan of action.

C. Establish a theory.

D. Identify the problem.

Answer: **A**

Answer: **A**

Question #218 (Topic: Exam A)

A network administrator needs to divide 192.168.1.0/24 into two equal halves. Which of the following subnet masks should the administrator use?

A. 255.255.0.0

B. 255.255.254.0

C. 255.255.255.0

D. 255.255.255.128

Answer: **D**

Answer: **D**

Question #219 (Topic: Exam A)

Which of the following protocols uses the Dijkstra's Link State Algorithm to establish routes inside its routing table?

A. OSPF

B. EIGRP

C. BGP

D. RIP

Answer: **A**

Answer: **A**

Question #220 (Topic: Exam A)

Which of the following traffic types does a DHCP Discover request from a PC use?

A. Broadcast

B. Anycast

C. Unicast

D. Multicast

Answer: **A**

Answer: **A**

Question #221 (Topic: Exam A)

A user's VoIP phone and workstation are connected through an inline cable. The user reports that the VoIP phone intermittently reboots, but the workstation is not having any network-related issues. Which of the following is the most likely cause?

A. The PoE power budget is exceeded.

B. Port security is violated.

C. The signal is degraded.

D. The Ethernet cable is not working.

Answer: **A**

Answer: **A**

Question #222 (Topic: Exam A)

A network administrator is installing a network for a small company. The switches will be installed in parallel to provide redundancy in case a specific device fails. Which of the following should be configured to prevent loops?

A. Spanning tree

B. CRC

C. Jumbo frames

D. Link aggregation

Answer: **A**

Answer: **A**

Question #223 (Topic: Exam A)

A company's network is experiencing high latency and packet loss during peak hours. Network monitoring tools show increased traffic on a switch. Which of the following should a network technician implement to reduce the network congestion and improve performance?

A. Load balancing

B. Port mirroring

C. Quality of service

D. Spanning Tree Protocol

Answer: **A**

Answer: **A**

Question #224 (Topic: Exam A)

Which of the following VPN types provides secure remote access to the network resources through a web portal?

A. Proxy

B. Clientless

C. Site-to-site

D. Direct connect

Answer: **B**

Answer: **B**

Question #225 (Topic: Exam A)

Which of the following best explains the role of confidentiality with regard to data at rest?

A. Data can be accessed by anyone on the administrative network.

B. Data can be accessed remotely with proper training.

C. Data can be accessed after privileged access is granted.

D. Data can be accessed after verifying the hash.

Answer: **C**

Answer: **C**

Question #226 (Topic: Exam A)

Several users in an organization report connectivity issues and lag during a video meeting. The network administrator performs a tcpdump and observes increased retransmissions for other non-video applications on the network. Which of the following symptoms describes the users' reported issues?

A. Latency

B. Packet loss

C. Bottlenecking

D. Jitter

Answer: **B**

Answer: **B**

Question #227 (Topic: Exam A)

An administrator enables DNS filtering on the firewall to block users from visiting malicious websites. Which of the following should the administrator also do? (Choose two.)

A. Disable DoH in users' internet browsers.

B. Update NS record to point to DNS filter servers.

C. Block port 443 to the malicious websites.

D. Block port 53 to servers on the internet.

E. Disable TLS v1.3 in users' internet browsers.

F. Implement DNSSEC for corporate records.

Answer: **AD**

Answer: **AD**

Question #228 (Topic: Exam A)

Which of the following is concerned with guaranteed availability of a cloud resource?

A. IPAM

B. SLA

C. MTBF

D. EOL

Answer: **B**

Answer: **B**

Question #229 (Topic: Exam A)

A network consultant needs to decide between running an Ethernet uplink or using the built-in 5GHz point-to-point functionality on a WAP. Which of the following documents provides the best information to assist the consultant with this decision?

A. Site survey results

B. Physical diagram

C. Service-level agreement

D. Logical diagram

Answer: **A**

Answer: **A**

Question #230 (Topic: Exam A)

While troubleshooting connectivity issues, a junior network administrator is given explicit instructions to test the host's TCP/IP stack first. Which of the following commands should the network administrator run?

A. ping 127.0.0.1

B. ping 169.254.1.1

C. ping 172.16.1.1

D. ping 192.168.1.1

Answer: **A**

Answer: **A**

Question #231 (Topic: Exam A)

Users at an organization report that the wireless network is not working in some areas of the building. Users report that other wireless network SSIDs are visible when searching for the network, but the organization's network is not displayed. Which of the following is the most likely cause?

A. Interference or channel overlap

B. Insufficient wireless coverage

C. Roaming misconfiguration

D. Client disassociation issues

Answer: **B**

Answer: **B**

Question #232 (Topic: Exam A)

Which of the following panels would be best to facilitate a central termination point for all network cables on the floor of a company building?

A. Patch

B. UPS

C. MDF

D. Rack

Answer: **A**

Answer: **A**

Question #233 (Topic: Exam A)

Which of the following is most commonly associated with many systems sharing one IP address in the public IP-addressing space?

A. PAT

B. NAT

C. VIP

D. NAT64

Answer: **A**

Answer: **A**

Question #234 (Topic: Exam A)

A network administrator is reviewing a production web server and observes the following output from the netstat command:

Proto	LocalAddress	ForeignAddress	State	PID
TCP	0.0.0.0:20	0.0.0.0:0	LISTENING	1308
TCP	0.0.0.0:22	0.0.0.0:0	LISTENING	329
TCP	0.0.0.0:23	0.0.0.0:0	LISTENING	758
TCP	0.0.0.0:25	0.0.0.0:0	LISTENING	1459
TCP	0.0.0.0:53	0.0.0.0:0	LISTENING	1243
TCP	0.0.0.0:69	0.0.0.0:0	LISTENING	392
TCP	0.0.0.0:80	0.0.0.0:0	LISTENING	43
TCP	0.0.0.0:443	0.0.0.0:0	LISTENING	43

Which of the following actions should the network administrator take to harden the security of the web server?

A. Disable the unused ports.

B. Enforce access control lists.

C. Perform content filtering.

D. Set up a screened subnet.

Answer: **A**

Answer: **A**

Question #235 (Topic: Exam A)

A network technician is examining the configuration on an access port and notices more than one VLAN has been set. Which of the following best describes how the port is configured?

A. With a voice VLAN

B. With too many VLAN

C. With a default VLAN

D. With a native VLAN

Answer: **A**

Answer: **A**

Question #236 (Topic: Exam A)

A small company has the following IP addressing strategy:

User subnet: 192.168.1.0/24

Voice subnet: 192.168.2.0/24

Server subnet: 192.168.10.0/24

A user is unable to connect to the company fileshare server located at 192.168.10.1. The user's networking configuration is:

```
Physical Address. . . . . : 7A-01-7A-21-12-01
IPv4 Address. . . . . : 192.168.1.50
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . : 192.168.1.200
DNS Servers . . . . . : 192.168.10.10
                      : 192.168.10.11
```

Which of the following will most likely correct the issue?

A. Changing the IPv4 address to 192.168.10.1

B. Changing the subnet mask to 255.255.255.0

C. Changing the DNS servers to internet IPs

D. Changing the physical address to 7A-01-7A-21-01-50

Answer: **B**

Answer: **B**

Question #237 (Topic: Exam A)

A network administrator needs to connect a department to a new network segment. They need to use a DHCP server located on another network.

Which of the following can the administrator use to complete this task?

A. IP Helper

B. Reservation

C. Exclusion

D. Scope

Answer: **A**

Answer: **A**

Question #238 (Topic: Exam A)

A network engineer is implementing a solution by which on-premises servers are migrated to a cloud service provider. Which of the following service models will provide a lift and shift solution to meet this requirement?

A. Infrastructure as a service

B. Infrastructure as code

C. Software as a service

D. Software-defined network

Answer: **A**

Answer: **A**

Question #239 (Topic: Exam A)

A network administrator is conducting an assessment and found network devices that do not meet standards. Which of the following configurations is considered a set of rules that devices should adhere to?

A. Production

B. Backup

C. Candidate

D. Golden

Answer: **D**

Answer: **D**

Question #240 (Topic: Exam A)

An IT department asks a newly hired employee to use a personal laptop until the company can provide one. Which of the following policies is most applicable to this situation?

A. IAM

B. BYOD

C. DLP

D. AUP

Answer: **B**

Answer: **B**

Question #241 (Topic: Exam A)

Which of the following offers the ability to manage access at the cloud VM instance?

A. Security group

B. Internet gateway

C. Direct Connect

D. Network ACL

Answer: **A**

Answer: **A**

Question #242 (Topic: Exam A)

In an environment with one router, which of the following will allow a network engineer to communicate between VLANs without purchasing additional hardware?

A. Subinterfaces

B. VXLAN

C. Layer 3 switch

D. VIP

Answer: **A**

Answer: **A**

Question #243 (Topic: Exam A)

A network administrator is concerned with on-path attacks and wants to provide encryption of fully qualified domain names in outbound communications. Which of the following best describes the protocol that provides the proper security layer for internet communication?

A. SSL

B. DoH

C. TLS

D. DNSSEC

Answer: **B**

Answer: **B**

Question #244 (Topic: Exam A)

A network administrator wants to restrict inbound traffic to allow only HTTPS to the company website, denying all other inbound traffic from the internet. Which of the following would best accomplish this goal?

A. ACL on the edge firewall

B. Port security on an access switch

C. Content filtering on a web gateway

D. URL filtering on an outbound proxy

Answer: **A**

Answer: **A**

Question #245 (Topic: Exam A)

A network engineer is deploying switches at a new remote office. The switches have been preconfigured with hostnames and STP priority values.

Based on the following table:

Hostname	MAC address	STP priority
core-sw01	0000.0000.005f	4096
distribution-sw01	0000.0000.001a	32768
access-sw01	0000.0000.005e	4096
access-sw02	0000.0000.001b	32768

Which of the following switches will become the root bridge?

A. core-sw01

B. access-sw01

C. distribution-sw01

D. access-sw02

Answer: **B**

Answer: **B**

Question #246 (Topic: Exam A)

Which of the following must be implemented to securely connect a company's headquarters with a branch location?

A. Split-tunnel VPN

B. Clientless VPN

C. Full-tunnel VPN

D. Site-to-site VPN

Answer: **D**

Answer: **D**

Question #247 (Topic: Exam A)

A network engineer needs to virtualize network services, including a router, at a remote branch location. Which of the following solutions meets the requirements?

A. NFV

B. VRF

C. VLAN

D. VPC

Answer: **A**

Answer: **A**

Question #248 (Topic: Exam A)

A group of users cannot connect to network resources. The technician runs ipconfig from one user's device and is able to ping the gateway shown from the command. Which of the following is most likely preventing the users from accessing network resources?

A. VLAN hopping

B. Rogue DHCP

C. Distributed DoS

D. Evil twin

Answer: **B**

Answer: **B**

Question #249 (Topic: Exam A)

Which of the following disaster recovery concepts is calculated by dividing the total hours of operation by the total number of units?

A. MTTR

B. MTBF

C. RPO

D. RTO

Answer: **B**

Answer: **B**

Question #250 (Topic: Exam A)

A network administrator has been monitoring the company's servers to ensure that they are available. Which of the following should the administrator use for this task?

A. Packet capture

B. Data usage reports

C. SNMP traps

D. Configuration monitoring

Answer: **C**

Answer: **C**

Question #251 (Topic: Exam A)

A network engineer is completing a wireless installation in a new building. A requirement is that all clients be able to automatically connect to the fastest supported network. Which of the following best supports this requirement?

A. Enabling band steering

B. Disabling the 5GHz SSID

C. Adding a captive portal

D. Configuring MAC filtering

Answer: **A**

Answer: **A**

Question #252 (Topic: Exam A)

A network technician notices a device is at EOL. For which of the following should the technician plan?

A. Retrofitting

B. Decommissioning

C. Restarting

D. Upgrading

Answer: **B**

Answer: **B**

Question #253 (Topic: Exam A)

A company implements a video streaming solution that will play on all computers that have joined a particular group, but router ACLs are blocking the traffic. Which of the following is the most appropriate IP address that will be allowed in the ACL?

A. 127.0.0.1

B. 172.17.1.1

C. 224.0.0.1

D. 240.0.0.1

Answer: **C**

Answer: **C**

Question #254 (Topic: Exam A)

Which of the following are the best device-hardening techniques for network security? (Choose two.)

A. Disabling unused ports

B. Performing regular scanning of unauthorized devices

C. Monitoring system logs for irregularities

D. Enabling logical security such as SSO

E. Changing default passwords

F. Ensuring least privilege concepts are in place

Answer: **AE**

Answer: **AE**

Question #255 (Topic: Exam A)

Which of the following allows for the interception of traffic between the source and destination?

A. Self-signed certificate

B. VLAN hopping

C. On-path attack

D. Phishing

Answer: **C**

Answer: **C**

Question #256 (Topic: Exam A)

A company discovers on video surveillance recordings that an unauthorized person installed a rogue access point in its secure facility. Which of the following allowed the unauthorized person to do this?

A. Evil twin

B. Honeytrap

C. Wardriving

D. Tailgating

Answer: **D**

Answer: **D**

Question #257 (Topic: Exam A)

A secure communication link needs to be configured between data centers via the internet. The data centers are located in different regions. Which of the following is the best protocol for the network administrator to use?

A. DCI

B. GRE

C. VXLAN

D. IPSec

Answer: **D**

Answer: **D**

Question #258 (Topic: Exam A)

Which of the following can be implemented to add an additional layer of security between a corporate network and network management interfaces?

A. Jump box

B. Console server

C. API interface

D. In-band management

Answer: **A**

Answer: **A**

Question #259 (Topic: Exam A)

A network administrator performed upgrades on a server and installed a new NIC to improve performance. Following the upgrades, users are unable to reach the server.

Which of the following is the most likely reason?

A. The PoE power budget was exceeded.

B. TX/RX was transposed.

C. A port security violation occurred.

D. An incorrect cable type was installed.

Answer: **C**

Answer: **C**

Question #260 (Topic: Exam A)

A junior network technician at a large company needs to create networks from a Class C address with 14 hosts per subnet. Which of the following numbers of host bits is required?

A. One

B. Two

C. Three

D. Four

Answer: **D**

Answer: **D**

Question #261 (Topic: Exam A)

Which of the following is a company most likely enacting if an accountant for the company can only see the financial department's shared folders?

A. General Data Protection Regulation

B. Least privilege network access

C. Acceptable use policy

D. End user license agreement

Answer: **B**

Answer: **B**

Question #262 (Topic: Exam A)

Which of the following connector types would most likely be used to connect to an external antenna?

A. BNC

B. ST

C. LC

D. MPO

Answer: **A**

Answer: **A**

Question #263 (Topic: Exam A)

A company recently converted most of the office laptops to connect wirelessly to the corporate network. After a recent high-traffic malware attack, narrowing the event to a specific user was difficult because of the wireless configuration. Which of the following actions should the company take?

A. Restrict users to the 5GHz frequency.

B. Upgrade to a mesh network.

C. Migrate from PSK to Enterprise.

D. Implement WPA2 encryption.

Answer: **C**

Answer: **C**

Question #264 (Topic: Exam A)

Which of the following cloud service models most likely requires the greatest up-front expense by the customer when migrating a data center to the cloud?

A. Infrastructure as a service

B. Software as a service

C. Platform as a service

D. Network as a service

Answer: **A**

Answer: **A**

Question #265 (Topic: Exam A)

After providing a username and password, a user must input a passcode from a phone application. Which of the following authentication technologies is used in this example?

A. SSO

B. LDAP

C. MFA

D. SAML

Answer: **C**

Answer: **C**

Question #266 (Topic: Exam A)

Which of the following facilities is the best example of a warm site in the event of information system disruption?

A. A combination of public and private cloud services to restore data

B. A partial infrastructure, software, and data on site

C. A full electrical infrastructure in place, but no customer devices on site

D. A full infrastructure in place, but no current data on site

Answer: **B**

Answer: **B**

Question #267 (Topic: Exam A)

Which of the following services runs on port 636?

A. SMTP

B. Syslog

C. TFTP

D. LDAPS

Answer: **D**

Answer: **D**

Question #268 (Topic: Exam A)

A company wants to implement data loss prevention by restricting user access to social media platforms and personal cloud storage on workstations. Which of the following types of filtering should the company deploy to achieve these goals?

A. Port

B. DNS

C. MAC

D. Content

Answer: **D**

Answer: **D**

Question #269 (Topic: Exam A)

An organization is struggling to get effective coverage using the wireless network. The organization wants to implement a solution that will allow for continuous connectivity anywhere in the facility. Which of the following should the network administrator suggest to ensure the best coverage?

A. Implementing additional ad hoc access points

B. Providing more Ethernet drops for user connections

C. Deploying a mesh network in the building

D. Changing the current frequency of the Wi-Fi

Answer: **C**

Answer: **C**

Question #270 (Topic: Exam A)

Which of the following protocols allows automatic routing failover for an active-passive firewall pair?

A. EIGRP

B. OSPF

C. VRRP

D. BGP

Answer: **C**

Answer: **C**

Question #271 (Topic: Exam A)

Which of the following is a major difference between an IPS and IDS?

A. An IPS needs to be installed in line with traffic and an IDS does not.

B. An IPS is signature-based and an IDS is not.

C. An IPS is less susceptible to false positives than an IDS.

D. An IPS requires less administrative overhead than an IDS.

Answer: **A**

Answer: **A**

Question #272 (Topic: Exam A)

Which of the following describes the best reason for using BGP?

A. Preventing a loop within a LAN

B. Improving reconvergence times

C. Exchanging router updates with a different ISP

D. Sharing routes with a Layer 3 switch

Answer: **C**

Answer: **C**

Question #273 (Topic: Exam A)

Which of the following steps of the troubleshooting methodology comes after testing the theory to determine cause?

A. Verify full system functionality.

B. Document the findings and outcomes.

C. Establish a plan of action.

D. Identify the problem.

Answer: **C**

Answer: **C**

Question #274 (Topic: Exam A)

A network administrator installed a new VLAN to the network after a company added an additional floor to the office. Users are unable to obtain an IP address on the new VLAN, but ports on existing VLANs are working properly. Which of the following configurations should the administrator update?

A. Scope size

B. Address reservations

C. Lease time

D. IP helper

Answer: **D**

Answer: **D**

Question #275 (Topic: Exam A)

A network engineer needs to add a boundary network to isolate and separate the internal network from the public-facing internet. Which of the following security defense solutions would best accomplish this task?

A. Trusted zones

B. URL filtering

C. ACLs

D. Screened subnet

Answer: **D**

Answer: **D**

Question #276 (Topic: Exam A)

Which of the following can help prevent unauthorized modifications to hardware when designing an MDF closet?

A. Lockable cabinets

B. Uninterruptable power supply

C. Port security

D. Microsegmentation

Answer: **A**

Answer: **A**

Question #277 (Topic: Exam A)

After installing a new 6E wireless router in a small office, a technician notices that some wireless devices are not able to achieve the rated speeds. Which of the following should the technician check to troubleshoot the issue? (Choose two.)

A. Client device compatibility

B. Back-end cabling

C. Weather phenomena

D. Voltage source requirements

E. Interference levels

F. Processing power

Answer: **A**

Answer: **A**

Question #278 (Topic: Exam A)

Which of the following routing protocols needs to have an autonomous system set in order to establish communication with neighbor devices?

A. OSPF

B. EIGRP

C. FHRP

D. RIP

Answer: **B**

Answer: **B**

Question #279 (Topic: Exam A)

An investment bank is seeking a DR backup solution. Which of the following provides the most cost-effective backup site?

A. Hot

B. Cold

C. Cluster

D. Warm

Answer: **B**

Answer: **B**

Question #280 (Topic: Exam A)

A firewall administrator is mapping a server's internal IP address to an external IP address for public use. Which of the following is the name of this function?

A. NAT

B. VIP

C. PAT

D. BGP

Answer: **A**

Answer: **A**

Question #281 (Topic: Exam A)

A network engineer is completing a new VoIP installation, but the phones cannot find the TFTP server to download the configuration files. Which of the following DHCP features would help the phone reach the TFTP server?

A. Exclusions

B. Lease time

C. Options

D. Scope

Answer: **C**

Answer: **C**

Question #282 (Topic: Exam A)

Which of the following best represents north-south traffic?

A. A connection between a computer and a public web server

B. Data moving between a data center and a DR site

C. Traffic between a production server and a backup server

D. Routing updates between OSPF routers

Answer: **A**

Answer: **A**

Question #283 (Topic: Exam A)

Which of the following provides an opportunity for an on-path attack?

A. Phishing

B. Dumpster diving

C. Evil twin

D. Tailgating

Answer: **C**

Answer: **C**

Question #284 (Topic: Exam A)

A network administrator is configuring a network for a new site that will have 150 users. Within the next year, the site is expected to grow by ten uses. Each user will have two IP addresses, one computer, and one phone connected to the network. Which of the following classful IPv4 address ranges will be best-suited for the network?

A. Class D

B. Class B

C. Class A

D. Class C

Answer: **B**

Answer: **B**

Question #285 (Topic: Exam A)

A network administrator needs to assign IP addresses to a newly installed network. They choose 192.168.1.0/24 as their network address and need to create three subnets

with 30 hosts on each subnet. Which of the following is a valid subnet mask that will meet the requirements?

A. 255.255.255.128

B. 255.255.255.192

C. 255.255.255.224

D. 255.255.255.240

Answer: **C**

Answer: **C**

Question #286 (Topic: Exam A)

The network engineering team needs to implement a wireless network within two separate buildings. Once the network is set up, users will need to authenticate using RADIUS in order to access internal resources. Which of the following wireless technologies should the team implement?

A. Mesh

B. Enterprise

C. Ad hoc

D. Point to point

Answer: **B**

Answer: **B**

Question #287 (Topic: Exam A)

A network engineer discovers network traffic that is sending confidential information to an unauthorized and unknown destination. Which of the following best describes the cause of this network traffic?

A. Ransomware

B. Darkware

C. Malware

D. Adware

Answer: **C**

Answer: **C**

Question #288 (Topic: Exam A)

A security administrator is looking for rogue servers on the network and would also like to determine which services are running on those servers. Which of the following software tools should the administrator use?

A. tracet

B. nmap

C. netstat

D. ping

Answer: **B**

Answer: **B**

Question #289 (Topic: Exam A)

A network administrator needs to secure SNMP access with authentication and encryption. Which of the following should the administrator use?

A. SNMPv3

B. SNMP community string

C. SNMP ACL access group

D. SNMPv2c

Answer: **A**

Answer: **A**

Question #290 (Topic: Exam A)

A user's desk has a workstation and an IP phone. The user is unable to browse the internet on the workstation, but the phone works. Which of the following configurations is required?

A. Voice VLAN

B. Native VLAN

C. Data VLAN

D. Trunk port

Answer: **C**

Answer: **C**

Question #291 (Topic: Exam A)

A network technician is configuring the company's network 100Mbps port Layer 2 switches. The technician wants increased throughput for the uplinks between switches. The technician connects multiple redundant links between the switches. Which of the following should the technician configure?

A. Spanning Tree Protocol

B. Switch Virtual Interfaces

C. Native VLAN

D. First Hop Redundancy Protocol

Answer: **A**

Answer: **A**

Question #292 (Topic: Exam A)

Which of the following would most likely be utilized to implement encryption in transit when using HTTPS?

A. SSH

B. TLS

C. SCADA

D. RADIUS

Answer: **B**

Answer: **B**

Question #293 (Topic: Exam A)

Some of the 20 employees who use the wireless network report they are unable to access network resources even though the wireless network is available. An administrator recently made the following configuration changes to the wireless DHCP server:

Network: 192.168.100.0

Mask: 255.255.255.240

Gateway: 192.168.100.1

Which of the following is the cause of this issue?

A. Incorrect VLAN assignment

B. Incorrect ACL configuration

C. Incorrect subnet mask

D. Incorrect default gateway

Answer: **C**

Answer: **C**

Question #294 (Topic: Exam A)

Which of the following typically uses compromised systems that become part of a bot network?

A. Evil twin attack

B. DDoS attack

C. XML injection

D. Brute-force password attack

Answer: **B**

Answer: **B**

Question #295 (Topic: Exam A)

Which of the following network ports is used when a client accesses an SFTP server?

A. 22

B. 80

C. 443

D. 3389

Answer: **A**

Answer: **A**

Question #296 (Topic: Exam A)

Two companies successfully merged. Following the merger, a network administrator identified a connection bottleneck. The newly formed company plans to acquire a high-end 40GB switch and redesign the network from a three-tier model to a collapsed core. Which of the following should the administrator do until the new devices are acquired?

A. Implement the FHRP.

B. Configure a route selection metric change.

C. Install a load balancer.

D. Enable link aggregation.

Answer: **D**

Answer: **D**

Question #297 (Topic: Exam A)

Which of the following is the step that a troubleshooter should take immediately after implementing a solution?

A. Review lessons learned during the process.

B. Establish a plan of action.

C. Verify full system functionality.

D. Document actions and outcomes.

Answer: **C**

Answer: **C**

Question #298 (Topic: Exam A)

Users can usually use RDP to connect to a terminal server with hostname TS19 that points to 10.0.100.19. However, users recently have been unable to connect to TS19. The technician pings 10.0.100.19 and gets an unreachable error. Which of the following is the most likely cause?

A. The users are on the wrong subnet.

B. The DHCP server renewed the lease.

C. The IP address was not reserved.

D. The hostname was changed.

Answer: **C**

Answer: **C**

Question #299 (Topic: Exam A)

Which of the following is the OSI model layer that uses a connectionless protocol for packet delivery?

A. Physical

B. Network

C. Transport

D. Application

Answer: **B**

Answer: **B**

Question #300 (Topic: Exam A)

A company experiences an incident involving a user who connects an unmanaged switch to the network. Which of the following technologies should the company implement to help avoid similar incidents without conducting an asset inventory?

A. Screened subnet

B. 802.1X

C. MAC filtering

D. Port security

Answer: **D**

Answer: **D**

Question #301 (Topic: Exam A)

Which of the following are benefits of implementing MFA and SSO in an enterprise environment? (Choose two.)

- A. Eliminating the number of password resets
- B. Centralizing user account management activities
- C. Adding an extra layer of security against compromised credentials
- D. Reducing the overhead of security tokens
- E. Enforcing conditional access policies
- F. Leveraging threat feed information from external sources

Answer: **BC**

Answer: **BC**

Question #302 (Topic: Exam A)

Users report connectivity issues after a network switch is replaced. There are three separate subnets for users, voice, and servers. The users cannot reach the resources in the server subnet. Upon investigation, a separate voice switch has thousands of discarded packets, but no users report issues making or receiving calls. Which of the following is most likely the issue?

- A. Incorrect cable
- B. Port administratively disabled
- C. VLAN misconfiguration
- D. Duplex setting

Answer: **C**

Answer: **C**

Question #303 (Topic: Exam A)

A network administrator needs to deploy a subnet using an IP address range that can support at least 260 devices with the fewest wasted addresses. Which of the following subnets should the administrator use?

A. 172.16.0.0/24

B. 172.25.2.0/23

C. 172.30.1.0/22

D. 172.33.0.0/21

Answer: **B**

Answer: **B**

Question #304 (Topic: Exam A)

Which of the following will allow secure, remote access to internal applications?

A. VPN

B. CDN

C. SAN

D. IDS

Answer: **A**

Answer: **A**

Question #305 (Topic: Exam A)

A junior network administrator gets a text message from a number posing as the domain registrar of the firm. The administrator is tricked into providing global administrator credentials. Which of the following attacks is taking place?

A. DNS poisoning

B. ARP spoofing

C. Vishing

D. Smishing

Answer: **D**

Answer: **D**

Question #306 (Topic: Exam A)

A network engineer wants to implement a secure solution that provides authentication and encryption to gather and monitor logs within the company network. Which of the following tools will accomplish this task?

A. Syslog

B. RADIUS

C. SNMPv3

D. API

Answer: **C**

Answer: **C**

Question #307 (Topic: Exam A)

A tourist attempts to access an online banking site on a mobile device while in a public place. Which of the following is the tourist most likely to experience?

A. Vishing

B. Tailgating

C. Shoulder surfing

D. Peripheral attack

Answer: **C**

Answer: **C**

Question #308 (Topic: Exam A)

Which of the following is used to separate a corporate network into multiple logical networks?

A. Load balancer

B. VLAN

C. CDN

D. IPS

Answer: **B**

Answer: **B**

Question #309 (Topic: Exam A)

A medical clinic recently configured a guest wireless network on the existing router. Since then, guests have been changing the music on the speaker system. Which of the following actions should the clinic take to prevent unauthorized access? (Choose two.)

A. Isolate smart devices to their own network segment.

B. Configure IPS to prevent guests from making changes.

C. Install a new AP on the network.

D. Set up a syslog server to log who is making changes.

E. Change the default credentials.

F. Configure GRE on the wireless router.

Answer: **AE**

Answer: **AE**

Question #310 (Topic: Exam A)

A network administrator needs to disable all inbound and outbound telnet traffic in the network. Which of the following ports should the administrator disable?

A. 23

B. 53

C. 123

D. 443

Answer: **A**

Answer: **A**

Question #311 (Topic: Exam A)

Which of the following stages of the troubleshooting methodology involves considering multiple approaches?

A. Verify full functionality.

B. Implement the solution.

C. Document the findings and actions.

D. Establish a theory of probable cause.

Answer: **D**

Answer: **D**

Question #312 (Topic: Exam A)

After a recent merger, a large number of alerts are coming in regarding extremely high utilization. Which of the following should be generated to help inform new alerting requirements?

A. SLA

B. Network diagram

C. Baseline

D. Heat map

Answer: **C**

Answer: **C**

Question #313 (Topic: Exam A)

Which of the following is concerned with guaranteed availability of a cloud resource?

A. IPAM

B. SLA

C. MTBF

D. EOL

Answer: **B**

Answer: **B**

Question #314 (Topic: Exam A)

A technician is designing a cloud service solution that will accommodate the company's current size, compute capacity, and storage capacity. Which of the following cloud deployment models will fulfill these requirements?

A. SaaS

B. PaaS

C. IaaS

D. IaC

Answer: **C**

Answer: **C**

Question #315 (Topic: Exam A)

A network administrator deploys several new desk phones and workstation cubicles. Each cubicle has one assigned switchport. The administrator run the following commands:

```
switchport mode access
```

```
switchport voice vlan 69
```

With which of the following VLANs will the workstation traffic be tagged?

A. Private VLAN

B. Voice VLAN

C. Native VLAN

D. Data VLAN

Answer: **D**

Answer: **D**

Question #316 (Topic: Exam A)

Which of the following is associated with avoidance, acceptance, mitigation, and transfer?

A. Risk

B. Exploit

C. Threat

D. Vulnerability

Answer: **A**

Answer: **A**

Question #317 (Topic: Exam A)

Several users report poor performance when accessing shared drives on a file server named Filed. While troubleshooting the issue, the network administrator swaps the Ethernet connections between File01 and another server named File02. Shortly thereafter, users report issues with File02, but Filed is no longer an issue. Which of the following is most likely impacting performance?

A. Port security

B. Speed/duplex mismatch

C. PoE power budget reduction

D. Keep-alive beacon failure

Answer: **B**

Answer: **B**

Question #318 (Topic: Exam A)

A network technician is requesting a fiber patch cord with a connector that is round and twists to install. Which of the following is the proper name of this connector type?

A. ST

B. BNC

C. SC

D. LC

Answer: **A**

Answer: **A**

Question #319 (Topic: Exam A)

A data center administrator is evaluating the use of jumbo frames within a storage environment. Which of the following describes the best reason to use jumbo frames in the storage environment?

A. To reduce device overhead

B. To report on the current root switch in the STP

C. To improve routing convergence

D. To increase drive throughput

Answer: **A**

Answer: **A**

Question #320 (Topic: Exam A)

A network administrator needs to efficiently deploy 200 systems using RFC1918 addresses. Which of the following networks should the administrator use?

A. 10.10.10.0/8

B. 150.200.1.0/24

C. 192.168.1.0/24

D. 254.169.0.0/24

Answer: **C**

Answer: **C**

Question #321 (Topic: Exam A)

An administrator requests a certificate for the company's domain name. In order to prove ownership of the domain, the certificate authority asks the administrator to create a specific DNS record. Which of the following record types is required?

A. SOA

B. MX

C. NS

D. TXT

Answer: **D**

Answer: **D**

Question #322 (Topic: Exam A)

An organization has four departments that each need access to different resources that do not overlap. Which of the following should a technician configure in order to implement and assign an ACL?

A. VLAN

B. DHCP

C. VPN

D. STP

Answer: **A**

Answer: **A**

Question #323 (Topic: Exam A)

A network analyst is installing a wireless network in a corporate environment. Employees are required to use their domain identities and credentials to authenticate and connect to the WLAN. Which of the following actions should the analyst perform on the AP to fulfill the requirements?

A. Enable MAC security.

B. Generate a PSK for each user.

C. Implement WPS.

D. Set up WPA3 protocol.

Answer: **D**

Answer: **D**

Question #324 (Topic: Exam A)

A network administrator is configuring a wireless connection between two warehouses. Which of the following wireless network types would best accomplish this task?

A. Ad hoc

B. Infrastructure

C. Point-to-point

D. Mesh

Answer: **C**

Answer: **C**

Question #325 (Topic: Exam A)

A network engineer is designing an internal network that needs to support both IPv4 and IPv6 routing. Which of the following routing protocols is capable of supporting both IPv4 and IPv6?

A. OSPFv3

B. RIPv2

C. BGP

D. EIGRP

Answer: **D**

Answer: **D**

Question #326 (Topic: Exam A)

While conducting a network audit, an administrator notices the spanning tree root in VLAN 300 does not match the configuration baseline, which could cause network outages due to spanning tree update failures. Which of the following should the administrator do to restore consistency?

A. Change the designated port on the misconfigured switch to a root port.

B. Change the root port to a listening port on the misconfigured switch.

C. Change the blocked port of the misconfigured switch to a learning port.

D. Increase the bridge ID/priority on the misconfigured switch.

Answer: **D**

Answer: **D**

Question #327 (Topic: Exam A)

A network administrator needs to connect a multimode fiber cable from the MDF to the server room. The administrator connects the cable to Switch 2, but there is no link light. The administrator tests the fiber and finds it does not have any issues. Swapping the connection to Switch 1 in a working port is successful, but the swapped connection does not work on Switch 2. Which of the following should the administrator verify next?

A. Fiber length

B. Transceiver model

C. Connector type

D. Port speed

Answer: **B**

Answer: **B**

Question #328 (Topic: Exam A)

Which of the following architecture types is most commonly associated with east-west traffic?

A. Colocated VMs

B. Hybrid cloud

C. Client-server

D. Hub and spoke

Answer: **A**

Answer: **A**

Question #329 (Topic: Exam A)

A company has been added to an unapproved list because of spam. The network administrator confirmed that a workstation was infected by malware. Which of the following processes did the administrator use to identify the root cause?

A. Traffic analysis

B. Availability monitoring

C. Baseline metrics

D. Network discovery

Answer: **A**

Answer: **A**

Question #330 (Topic: Exam A)

Which of the following actions happen during the first step of the troubleshooting methodology? (Choose two.)

A. Question the obvious.

B. Divide and conquer.

C. Identify symptoms.

D. Establish a theory.

E. Use a top-to-bottom approach.

F. Question users.

Answer: **CF**

Answer: **CF**

Question #331 (Topic: Exam A)

Which of the following cable types allows the use of QSFP ports without requiring transceivers?

A. Multimode

B. Twinaxial

C. RG11

D. Category 6

Answer: **B**

Answer: **B**

Question #332 (Topic: Exam A)

A network administrator determines that some switch ports have more errors present than expected. The administrator traces the cabling associated with these ports. Which of the following would most likely be causing the errors?

A. Shielded conduit

B. Plenum spaces

C. Voltage transformers

D. Hydraulic lifts

Answer: **C**

Answer: **C**

Question #333 (Topic: Exam A)

A network administrator is troubleshooting an issue with a user's IP address. Which of the following layers of the OSI model should be considered?

A. Data link

B. Network

C. Session

D. Presentation

Answer: **B**

Answer: **B**

Question #334 (Topic: Exam A)

Which of the following will most likely allow a network administrator to work with current data after a disaster?

A. Review backup tapes.

B. Recover from cloud storage.

C. Fail over to a hot site.

D. Restore from a snapshot.

Answer: **C**

Answer: **C**

Question #335 (Topic: Exam A)

A network technician is attempting to map networking devices. When logged in to the local switch, the technician uses LLDP and discovers only those devices that are connected directly to the switch. Which of the following should the technician perform to discover all the networking devices?

A. Reconfigure the management IP addresses of the networking devices on the same subnet.

B. Enable Spanning Tree Protocol to allow the switches on the network to be discovered.

C. Log in to each of the connected networking devices and perform the process again.

D. Upgrade the OS on the networking devices to the current version of LLDP.

Answer: **C**

Answer: **C**

Question #336 (Topic: Exam A)

Which of the following can be used when a server at a remote site is physically unreachable?

A. OOB management

B. Crash cart

C. Jump box

D. Console

Answer: **A**

Answer: **A**

Question #337 (Topic: Exam A)

Which of the following source control features allows an administrator to test a new configuration without changing the primary configuration?

A. Central repository

B. Conflict identification

C. Branching

D. Version control

Answer: **C**

Answer: **C**

Question #338 (Topic: Exam A)

A network engineer needs to change, update, and control APs remotely, with real-time visibility over HTTPS. Which of the following will best allow these actions?

A. Web interface

B. Command line

C. SNMP console

D. API gateway

Answer: **A**

Answer: **A**

Question #339 (Topic: Exam A)

Which of the following is a location where a plenum cable is recommended?

A. Frequently traveled areas

B. Outdoor environments

C. Inside conduits

D. Airspaces above a ceiling

Answer: **D**

Answer: **D**

Question #340 (Topic: Exam A)

A customer calls the help desk to report issues connecting to the internet. The customer can reach a local database server. A technician goes to the site and examines the configuration:

```
IP Address.....192.168.1.223
Subnet Mask.....255.255.255.192
Default Gateway..192.168.1.190
DNS 1.....192.168.1.196
DNS 2.....8.8.8.8
```

Which of the following is causing the user's issue?

A. Incorrect DNS

B. Unreachable gateway

C. Failed root bridge

D. Poor upstream routing

Answer: **B**

Answer: **B**

Question #341 (Topic: Exam A)

A company recently implemented a videoconferencing system that utilizes large amounts of bandwidth. Users start reporting slow internet speeds and an overall decrease in network performance. Which of the following are most likely the causes of the network performance issues? (Choose two.)

A. DNS misconfiguration

B. Inadequate network security

C. Malware or a virus

D. Outdated software

E. Incorrect QoS settings

F. Network congestion

Answer: **EF**

Answer: **EF**

Question #342 (Topic: Exam A)

A technician needs to identify a computer on the network that is reportedly downloading unauthorized content. Which of the following should the technician use?

A. Anomaly alerts

B. Port mirroring

C. Performance monitoring

D. Packet capture

Answer: **D**

Answer: **D**

Question #343 (Topic: Exam A)

Which of the following troubleshooting steps would provide a change advisory board with the information needed to make a decision?

A. Identify the problem.

B. Develop a theory of probable cause.

C. Test the theory to determine cause.

D. Establish a plan of action.

Answer: **D**

Answer: **D**

Question #344 (Topic: Exam A)

Application software systems can no longer receive updates and security patches, and continuing to use the application software might expose the system to attacks. Which of the following best describes this scenario?

A. SLA

B. Licensing

C. SIEM

D. EOL

Answer: **D**

Answer: **D**

Question #345 (Topic: Exam A)

Which of the following technologies uses a VIP to provide gateway redundancy between two routers?

A. LACP

B. PAT

C. FHRP

D. BGP

Answer: **C**

Answer: **C**

Question #346 (Topic: Exam A)

Which of the following is the best way to reduce the likelihood of electrostatic discharge?

A. Uninterruptable power supply

B. Surge protector

C. Power distribution units

D. Temperature and humidity control

Answer: **D**

Answer: **D**

Question #347 (Topic: Exam A)

Several unauthorized APs near the corporate headquarters are broadcasting similar SSIDs. Which of the following types of attacks is being described?

A. Evil twin

B. Wi-Fi jamming

C. On-path

D. Wardriving

Answer: **A**

Answer: **A**

Question #348 (Topic: Exam A)

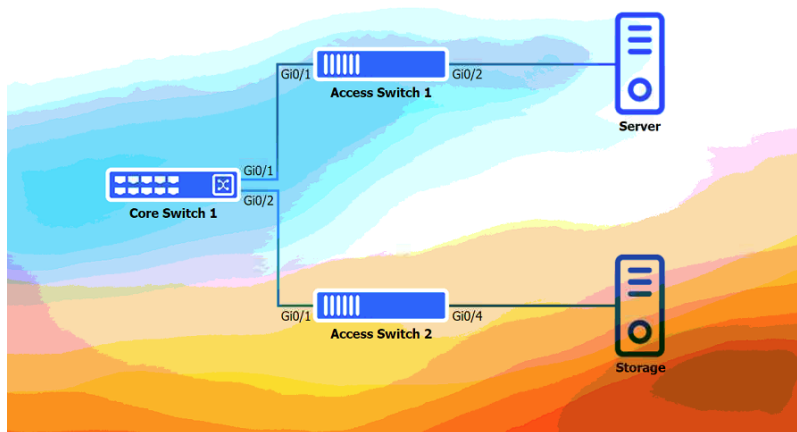
SIMULATION

A network administrator receives a ticket stating there is a connectivity issue between the server and the storage devices. The administrator needs to identify any issues and implement the appropriate solutions.

INSTRUCTIONS

Click on each access and core switch. Enter commands into the terminal to identify any problems and select the appropriate resolutions. Type help in each terminal to view a list of available commands.

If at any time you would like to bring back the initial state of the simulation, please click the Reset All button.



Access Switch 1	Solution
<pre>access-switch-1#</pre>	What problem was discovered? No problem found No problem found Incorrect VLAN Disabled interface CRC errors on the interface Giants and input errors Err-disabled interface Disconnected interface What should be done to fix the problem? Select a solution Select a solution Enable interface Assign correct VLAN to interface Check and replace cable/SFP Disable BPDU Guard Enable jumbo frames

Core Switch 1	Solution
<pre>core-switch-1#</pre>	What problem was discovered? No problem found No problem found Incorrect VLAN Disabled interface CRC errors on the interface Giants and input errors Err-disabled interface Disconnected interface What should be done to fix the problem? Select a solution Select a solution Enable interface Assign correct VLAN to interface Check and replace cable/SFP Disable BPDU Guard Enable jumbo frames

Access Switch 2	Solution
access-switch-2#	What problem was discovered? No problem found ☒ No problem found ☐ Incorrect VLAN ☐ Disabled interface ☐ CRC errors on the interface ☐ Giants and input errors ☐ Err-disabled interface ☐ Disconnected interface What should be done to fix the problem? Select a solution ☒ Select a solution ☐ Enable interface ☐ Assign correct VLAN to interface ☐ Check and replace cable/SFP ☐ Disable BPDU Guard ☐ Enable jumbo frames

Answer: Access Switch 1

Commands entered:

```
show interface GigabitEthernet0/2 status
show interface GigabitEthernet0/2
show interface GigabitEthernet0/2 switchport
```

Problem discovered: Err-disabled interface

Solution: Disable BPDU Guard

Core Switch 1

Commands entered:

```
show interface GigabitEthernet0/1 status
show interface GigabitEthernet0/2 status
```

Problem discovered: No problem found

Solution: None required

Access Switch 2

Commands entered:

```
show interface GigabitEthernet0/4 status
show interface GigabitEthernet0/4
show interface GigabitEthernet0/4 switchport
```

Problem: Giants and input errors

Solution: Enable jumbo frames

Answer: Access Switch 1
Commands entered:
`show interface GigabitEthernet0/2 status`
`show interface GigabitEthernet0/2`
`show interface GigabitEthernet0/2 switchport`

Problem discovered: Err-disabled interface
Solution: Disable BPDU Guard

Core Switch 1
Commands entered:
`show interface GigabitEthernet0/1 status`
`show interface GigabitEthernet0/2 status`

Problem discovered: No problem found
Solution: None required

Access Switch 2
Commands entered:
`show interface GigabitEthernet0/4 status`
`show interface GigabitEthernet0/4`
`show interface GigabitEthernet0/4 switchport`

Problem: Giants and input errors
Solution: Enable jumbo frames

Question #349 (Topic: Exam A)

A security engineer is trying to connect cameras to a 12-port PoE switch, but only eight cameras turn on. Which of the following should the engineer check first?

A. Ethernet cable type

B. Voltage

C. Transceiver compatibility

D. DHCP addressing

Answer: B

Answer: B

Question #350 (Topic: Exam A)

A network administrator is setting up a firewall to protect the organization's network from external threats. Which of the following should the administrator consider first when configuring the firewall?

A. Required ports, protocols, and services

B. Inclusion of a deny all rule

C. VPN access

D. Outbound access originating from customer-facing servers

Answer: **A**

Answer: **A**

Question #351 (Topic: Exam A)

A network administrator is unable to ping a remote server from a newly connected workstation that has been added to the network, Ping to 127.0.0.1 on the workstation is failing. Which of the following should the administrator perform to diagnose the problem?

A. Verify the NIC interface status.

B. Verify the network is not congested.

C. Verify the router is not dropping packets.

D. Verify that DNS is resolving correctly.

Answer: **A**

Answer: **A**

Question #352 (Topic: Exam A)

A company is installing network cables above the ceiling tiles. The cables must be fire rated. Which of the following cable types should the company use?

A. Coaxial

B. Plenum

C. Crossover

D. STP

Answer: **B**

Answer: **B**

Question #353 (Topic: Exam A)

A VoIP phone powers up, but it is unable to receive or place calls. After connecting to another switchport, the issue is resolved. Which of the following was most likely the issue with the previous port?

A. The port had a half-duplex configuration.

B. The port was in an error-disabled state.

C. The port had an incorrect VLAN assignment.

D. The port exceeded its PoE budget.

Answer: **C**

Answer: **C**

Question #354 (Topic: Exam A)

A network administrator recently configured an autonomous wireless AP and performed a throughput test via [comptiaspeedtester.com](https://www.comptiaspeedtester.com). The result was 75Mbps. When connected to other APs, the results reached 500Mbps. Which of the following is most likely the reason for this difference?

A. Channel width configuration

B. DNS server issues

C. Authentication failure

D. Incorrect DHCP settings

Answer: **A**

Answer: **A**

Question #355 (Topic: Exam A)

Which of the following technologies is most appropriate for a business that requires high-speed access to frequently used web content, such as images and videos?

A. CDN

B. SAN

C. Firewall

D. Switch

Answer: **A**

Answer: **A**

Question #356 (Topic: Exam A)

Which of the following should be included in an SLA?

A. Patches and bug fixes

B. Licensing

C. Inventory management

D. Stakeholders

Answer: **A**

Answer: **A**

Question #357 (Topic: Exam A)

A network administrator thinks a change was made to mail servers at comptia.org. Which of the following should the administrator execute to verify the change?

A. ping www.comptia.org

B. dig comptia.org ANY

C. nslookup -type=ns coir.ptia.org

D. tracert comptia.org

Answer: **B**

Answer: **B**

Question #358 (Topic: Exam A)

A network technician receives a new ticket while working on another issue. The new ticket is critical to business operations. Which of the following documents should the technician reference to determine which ticket to complete first?

A. NDA

B. AUP

C. SLA

D. MOU

Answer: **C**

Answer: **C**

Question #359 (Topic: Exam A)

A network engineer is troubleshooting connectivity for a newly installed server on an existing VLAN. The engineer reviews the following requested output:

```
c:\>ipconfig  
IP Address: 192.168.100.225  
Mask: 255.255.255.224  
Gateway: 192.168.100.254
```

The engineer then logs in to the router and runs the following:

```
Router# show ip route
Codes: I - IGRP derived, R - RIP derived, O - OSPF derived
       C - connected, S - static, E - EGP derived, B - BGP derived
       * - candidate default route, IA - OSPF inter area route
       E1 - OSPF external type 1 route, E2 - OSPF external type 2 route
Gateway of last resort is not set
C 192.168.100.0/24 is directly connected, GigabitEthernet0/0
```

Which of the following describes the issue?

- A. The server has an incorrect subnet mask.
- B. There is a duplicate IP address on the network.
- C. The DHCP address pool is exhausted.
- D. The router is missing a default route.

Answer: **A**

Answer: **A**

Question #360 (Topic: Exam A)

A network administrator installs new cabling to connect new computers and access points. After deploying the equipment, the administrator notices a few of the devices are not connecting properly. The administrator moves the devices to a different port, but it does not resolve the issue. Which of the following should the administrator verify next?

- A. Power budget
- B. Device requirements
- C. Port status
- D. Cable termination

Answer: **D**

Answer: **D**

Question #361 (Topic: Exam A)

Which of the following could provide a lightweight and private connection to a remote box?

A. Site-to-site VPN

B. Telnet

C. Console

D. Secure Shell

Answer: **D**

Answer: **D**

Question #362 (Topic: Exam A)

Which of the following routing protocols is most commonly used to interconnect WANs?

A. IGP

B. EIGRP

C. BGP

D. OSPF

Answer: **C**

Answer: **C**

Question #363 (Topic: Exam A)

A server administrator deploys a new web server with an IP address of 192.168.16.100. The security policy dictates that all non-secure connection methods should be blocked. Which of the following security rules will satisfy the request without prohibiting other traffic?

A. Deny any 192.168.16.0 255.255.255.0 80

B. Deny 192.168.16.100 255.255.255.255 any

C. Deny any 192.168.16.100 255.255.255.255 80

D. Permit any 192.168.16.100 255.255.255.255 80

E. Permit 192.168.16.100 255.255.255.255 any

Answer: **C**

Answer: **C**

Question #364 (Topic: Exam A)

Which of the following protocol ports should be used to securely transfer a file?

A. 22

B. 69

C. 80

D. 3389

Answer: **A**

Answer: **A**

Question #365 (Topic: Exam A)

A company reports that their facsimile machine no longer has a dial tone when trying to send a fax. The phone cable is damaged on one end. Which of the following types of connectors should a technician replace?

A. F-type

B. RJ45

C. SC

D. RJ11

Answer: **D**

Answer: **D**

Question #366 (Topic: Exam A)

A network technician is installing a new switch that does not support STP at the access layer of a network. The technician wants a redundant connection to the distribution switch. Which of the following should the technician use?

A. Link aggregation

B. Subinterfaces

C. Switch virtual interfaces

D. Half-duplex connections

Answer: **A**

Answer: **A**

Question #367 (Topic: Exam A)

A company recently rearranged some users' workspaces and moved several users to previously used workspaces. The network administrator receives a report that all of the users who were moved are having connectivity issues. Which of the following is the most likely reason?

A. Ports are error disabled.

B. Ports have an incorrect native VLAN.

C. Ports are having an MDIX issue.

D. Ports are trunk ports.

Answer: **B**

Answer: **B**

Question #368 (Topic: Exam A)

A newly opened retail shop uses a combination of new tablets, PCs, printers, and legacy card readers. Which of the following wireless encryption types is the most secure and compatible?

A. WPA3

B. WPA2

C. WPA2 / WPA3 mixed mode

D. WPA / WPA2 mixed mode

Answer: **C**

Answer: **C**

Question #369 (Topic: Exam A)

A network administrator changed an external DNS to point customers to a new server. Which of the following tools should the administrator use to test the new server's configuration?

A. ping

B. tracert

C. tcpdump

D. nslookup

Answer: **D**

Answer: **D**

Question #370 (Topic: Exam A)

A company is expanding to another floor in the same building. The network engineer configures a new switch with the same VLANs as the existing stack. When the network engineer connects the new switch to the existing stack, all users lose connectivity. Which of the following is the most likely reason?

A. The new switch has unused ports disabled.

B. The new switch does not have a default gateway.

C. The new switch is connected to an access port.

D. The new switch is in a spanning tree loop.

Answer: **D**

Answer: **D**

Question #371 (Topic: Exam A)

Which of the following is the most likely benefit of installing server equipment in a rack?

A. Simplified troubleshooting process

B. Decreased power consumption

C. Improved network performance

D. Increased compute density

Answer: **D**

Answer: **D**

Question #372 (Topic: Exam A)

A technician is implementing a new SD-WAN device with a default configuration. The technician receives a URL via email and connects the new device to the internet to complete the installation. Which of the following is this an example of?

A. Voice VLAN

B. Native VLAN

C. Data VLAN

D. Trunk port

Answer: **C**

Answer: **C**

Question #373 (Topic: Exam A)

A help desk technician receives a report that users cannot access internet URLs. The technician does some ping tests and finds that sites fail when a URL is used but work when an IP is used. Which of the following tools should the technician utilize next?

A. tcpdump

B. tracert

C. nmap

D. dig

Answer: **D**

Answer: **D**

Question #374 (Topic: Exam A)

Which of the following is the part of a DR plan that identifies the critical systems that should be recovered first after an incident?

A. RTO

B. SLA

C. MTBF

D. SIEM

Answer: **A**

Answer: **A**

Question #375 (Topic: Exam A)

Which of the following is the best VPN to use for reducing data bandwidth requirements of the corporate network?

A. Split-tunnel

B. Site-to-site

C. Full tunnel client

D. GRE tunnel

Answer: **A**

Answer: **A**

Question #376 (Topic: Exam A)

An ISP provided a company with a pre-configured modem and five public static IP addresses. Which of the following does the company's firewall require to access the internet? (Choose two.)

A. NTP server

B. Default gateway

C. The modem's IP address

D. One static IP address

E. DNS servers

F. DHCP server

Answer: **BD**

Answer: **BD**

Question #377 (Topic: Exam A)

During a recent security assessment, an assessor attempts to obtain user credentials by pretending to be from the organization's help desk. Which of the following attacks is the assessor using?

A. Social engineering

B. Tailgating

C. Shoulder surfing

D. Smishing

E. Evil twin

Answer: **A**

Answer: **A**

Question #378 (Topic: Exam A)

Which of the following dynamic routing protocols is used on the internet?

A. EIGRP

B. BGP

C. RIP

D. OSPF

Answer: **B**

Answer: **B**

Question #379 (Topic: Exam A)

A technician is implementing a new SD-WAN device with a default configuration. The technician receives a URL via email and connects the new device to the internet to complete the installation. Which of the following is this an example of?

A. SASE device installation

B. Zero-touch provisioning

C. Infrastructure as code

D. Configuration management

Answer: **B**

Answer: **B**

Question #380 (Topic: Exam A)

SIMULATION

A network technician needs to resolve some issues with a customer's SOHO network. The customer reports that some of the PCs are not connecting to the network, while others appear to be working as intended.

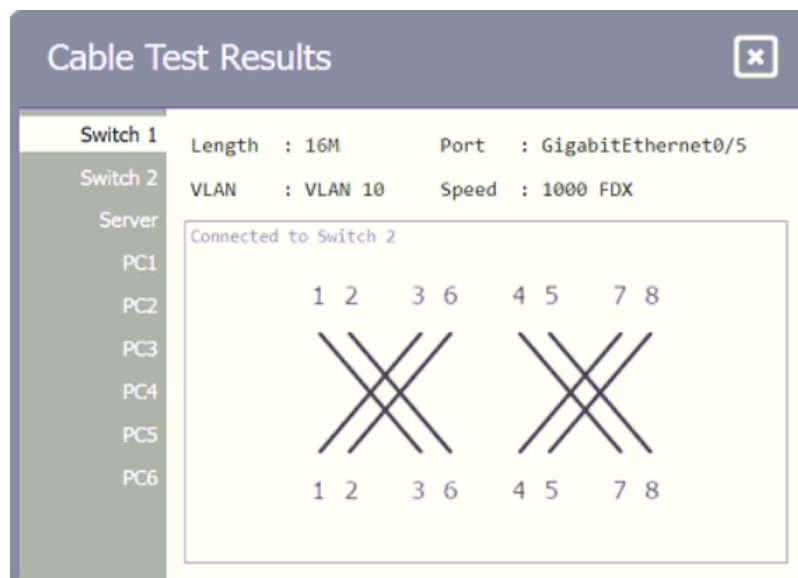
INSTRUCTIONS

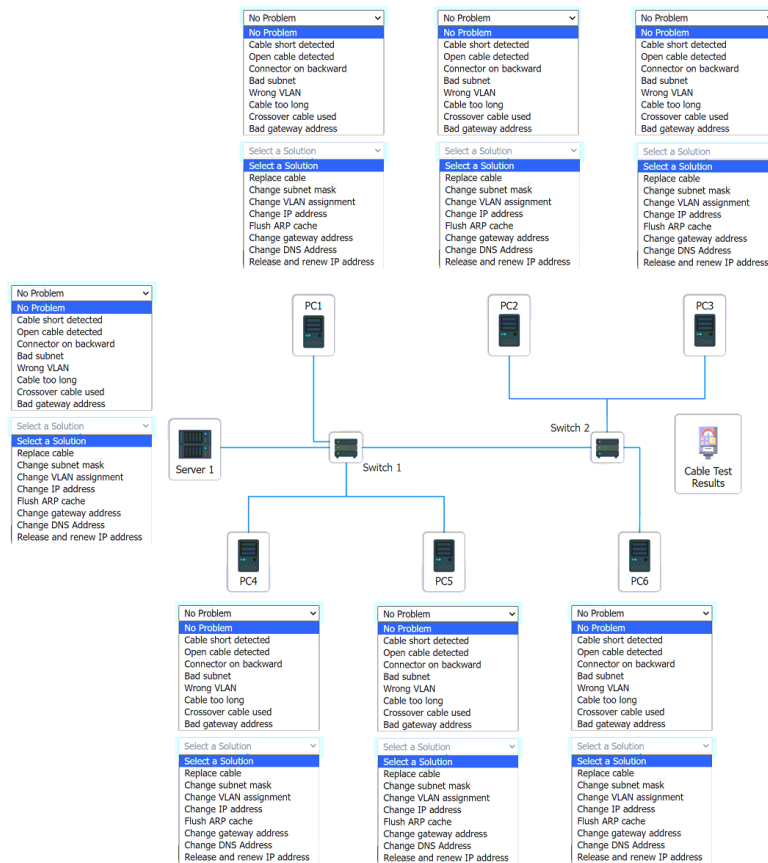
Troubleshoot all the network components.

Review the cable test results first, then diagnose by clicking on the appropriate PC and server. Type help in each terminal to view a list of available commands.

Identify any components with a problem and recommend a solution to correct each problem.

If at any time you would like to bring back the initial state of the simulation, please click the Reset All button.





Answer: **Cable Test (Patch Cable between Switch 1 Gi0/5 and Switch 2):**

- Problem discovered: Connector on backward
- Solution: Replace cable

Server 1:

- Problem discovered: Bad gateway address
- Solution: Change gateway address

PC2:

- Problem discovered: Wrong VLAN
- Solution: Change VLAN assignment

PC3:

- Problem discovered: Bad subnet
- Solution: Change the subnet mask

PC1, PC4, PC5, PC6:

- No problem found

Answer: **Cable Test (Patch Cable between Switch 1 Gi0/5 and Switch 2):**

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- Problem discovered: Wrong VLAN
- Solution: Change VLAN assignment

PC3:

- Problem discovered: Bad subnet
- Solution: Change the subnet mask

PC1, PC4, PC5, PC6:

- No problem found

Question #381 (Topic: Exam A)

Which of the following would allow a network administrator to remotely access a secure subnet from a shared, secure workstation?

A. Enable SSH through the firewall.

B. Dial into an in-band modem.

C. Connect through a jump host.

D. Configure a site-to-site VPN.

Answer: **C**

Answer: **C**

Question #382 (Topic: Exam A)

Which of the following kinds of targeted attacks uses multiple computers or bots to request the same resource repeatedly?

A. On-path

B. DDoS

C. ARP spoofing

D. MAC flooding

Answer: **B**

Answer: **B**

Question #383 (Topic: Exam A)

Which of the following is the most cost-effective way to safely expand outlet capacity in an IDF?

A. PDU

B. Surge protector

C. UPS

D. Power strip

Answer: **A**

Answer: **A**

Question #384 (Topic: Exam A)

A network administrator is creating a subnet that will include 45 separate hosts on a small private network within a large network architecture. Which of the following options is the most efficient use of network addresses when assigning this network?

A. 10.0.50.128/25

B. 10.7.142.128/27

C. 10.152.4.192/26

D. 10.192.1.64/28

Answer: **C**

Answer: **C**

Question #385 (Topic: Exam A)

A network technician is troubleshooting an issue between a web server and computers within a LAN. The technician needs to gather information about the IP addresses, protocols, and content of the traffic. Which of the following will best help the technician troubleshoot the issue?

A. Flow data

B. SNMP traps

C. Packet capture

D. Server logs

Answer: **C**

Answer: **C**

Question #386 (Topic: Exam A)

Which of the following OSI model layers can utilize a connectionless protocol for data transmission?

A. Physical

B. Network

C. Transport

D. Application

Answer: **C**

Answer: **C**

Question #387 (Topic: Exam A)

Which of the following layers in the OSI model is responsible for establishing, maintaining, and terminating connections between nodes?

A. Physical

B. Network

C. Session

D. Transport

Answer: **C**

Answer: **C**

Question #388 (Topic: Exam A)

Which of the following protocols is commonly associated with TCP port 443?

A. Telnet

B. SMTP

C. HTTPS

D. SNMP

Answer: **C**

Answer: **C**

Question #389 (Topic: Exam A)

A network engineer configures the network settings in a new server as follows:

IP address = 192.168.1.15

Subnet mask = 255.255.255.0

Gateway = 192.168.1.255

The server can reach other hosts on the same subnet successfully, but it cannot reach hosts on different subnets. Which of the following is most likely configured incorrectly?

A. Subnet mask

B. Gateway

C. Default route

D. IP address

Answer: **B**

Answer: **B**

Question #390 (Topic: Exam A)

A network administrator needs to implement a solution to filter access to the internet. Which of the following should the administrator most likely implement?

A. Router

B. Cloud gateway

C. Proxy

D. Intrusion detection system

Answer: **C**

Answer: **C**

Question #391 (Topic: Exam A)

A user in the finance department reports that they cannot access the internet. Other finance department users are not having issues.

A network administrator runs a route print command on the user's machine:

```
IPv4 route table
=====
Active routes:
Network Destination  Netmask          Gateway          Interface        Metric
10.66.0.0            255.255.0.0      On-link          10.66.1.97       301
10.66.1.97           255.255.255.255  On-link          10.66.1.97       301
10.66.255.255        255.255.255.255  On-link          10.66.1.97       301
192.168.0.0          255.255.0.0      192.168.0.1     192.168.0.1     101
192.168.0.0          255.255.255.0    192.168.40.1    192.168.0.1     101
127.0.0.0            255.0.0.0        On-link          127.0.0.1        331
127.0.0.1            255.255.255.255  On-link          127.0.0.1        301
127.0.0.1            255.255.255.255  On-link          127.0.0.1        301
127.255.255.255      255.255.255.255  On-link          127.0.0.1        331
224.0.0.0            240.0.0.0        On-link          127.0.0.1        331
224.0.0.0            240.0.0.0        On-link          10.66.1.97       331
255.255.255.255      255.255.255.255  On-link          127.0.0.1        301
255.255.255.255      255.255.255.255  On-link          10.66.1.97       301
```

Which of the following is most likely the root cause of the issue?

- A. The default route is missing.
- B. The route metric value is incorrect.
- C. The interface is pointing to the localhost.
- D. The default gateway is configured with an incorrect subnet mask.

Answer: **A**

Answer: **A**

Question #392 (Topic: Exam A)

Which of the following explains what happens if a packet is lost in transit when using UDP?

- A. The data link layer will recognize the error and resend the packet.
- B. IP uses the TTL field to track packet hops and will resend the packet if necessary.
- C. If the sender does not receive a UDP acknowledgement, the packet will be resent.
- D. Some applications will recognize the loss and initiate a resend of the packet if necessary.

Answer: **D**

Answer: **D**

Question #393 (Topic: Exam A)

After a networking intern plugged in a switch, a significant number of the users in a building lost connectivity. Which of the following is the most likely root cause?

A. VTP update

B. Port security issue

C. LLDP misconfiguration

D. Native VLAN mismatch

Answer: **A**

Answer: **A**

Question #394 (Topic: Exam A)

A network administrator upgraded the wireless access points and wants to implement a configuration that will give users higher speed and less channel overlap based on device compatibility. Which of the following will accomplish this goal?

A. 802.1X

B. MIMO

C. ESSID

D. Band steering

Answer: **D**

Answer: **D**

Question #395 (Topic: Exam A)

A technician is deploying new networking hardware for company branch offices. The bridge priority must be properly set. Which of the following should the technician configure?

A. Spanning tree protocol

B. Jumbo frames

C. Perimeter network

D. Port security

Answer: **A**

Answer: **A**

Question #396 (Topic: Exam A)

Which of the following should a company implement in order to share a single IP address among all the employees in the office?

A. STP

B. BGP

C. PAT

D. VXLAN

Answer: **C**

Answer: **C**

Question #397 (Topic: Exam A)

Which of the following is the greatest advantage of maintaining a cold DR site compared to other DR sites?

A. Redundancy

B. Availability

C. Security

D. Cost

Answer: **D**

Answer: **D**

Question #398 (Topic: Exam A)

A network engineer needs to deploy an access point at a remote office so that it will not communicate back to the wireless LAN controller. Which of the following deployment methods must the engineer use to accomplish this task?

A. ESSID

B. Lightweight

C. Autonomous

D. Ad hoc

Answer: **C**

Answer: **C**

Question #399 (Topic: Exam A)

A technician wants to experiment with NAC in a wireless environment. Which of the following NAC techniques should be implemented for wireless networking if ease of implementation is the most important factor?

A. Port security

B. EAPoL

C. 802.1X

D. MAC filtering

Answer: **D**

Answer: **D**

Question #400 (Topic: Exam A)

A junior network administrator is auditing the company network and notices incrementing input errors on a long-range microwave interface. Which of the following is the most likely reason for the errors?

A. The parabolic signal is misaligned.

B. The omnidirectional signal is being jammed.

C. The omnidirectional signal is not strong enough to receive properly.

D. The parabolic signal uses improper routing protocols.

Answer: **A**

Answer: **A**

Question #401 (Topic: Exam A)

A network administrator wants to update a geofencing policy to limit access to the corporate network based on location. Which of the following will the administrator most likely leverage?

A. MAC filtering

B. Administrative distance

C. Bluetooth beacon signals

D. IP address blocks

Answer: **D**

Answer: **D**

Question #402 (Topic: Exam A)

Which of the following used to store and deliver content to clients in a geographically distributed manner using edge servers?

A. Load balancer

B. CDN

C. DNS server

D. SAN

Answer: **B**

Answer: **B**

Question #403 (Topic: Exam A)

Which of the following connectors allows a singular QSFP transceiver to have several physical connections?

A. RJ45

B. ST

C. LC

D. MPO

Answer: **D**

Answer: **D**

Question #404 (Topic: Exam A)

An organization is concerned about rogue devices connecting to the network and wants the network team to implement a solution. Which of the following types of filtering should the network team use to address this concern?

A. MAC

B. Content

C. URL

D. IP

Answer: **A**

Answer: **A**

Question #405 (Topic: Exam A)

A network administrator is redesigning a wireless network in which corporate and guest users share a single SSID that is configured with WPA2 PSK. Which of the following capabilities should be configured in order to improve security? (Choose two.)

A. Wi-Fi Protected Access

B. New SSID for contractors

C. Password authentication

D. Wired Equivalent Privacy

E. Captive portal

F. Wi-Fi Protected Setup

Answer: **BE**

Answer: **BE**

Question #406 (Topic: Exam A)

A network administrator is migrating the existing phone system to VoIP. The vendor requires a separate voice network for the phones to function. Which of the following technologies should the administrator use to allow for this configuration?

A. VLAN

B. VXLAN

C. VPN

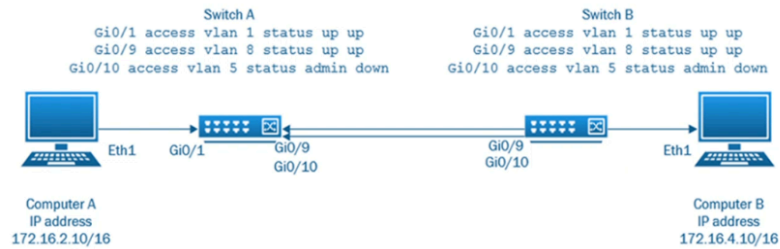
D. VPC

Answer: **A**

Answer: **A**

Question #407 (Topic: Exam A)

Computer A cannot reach Computer B in this network:



Which of the following will resolve the issue?

A. Set up link aggregation between Switch A and Switch B on Gi0/9 and Gi0/10.

B. Configure spanning tree on Switch A and Switch B.

C. Establish a trunk connection between Switch A and Switch B.

D. Put Computer A and Computer B in the same /24 subnet.

Answer: **C**

Answer: **C**

Question #408 (Topic: Exam A)

Which of the following most directly secures sensitive information on a network?

A. Data-in-transit encryption

B. Principle of least privilege

C. Role-based access controls

D. Multifactor authentication

Answer: **A**

Answer: **A**

Question #409 (Topic: Exam A)

A user is unable to reach a website and wants to verify its IP information. Which of the following tools is used to resolve the authoritative server to its FQDN?

A. ipconfig

B. arp

C. nslookup

D. ping

Answer: **C**

Answer: **C**

Question #410 (Topic: Exam A)

Users in one building on the campus network report very slow network access. A network technician logs in to the router to see if there are any errors associated with traffic to and from that particular building. Which of the following commands will the technician most likely run?

A. show arp

B. show config

C. show route

D. show interface

Answer: **D**

Answer: **D**

Question #411 (Topic: Exam A)

Which of the following is enforced through legislation?

A. AUP

B. GDPR

C. Code of conduct

D. EULA

Answer: **B**

Answer: **B**

Question #412 (Topic: Exam A)

Which of the following actions should be taken as part of the first step of the troubleshooting methodology?

A. Conduct tests to verify ideas.

B. Handle multiple problems individually.

C. Create a theory about the possible root cause.

D. Use a top-down approach.

Answer: **B**

Answer: **B**

Question #413 (Topic: Exam A)

After extremely high temperatures cause a power outage, the servers automatically shut down, even though the UPSs for the servers still have hours of battery life. Which of the following should a technician recommend?

A. Include backup power for air-conditioning units.

B. Configure door locks to automatically lock during power outages.

C. Increase UPS battery size.

D. Add an IoT-enabled thermostat.

Answer: **A**

Answer: **A**

Question #414 (Topic: Exam A)

While deploying a new fleet of computers on a DHCP network, a network administrator notices that new computers cannot connect to the internet. Which of the following is most likely the problem?

A. Incorrect default gateway

B. Address pool exhaustion

C. Duplicate IP address

D. Incorrect subnet mask

Answer: **B**

Answer: **B**

Question #415 (Topic: Exam A)

A network technician is designing a LAN for a new facility. The company is expecting more than 300 devices to connect to the network. Which of the following masks will provide the most efficient subnet?

A. 255.255.0.0

B. 255.255.192.0

C. 255.255.254.0

D. 255.255.255.254

Answer: **C**

Answer: **C**

Question #416 (Topic: Exam A)

A network administrator received reports that a 40Gb connection is saturated. The only server the administrator can use for data collection in that location has a 10Gb connection to the network. Which of the following is the best method to use on the server to determine the source of the saturation?

A. Port mirroring

B. Log aggregation

C. Flow data

D. Packet capture

Answer: **C**

Answer: **C**

Question #417 (Topic: Exam A)

A technician is alarmed after noticing more than 5,000 CRCs on a switchport. Which of the following should the technician do next?

A. Move to a different switchport.

B. Check for errdisable,

C. Swap out the patch cable.

D. Enable class-of-service.

Answer: **C**

Answer: **C**

Question #418 (Topic: Exam A)

A network engineer runs ipconfig and notices that the default gateway is 0.0.0.0. Which of the following address types is in use?

A. APIPA

B. Multicast

C. Class C

D. Experimental

Answer: **A**

Answer: **A**

Question #419 (Topic: Exam A)

A network administrator needs to deploy a new wireless solution at a customer site. All external users must provide credentials before being granted access to the wireless solution. Which of the following would best meet the requirements of this installation?

A. Adjust the bands to be non-overlapping.

B. Set up lightweight access points.

C. Enable enterprise authentication.

D. Add a captive portal for the guest network.

Answer: **D**

Answer: **D**

Question #420 (Topic: Exam A)

A network engineer adds a tunnel for a new branch network. Which of the following ensures that all data is encrypted inside the tunnel?

A. ESP

B. SSH

C. GRE

D. IKE

Answer: **A**

Answer: **A**

Question #421 (Topic: Exam A)

Which of the following routing technologies uses an attribute list for path selection?

A. BGP

B. RIP

C. EIGRP

D. OSPF

Answer: **A**

Answer: **A**

Question #422 (Topic: Exam A)

A network administrator is configuring a subnet that needs to support 13 usable IP addresses. Which of the following subnet masks best meets this requirement?

A. 255.255.255.128

B. 255.255.256.192

C. 255.285.256.224

D. 255.255.255.240

Answer: **D**

Answer: **D**

Question #423 (Topic: Exam A)

A systems administrator installs multiple web servers and intends to use the First Hop Redundancy Protocol to control routing to the servers. Which of the following types of IP addresses should the administrator use to support this routing approach?

A. Virtual

B. Gateway

C. Static

D. Dynamic

Answer: **A**

Answer: **A**

Question #424 (Topic: Exam A)

Which of the following is the best device to use to provide security and filtering between a private network and a public network?

A. Switch

B. Firewall

C. Bridge

D. Gateway

Answer: **B**

Answer: **B**

Question #425 (Topic: Exam A)

An employee has a new laptop and reports slow performance when using the wireless network. Switch firmware was updated the previous night. A network administrator logs in to the switch and sees the following statistics on the switch interface for that employee:

```
98469 packets input, 1681937 bytes, 0 no buffer
Received 1548 broadcasts (25285 multicasts)
65335 runts, 0 giants, 0 throttles
11546 input errors, 5 CRC, 0 frame, 0 overrun, 0 ignored
0 input packets with dribble condition detected
22781 packets output, 858040 bytes, 0 underruns
0 output errors, 89920 collisions, 0 interface resets
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out
```

Which of the following is most likely the cause of the issue?

A. The patch cord from the wall jack is faulty.

B. The switchport bandwidth needs to be increased.

C. Multicast is not configured correctly on the switch.

D. The NIC is set to half duplex.

Answer: **D**

Answer: **D**

Question #426 (Topic: Exam A)

Which of the following should an installer orient a port-side exhaust to when installing equipment?

A. The patch panel

B. The front of the IDF

C. The warm aisle

D. The administrator console

Answer: **C**

Answer: **C**

Question #427 (Topic: Exam A)

Which of the following ports is a secure protocol?

A. 20

B. 23

C. 443

D. 445

Answer: **C**

Answer: **C**

Question #428 (Topic: Exam A)

A detective is investigating an identity theft case in which the target had an RFID-protected payment card issued and compromised in the same day. The only place the target claims to have used the card was at a local convenience store. The detective notices a video camera at the store is placed in such a way that customers' credentials can be seen when they pay. Which of the following best explains this social engineering technique?

A. Shoulder surfing

B. Impersonation

C. Vishing

D. Tailgating

Answer: **A**

Answer: **A**

Question #429 (Topic: Exam A)

Which of the following can protect network equipment from voltage variations?

A. MDF

B. Power strips

C. PDU

D. UPS

Answer: **D**

Answer: **D**

Question #430 (Topic: Exam A)

Which of the following should a junior security administrator recommend implementing to mitigate malicious network activity?

A. IPS

B. Honeypot

C. SIEM

D. VPN

Answer: **A**

Answer: **A**

Question #431 (Topic: Exam A)

A company uses a solution to rotate the password that engineers use to connect to critical systems. Which of the following best describes this solution?

A. Network access control

B. TACACS+

C. Key management

D. Multifactor authentication

Answer: **C**

Answer: **C**

Question #432 (Topic: Exam A)

Which of the following is the best way to keep devices on during a loss of power?

A. UPS

B. Power load

C. PDU

D. Voltage

Answer: **A**

Answer: **A**

Question #433 (Topic: Exam A)

A network engineer needs to order cabling to connect two buildings within the same city. Which of the following media types should the network engineer use?

A. Coaxial

B. Twinaxial

C. Single-mode fiber

D. Cat 5

Answer: **C**

Answer: **C**

Question #434 (Topic: Exam A)

A server administrator needs to add a record to the company's DNS server to verify ownership of a web domain. The administrator has the record's name and value. Which of the following record types should the administrator use to add the record to the DNS server?

A. TXT

B. A

C. PTR

D. CNAME

Answer: **A**

Answer: **A**

Question #435 (Topic: Exam A)

A network technician is troubleshooting the connection to the company website. The traceroute command produces the following output:

```
Traceroute to www.mysite.com (8.8.8.8) over a maximum of 30 hops  
  1. 10.1.1.1      <1 ms    <1 ms    <1ms  
  2.              *         *         *  
  3.              *         *         *  
  4.              *         *         *  
  5.              *         *         *  
  
Traceroute complete
```

Which of the following should the technician do to identify the path to the server?

A. Review the router's ACL.

B. Execute netstat.

C. Perform an nslookup.

D. Enable LDP.

Answer: **A**

Answer: **A**

Question #436 (Topic: Exam A)

A major natural disaster strikes a company's headquarters, causing significant destruction and data loss. The company needs to quickly recover and resume operations. Which of the following will a network administrator need to do first?

A. Conduct a damage assessment.

B. Migrate to the cold site.

C. Notify customers of the disaster.

D. Establish a communication plan.

Answer: **A**

Answer: **A**

Question #437 (Topic: Exam A)

After installing a new wireless access point, an engineer tests the device and sees that it is not performing at the rated speeds. Which of the following should the engineer do to troubleshoot the issue? (Choose two.)

A. Ensure a bottleneck is not coming from other devices on the network.

B. Install the latest firmware for the device.

C. Create a new VLAN for the access point.

D. Make sure the SSID is not longer than 16 characters.

E. Configure the AP in autonomous mode.

F. Install a wireless LAN controller.

Answer: **AB**

Answer: **AB**

Question #438 (Topic: Exam A)

After a recent corporate merger, a network administrator needs to build reference documentation on multivendor infrastructure. Which of the following protocols will provide the information?

A. ICMP

B. CDP

C. LLDP

D. SIP

Answer: **C**

Answer: **C**

Question #439 (Topic: Exam A)

Which of the following steps of the troubleshooting methodology is most likely to involve talking to a user who opens a ticket?

A. Verifying full system functionality

B. Establishing a theory of probable cause

C. Identifying the problem

D. Implementing the solution

Answer: **C**

Answer: **C**

Question #440 (Topic: Exam A)

Which of the following is the most cost-effective way for a network administrator to establish a persistent, secure connection between two facilities?

A. Site-to-site VPN

B. GRE tunnel

C. VXLAN

D. Dedicated line

Answer: **A**

Answer: **A**

Question #441 (Topic: Exam A)

A network technician sets up a computer on the accounting department floor for a user from the marketing department. The user reports that they cannot access the marketing department's shared drives but can access the internet. Which of the following is the most likely cause of this issue?

- A. Mismatched switchport duplex
- B. Misconfigured gateway settings
- C. Incorrect VLAN assignment
- D. SVI is assigned to the wrong IP address

Answer: **C**

Answer: **C**

Question #442 (Topic: Exam A)

A network administrator is hosting a local application on a Type 2 hypervisor, but the application only has accessibility to the host machine. Which of the following address types would the administrator most likely visit to access the application?

- A. Public
- B. Loopback
- C. Private
- D. APIPA

Answer: **B**

Answer: **B**

Question #443 (Topic: Exam A)

A network manager connects two switches together and uses two connecting links. Which of the following configurations will prevent Layer 2 loops?

A. 802.1Q tagging

B. Full duplex

C. Link aggregation

D. QoS

Answer: **C**

Answer: **C**

Question #444 (Topic: Exam A)

Which of the following remote connectivity methods is used when data and management traffic use the same network infrastructure?

A. In-band

B. Native VLAN

C. Full-tunnel

D. Site-to-site

Answer: **A**

Answer: **A**

Question #445 (Topic: Exam A)

Which of the following steps of the troubleshooting methodology would most likely involve comparing current throughput tests to a baseline?

A. Implement the solution.

B. Verify full system functionality.

C. Document findings.

D. Test the theory.

Answer: **B**

Answer: **B**

Question #446 (Topic: Exam A)

Which of the following uses the longest prefix match to determine an exit interface?

A. ARP table

B. MAC address table

C. Routing table

D. Netstat table

Answer: **C**

Answer: **C**

Question #447 (Topic: Exam A)

A network administrator needs to securely connect to an Ubuntu server for management purposes. Which of the following protocols will most likely address this requirement?

A. HTTPS

B. SFTP

C. RDP

D. SSH

Answer: **D**

Answer: **D**

Question #448 (Topic: Exam A)

Which of the following allows for interactive, secure remote management of a network infrastructure device?

A. SSH

B. VNC

C. RDP

D. SNMP

Answer: **A**

Answer: **A**

Question #449 (Topic: Exam A)

After a recent security awareness phishing campaign, the cybersecurity team discovers that additional security measures need to be set up when users access potentially malicious websites. Which of the following security measures will best address this concern?

A. Implement DNS filtering.

B. Update ACLs to only allow HTTPS.

C. Configure new IPS hardware.

D. Deploy 802.1X security features.

Answer: **A**

Answer: **A**

Question #450 (Topic: Exam A)

During a security audit, a consulting firm notices inconsistencies between the documentation and the environment. Which of the following can keep a record of who made the changes and what the changes are?

A. Network access control

B. Configuration monitoring

C. Zero Trust

D. Syslog

Answer: **B**

Answer: **B**

Question #451 (Topic: Exam A)

Which of the following describes a malicious application that takes advantage of a software flaw?

A. Risk

B. Vulnerability

C. Exploit

D. Threat

Answer: **C**

Answer: **C**

Question #452 (Topic: Exam A)

Which of the following allows a standard user to log in to multiple resources with one account?

A. RADIUS

B. MFA

C. TACACS+

D. SSO

Answer: **D**

Answer: **D**

Question #453 (Topic: Exam A)

Which of the following describes the capability of a cloud service to adjust resources on demand?

A. Elasticity

B. Multitenancy

C. Containers

D. Scalability

Answer: **A**

Answer: **A**

Question #454 (Topic: Exam A)

A company's Chief Information Security Officer requires that servers and firewalls have accurate time stamps when creating log files so that security analysts can correlate events during incident investigations. Which of the following should be implemented?

A. Syslog server

B. SMTP

C. NTP

D. SNMP

Answer: **C**

Answer: **C**

Question #455 (Topic: Exam A)

Which of the following connector types is most commonly associated with Wi-Fi antennas?

A. BNC

B. SFP

C. MPO

D. RJ45

Answer: **A**

Answer: **A**

Question #456 (Topic: Exam A)

An organization recently connected a new computer to the LAN. The user is unable to ping the default gateway. The technician examines the configuration and sees a self-assigned IP address. Which of the following is the most likely cause?

A. The DHCP server is not available.

B. An RFC1918 address is being used.

C. The TCP/IP stack is disabled.

D. A static IP is assigned.

Answer: **A**

Answer: **A**

Question #457 (Topic: Exam A)

A network administrator is troubleshooting a connectivity issue between two devices on two different subnets. The administrator verifies that both devices can successfully ping other devices on the same subnet. Which of the following is the most likely cause of the connectivity issue?

A. Incorrect default gateway

B. Faulty Ethernet cable

C. Wrong duplex settings

D. VLAN mismatch

Answer: **A**

Answer: **A**

Question #458 (Topic: Exam A)

A network technician is attempting to harden a commercial switch that was recently purchased. Which of the following hardening techniques best mitigates the use of publicly available information?

A. Changing the default password

B. Blocking inbound SSH connections

C. Removing the gateway from the network configuration

D. Restricting physical access to the switch

Answer: **A**

Answer: **A**

Question #459 (Topic: Exam A)

A new backup system takes too long to copy files to the new SAN each night. A network administrator makes a simple change to the network and the devices to decrease backup times. Which of the following does the network administrator change?

A. QoS

B. SDN

C. MTU

D. VXLAN

E. TTL

Answer: **C**

Answer: **C**

Question #460 (Topic: Exam A)

Users are experiencing significant lag while connecting to a cloud-based application during peak hours. An examination of the network reveals that the bandwidth is being heavily utilized. Further analysis shows that only a few users are using the application at any given time. Which of the following is the most cost-effective solution for this issue?

A. Limit the number of users who can access the application.

B. Lease a Direct Connect connection to the cloud service provider.

C. Implement QoS to prioritize application traffic.

D. Use a CDN to service the application.

Answer: **C**

Answer: **C**

Question #461 (Topic: Exam A)

Users in a company report that after walking from one room to another, wireless connectivity is lost. The SSID is available, but the users have to manually reconnect every time they change rooms. Which of the following is most likely the cause of this issue?

A. Each room has wireless interference.

B. There is poor wireless coverage.

C. MAC filtering is applied in the WLAN.

D. Roaming configuration is disabled.

Answer: **D**

Answer: **D**

Question #462 (Topic: Exam A)

A network administrator needs to monitor data from recently installed firewalls in multiple locations. Which of the following solutions would best meet the administrator's needs?

A. IDS

B. IPS

C. SIEM

D. SNMPv2

Answer: **C**

Answer: **C**

Question #463 (Topic: Exam A)

Which of the following is an example of a split-tunnel VPN?

A. Only public resources are accessed through the user's internet connection.

B. Encrypted resources are accessed through separate tunnels.

C. All corporate and public resources are accessed through routing to on-site servers.

D. ACLs are used to balance network traffic through different connections.

Answer: **A**

Answer: **A**

Question #464 (Topic: Exam A)

A company's VoIP phone connection is cutting in and out. A senior network engineer is recommending the implementation of a voice VLAN. Which of the following should be configured?

A. 802.1Q tagging

B. Jumbo frames

C. Native VLAN

D. Link aggregation

Answer: **A**

Answer: **A**

Question #465 (Topic: Exam A)

During a VoIP call, a user notices inconsistent audio and logs an incident ticket. A network administrator notices inconsistent delays in arrival of the RTP packets. Which of the following troubleshooting tools should the network administrator use to determine the issue?

A. Toner and probe

B. Protocol analyzer

C. Cable tester

D. Spectrum reader

Answer: **B**

Answer: **B**

Question #466 (Topic: Exam A)

An administrator needs to configure an IoT device with a /21 subnet mask, but the device will only accept dotted decimal notation. Which of the following subnet masks should the administrator use?

A. 255.255.224.0

B. 255.255.240.0

C. 255.255.248.0

D. 255.255.252.0

Answer: **C**

Answer: **C**

Question #467 (Topic: Exam A)

Which of the following tools uses ICMP to help determine whether a network host is reachable?

A. tcpdump

B. netstat

C. nslookup

D. ping

Answer: **D**

Answer: **D**

Question #468 (Topic: Exam A)

Which of the following, in addition to a password, can be asked of a user for MFA?

A. PIN

B. Favorite color

C. Hard token

D. Mother's maiden name

Answer: **C**

Answer: **C**

Question #469 (Topic: Exam A)

A government entity wants to implement technology that can block websites based on country code. Which of the following will best enable this requirement?

A. URL filtering

B. Content filtering

C. DNS poisoning

D. MAC filtering

Answer: **A**

Answer: **A**

Question #470 (Topic: Exam A)

A network administrator needs to create a way to redirect a network resource that has been on the local network but is now hosted as a SaaS solution. Which of the following records should be used to accomplish the task?

A. TXT

B. AAA

C. PTR

D. CNAME

Answer: **D**

Answer: **D**

Question #471 (Topic: Exam A)

A network engineer configures an application server so that it automatically adjusts resource allocation as demand changes. This server will host a new application and demand is not predictable. Which of the following concepts does this scenario demonstrate?

A. Scalability

B. Software as a service

C. Hybrid cloud

D. Elasticity

Answer: **D**

Answer: **D**

Question #472 (Topic: Exam A)

A user submits an escalated ticket regarding failed logins on their laptop. The user states that the time displayed on the laptop is incorrect. An administrator thinks the issue is related to the NTP. Which of the following should the administrator do next?

A. Create a plan of action.

B. Implement a solution.

C. Identify the problem.

D. Test the theory.

Answer: **D**

Answer: **D**

Question #473 (Topic: Exam A)

Which of the following objectives does an evil twin achieve?

A. DNS poisoning

B. Login credentials

C. ARP spoofing

D. Denial of service

Answer: **B**

Answer: **B**

Question #474 (Topic: Exam A)

Which of the following server types is responsible for IP address management for clients?

A. NTP

B. FTP

C. DHCP

D. LDAP

Answer: **C**

Answer: **C**

Question #475 (Topic: Exam A)

Clients report that phones are down. The network technicians determine that PCs and printers are online. However, the phones display the message Searching for service. Which of the following should the technicians check first?

A. PoE statistics on the switch

B. DHCP options

C. Data VLANs on switchports

D. The default gateway on the switch

Answer: **B**

Answer: **B**

Question #476 (Topic: Exam A)

Which of the following physical installation factors would be the most important when a network switch is installed in a sealed enclosure?

A. Fire suppression

B. Power budget

C. Temperature

D. Humidity

Answer: **C**

Answer: **C**

Question #477 (Topic: Exam A)

A network engineer is installing new PoE wireless APs. The first five APs deploy successfully, but the sixth one fails to start. Which of the following should the engineer investigate first?

A. Signal strength

B. Duplex mismatch

C. Power budget

D. CRC

Answer: **C**

Answer: **C**

Question #478 (Topic: Exam A)

A network technician is working on a PC with a faulty NIC. The host is connected to a switch with secured ports. After testing the connection cables and using a known good NIC, the host is still unable to connect to the network. Which of the following is causing the connection issue?

A. MAC address of the new card

B. BPDU guard settings

C. Link aggregation settings

D. PoE over power budget

Answer: **A**

Answer: **A**

Question #479 (Topic: Exam A)

A network engineer added more APs to improve wireless coverage. However, users now report that the connectivity disconnects and reconnects repeatedly. Which of the following is causing the issue?

A. Throughput capacity

B. Roaming misconfiguration

C. Channel overlap

D. Packet loss

Answer: **C**

Answer: **C**

Question #480 (Topic: Exam A)

A network administrator is creating a copy of network traffic for later analysis. The flow of traffic should not be changed. Which of the following will most directly accomplish this task?

A. SPAN port

B. Network tap

C. Protocol analyzer

D. Link aggregation

Answer: **B**

Answer: **B**

Question #481 (Topic: Exam A)

Which of the following will a router prioritize first in the route selection process?

A. Prefix length

B. Metric

C. Administrative distance

D. Default route

Answer: **A**

Answer: **A**

Question #482 (Topic: Exam A)

A technician configures a new printer and places it on the network. Which of the following is the best way to ensure the printer will always keep the same IP configuration?

A. Add a new scope.

B. Place an exclusion.

C. Extend the lease time.

D. Create a reservation.

Answer: **D**

Answer: **D**

Question #483 (Topic: Exam A)

A user tries to visit www.abc.com, but the website that displays is www.cba.com. Which of the following should be done in order to reach the correct website?

A. Modify the CNAME record.

B. Update the PTR record.

C. Change the NTP settings.

D. Delete the TXT record.

Answer: **A**

Answer: **A**

Question #484 (Topic: Exam A)

Which of the following record types does an IP-to-name lookup use in DNS?

A. NS

B. PTR

C. A

D. CNAME

Answer: **B**

Answer: **B**

Question #485 (Topic: Exam A)

Which of the following is most closely associated with having a leased line to a public cloud provider?

A. VPN

B. Direct Connect

C. Internet gateway

D. Private cloud

Answer: **B**

Answer: **B**

Question #486 (Topic: Exam A)

A network engineer is implementing a new connection between core switches. The engineer deploys the following configurations to the switches:

core-sw01

```
vlan 100
  name server-vlan-100

interface Ethernet 1/1
  channel-group mode active

interface Ethernet 1/2
  channel-group mode active

interface port-channel 1
  switchport mode trunk
  switchport trunk allow vlan 100
```

core-sw02

```
vlan 100
  name server-vlan-100

interface Ethernet 1/1
  switchport mode trunk
  switchport trunk allow vlan 100

interface Ethernet 1/2
  switchport mode trunk
  switchport trunk allow vlan 100

interface port-channel 1
  switchport mode trunk
  switchport trunk allow vlan 100
```

Which of the following is the state of the core-sw01 port-channel interfaces?

A. Incrementing CRC errors

B. Error disabled

C. Administratively down

D. Suspended

Answer: **D**

Answer: **D**

Question #487 (Topic: Exam A)

A company is migrating a data center from on premises to the cloud. Which of the following tools will help maintain consistency, reliability, and efficiency of provisioning and management?

A. IaC

B. CDN

C. SASE

D. ZTA

Answer: **A**

Answer: **A**

Question #488 (Topic: Exam A)

A network administrator wants to update a geofencing policy to limit remote access to the corporate network based on country location. Which of the following would the administrator most likely leverage?

A. MAC filtering

B. Administrative distance

C. Bluetooth beacon signals

D. IP address blocks

Answer: **D**

Answer: **D**

Question #489 (Topic: Exam A)

Which of the following is the best outcome of documenting lessons learned after a complicated network issue is resolved?

A. Establishing action plans

B. Decreasing troubleshooting time

C. Identifying problematic employees

D. Increasing the network hardware budget

Answer: **B**

Answer: **B**

Question #490 (Topic: Exam A)

A network architect is implementing a VPN solution for remote workers. The architect wants to ensure that data transmitted through the VPN is protected from eavesdropping and tampering. Which of the following solutions meet these requirements? (Choose two.)

A. SSL

B. AES

C. TKIP

D. IPSec

E. EAP

F. SSH

Answer: **BD**

Answer: **BD**

Question #491 (Topic: Exam A)

A user runs ifconfig on a PC and sees the following address information:

```
IP address: 192.168.1.253
subnet mask: 255.255.255.0
default gateway: 192.168.1.1
```

The user then plugs a new device into the network and statically assigns it the address 192.168.1.255. The device does not work on the network. Which of the following explains the issue?

- A. The IP address is static.
- B. The IP address is a broadcast address.
- C. The IP address is outside the scope of the DHCP range.
- D. The IP address has a wrong subnet mask.

Answer: **B**

Answer: **B**

Question #492 (Topic: Exam A)

Which of the following Layer 1 devices is responsible for the termination of fiber-optic connections on an SFP-capable switch?

- A. Transceiver
- B. Modem
- C. Ethernet NIC
- D. Repeater

Answer: **A**

Answer: **A**

Question #493 (Topic: Exam A)

A small business is choosing between static and dynamic routing for its network. Which of the following is the best reason to use dynamic routing in a growing

network?

- A. Easier to configure compared to using manually entered routes
- B. Does not require additional network security controls
- C. Features enhanced network monitoring and visibility
- D. Includes automatic changes and updates in network topology

Answer: **D**

Answer: **D**

Question #494 (Topic: Exam A)

An employee connects to the company network from home using a VPN. Company data uses the VPN, while internet traffic uses the home ISP connection. Which of the following best describes the connectivity method in use?

- A. IPSec
- B. Split-tunnel
- C. Clientless
- D. Out-of-band

Answer: **B**

Answer: **B**

Question #495 (Topic: Exam A)

Three access points have Ethernet that runs through the ceiling. One of the access points cannot reach the internet. Which of the following tools can help identify the issue?

- A. Network tap
- B. Cable tester

C. Visual fault locator

D. Toner and probe

Answer: **B**

Answer: **B**

Question #496 (Topic: Exam A)

A network administrator deploys new network hardware. While configuring the network monitoring server, the server could authenticate but could not determine the specific status of the hardware. Which of the following would the administrator most likely do to resolve the issue?

A. Use the public community string.

B. Import the appropriate MIB.

C. Set up a switch port analyzer and forward traffic.

D. Configure SNMPv3 privacy.

Answer: **B**

Answer: **B**

Question #497 (Topic: Exam A)

A network engineer receives a new router to use for WAN connectivity. Which of the following best describes the layer the network engineer should connect the new router to?

A. Core

B. Leaf

C. Spine

D. Access

Answer: **A**

Answer: **A**

Question #498 (Topic: Exam A)

A network administrator is extending a network beyond the primary equipment location. Which of the following is where the administrator should install additional network switches?

A. MDF

B. VPC

C. VXLAN

D. IDF

Answer: **D**

Answer: **D**

Question #499 (Topic: Exam A)

A network administrator installs a wireless AP to an existing PoE switch on G1/0/1. The administrator notices that the new wireless AP is not receiving any power. The administrator logs in to the switch and views the following power statistics:



Which of the following is the most likely reason the wireless AP is not receiving power?

A. The switch is not configured to provide power over the connected port.

B. The wireless AP's power cable is damaged.

C. The wireless AP is not receiving power due to overutilization.

D. The switch's firmware needs to be updated.

Answer: **A**

Answer: **A**

Question #500 (Topic: Exam A)

A user cannot access an external server for a client after connecting to a VPN. Which of the following commands would a support agent most likely use to examine the issue? (Choose two.)

A. nslookup

B. tcpdump

C. arp

D. dig

E. tracert

F. route print

Answer: **EF**

Answer: **EF**

Question #501 (Topic: Exam A)

A network technician needs to use a private Class A RFC1918 addressing scheme for user laptops on the company's network. Which of the following subnets should the technician choose?

A. 10.1.1.0/24

B. 100.1.1.0/24

C. 172.16.1.0/24

D. 192.168.1.0/24

Answer: **A**

Answer: **A**

Question #502 (Topic: Exam A)

Which of the following cables is the most appropriate to use when running bulk cables in ceilings?

A. Plenum

B. Coaxial

C. Ethernet

D. DAC

Answer: **A**

Answer: **A**

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